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Service aaa

Overview

Relevant source files

- [

  README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md))

- [

  package.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json))

Purpose and Scope

This document provides a high-level introduction to the AAA (Authentication, Authorization, and Accounting) service, an OAuth2-compliant authentication and authorization server. It covers the service's core purpose, key features, technology stack, architectural components, and integration points within the microservices ecosystem.

For detailed information on specific subsystems:

- Deployment and runtime configuration: see Deployment Architecture and Configuration System
- Authentication methods and flows: see Authentication System and Grant Types and Login Methods
- Token lifecycle and management: see Token Management
- Data persistence and caching: see Data Layer Architecture and Redis Caching Service
- Message-based communication: see Kafka Message Consumers

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L1-L10)

README.md1-10 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L1-L10) Table of Contents

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Service Purpose

The AAA service is a comprehensive OAuth2-compliant authentication and authorization server designed for microservices architectures. It provides secure user authentication and authorization across multiple applications and services through standardized token-based flows.

Key Responsibilities:

- **Authentication**: Verify user identity through multiple methods (password, biometric, OTP, social login)
- **Authorization**: Manage access permissions through OAuth2 scopes and client credentials
- **Accounting**: Track authentication events, token lifecycle, and user sessions
- **Multi-Tenant Support**: Isolate configurations and cryptographic keys per domain
- **Service Integration**: Enable secure service-to-service communication via client credentials

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L1-L7)

README.md1-7 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L1-L7)[

package.json4](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L4-L4 (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L4-L4))

Technology Stack

Layer	Technology	Version	Purpose
Runtime	Node.js	16+ Alpine	JavaScript execution environment
Language	TypeScript	5.2+	Type-safe application code
Database	MySQL	8.0+	Persistent storage for users, tokens, clients
Cache	Redis	3.0+	High-performance session and token caching
Message Broker	Apache Kafka	2.8+	Asynchronous microservice communication
Service Discovery	Apache ZooKeeper	3.6+	Distributed coordination
Container	Docker	-	Multi-stage containerized deployment
Cryptography	JWT, RSA	-	Token signing and password encryption
Dependency Injection	TypeDI	0.8.0	Service container management

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L64-L75)

README.md64-75 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L64-L75)[

package.json18-40](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L18-L40 (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L18-L40))

Core Authentication Methods

The service implements 18+ OAuth2 grant types to support diverse authentication scenarios:

Grant Type	Use Case	Token Expiry
`password`	Standard username/password login	15 minutes (access)
`password_otp`	Password + SMS/Mobile OTP verification	90 seconds (OTP token)
`password_faceid`	Password + Face ID biometric	15 minutes
`biometric`	Biometric-only authentication	15 minutes
`access_google`	Google OAuth social login	15 minutes
`access_facebook`	Facebook OAuth social login	15 minutes
`access_apple`	Apple Sign-In integration	15 minutes
`client_credentials`	Service-to-service authentication	24 hours (refresh)
`demo`	Demo account access for testing	15 minutes
`link_account`	Multi-provider account linking	15 minutes

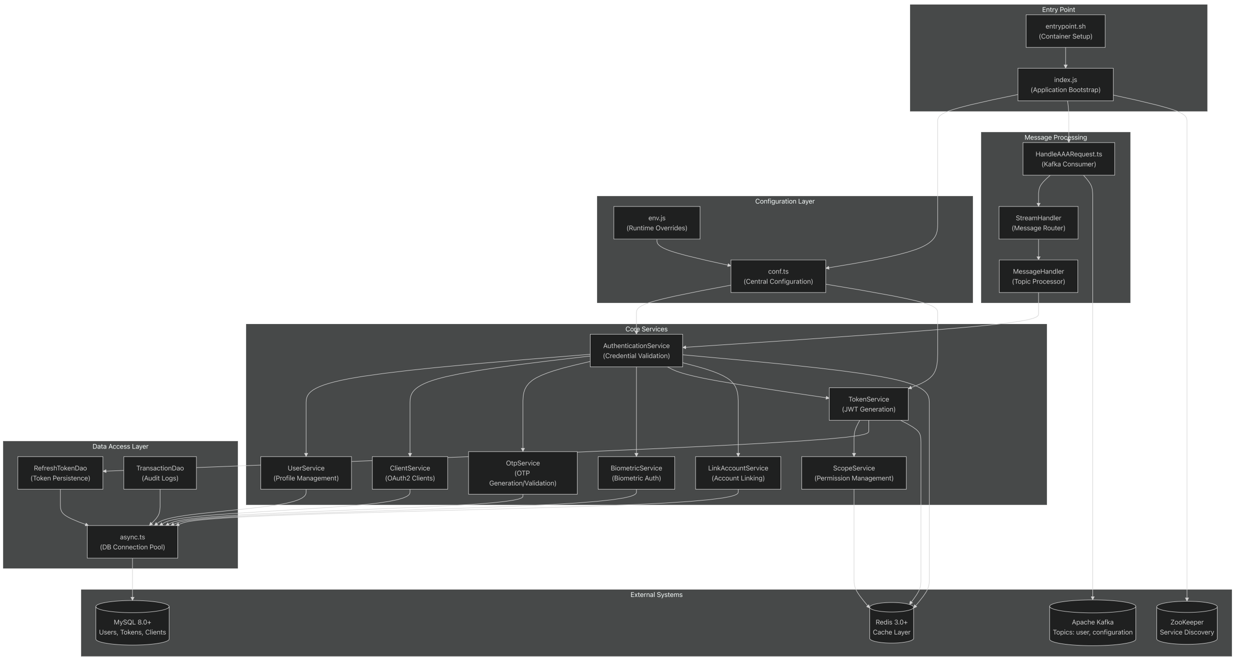
Additional variations support organization login, domain-based access, and partner-specific credential flows.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L11-L211)

README.md11-211 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L11-L211)

Service Architecture Components

The following diagram maps the high-level architectural concepts to their concrete implementations in the codebase:

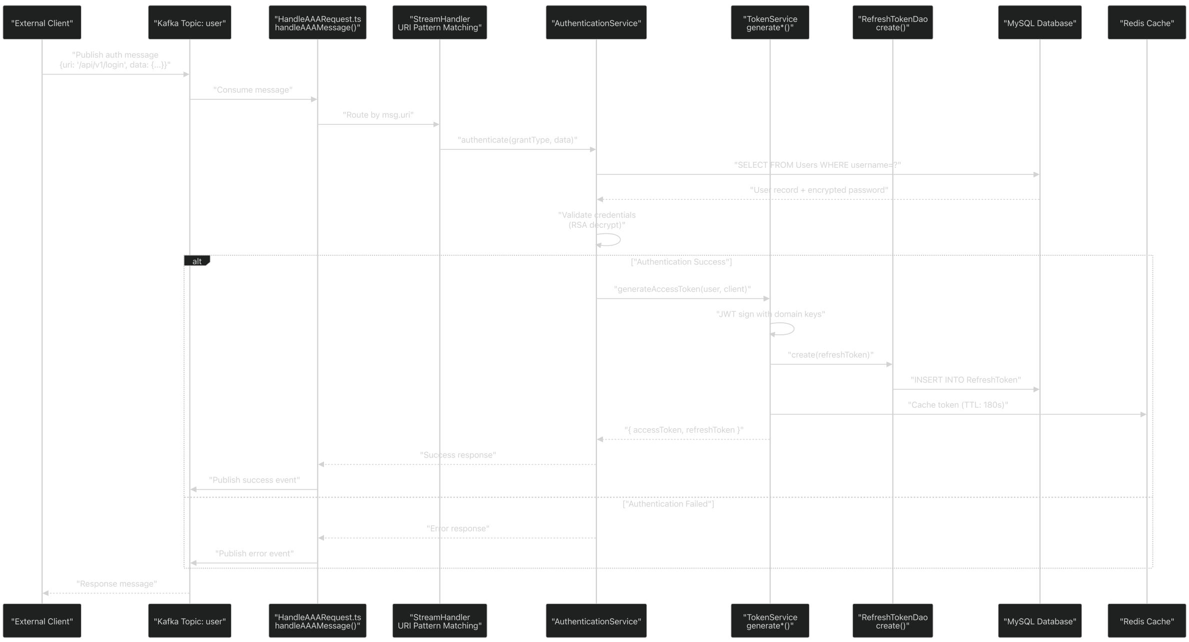


Sources: High-Level Diagram 1, (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L53-L62)

README.md53-62 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L53-L62)

Message-Driven Request Flow

The AAA service operates as a Kafka consumer, processing authentication requests asynchronously. The following sequence diagram illustrates how a login request flows through the system, mapping to actual code entry points:



Sources: High-Level Diagram 3, (https://github.com/nhsv-tradex/aaa/blob/59961d60/HandleAAARequest.ts)

[HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/HandleAAARequest.ts) (https://github.com/nhsv-tradex/aaa/blob/59961d60/HandleAAARequest.ts) (implied),[

async.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/async.ts (https://github.com/nhsv-tradex/aaa/blob/59961d60/async.ts)) (implied)

Security Architecture

Cryptographic Key Management

The service implements domain-specific cryptographic isolation:

Key Type	Purpose	Location Pattern	Algorithm
RSA Private Key	Password encryption	`keys/aaa/{domain}/rsa-private.key`	RSA 2048-bit
RSA Public Key	Password decryption	`keys/aaa/{domain}/rsa-public.key`	RSA 2048-bit
JWT Private Key	Token signing	`keys/{domain}/jwt-private.key`	RS256
JWT Public Key	Token verification	`keys/{domain}/jwt-public.key`	RS256

Keys are loaded from external repositories during build time and mounted into containers at runtime, never committed to source control.

Token Lifecycle

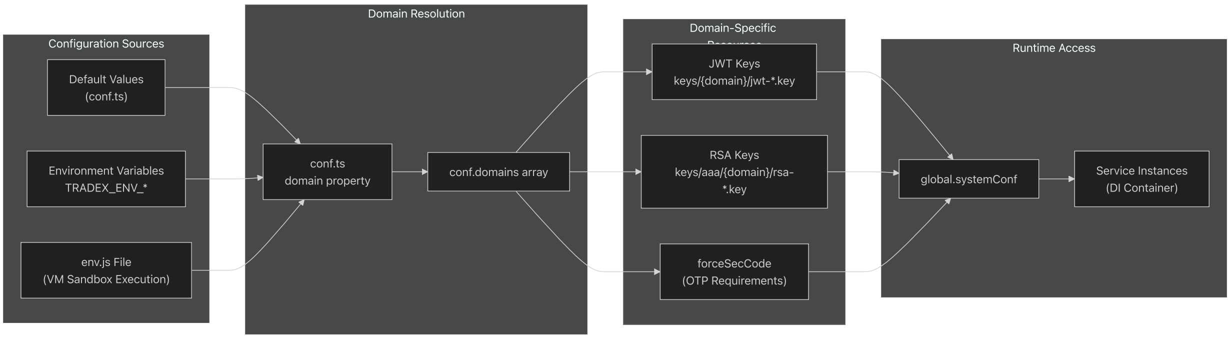
Token Type	Use Case	Expiration	Storage
Access Token	API authorization	15 minutes	Redis cache only
Refresh Token	Token renewal	24 hours	MySQL + Redis
Refresh Token (Remember)	Extended sessions	30 days	MySQL + Redis
OTP Token	Two-factor auth	90 seconds	MySQL + Redis

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L21-L160)

README.md21-160 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L21-L160) High-Level Diagram 5

Multi-Tenant Configuration

The service supports multiple tenant domains with isolated configurations:



Sources: High-Level Diagram 5, (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L102-L172)

README.md102-172 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L102-L172)

Database Schema Overview

The MySQL database implements the following core entity relationships:

Table	Primary Purpose	Key Relationships
`Users`	User accounts and authentication data	→ `RefreshToken`, `OTP`, `Biometric`, `LinkAccount`
`Clients`	OAuth2 client applications	→ `RefreshToken` (via clientId)
`LoginMethods`	Available authentication methods per user	→ `Users`
`Scopes`	Permission definitions	→ `ScopeGroups`
`ScopeGroups`	Permission groupings	← `Scopes`
`RefreshToken`	Token persistence with cascading deletes	← `Users`, `Clients`
`OTP`	One-time password records	← `Users`
`Biometric`	Biometric authentication data	← `Users`
`LinkAccount`	Multi-provider account links	← `Users` (userId + partnerId)
`Transaction`	Authentication event audit log	← `Users`

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L213-L225)

README.md213-225 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L213-L225)

Integration Points

Kafka Topics

Topic	Direction	Message Types
`user`	Consumer	Authentication requests, user operations
`configuration`	Consumer	Configuration updates, domain changes
`events`	Producer	Authentication events, token lifecycle events

External OAuth Providers

Provider	Grant Type	Configuration
Google	`access_google`	OAuth 2.0 client credentials
Facebook	`access_facebook`	Facebook App ID + Secret
Apple	`access_apple`	Apple Sign-In credentials
TechX	`access_techx`	Custom partner integration

Redis Caching Strategy

Data Type	Key Pattern	TTL	Invalidation
Access Tokens	`token:access:{token}`	900s (15 min)	Expiry only
Refresh Tokens	`token:refresh:{token}`	86400s (24 hrs)	Manual + Expiry
User Scopes	`scope:user:{userId}`	180s (3 min)	Manual on update
Client Data	`client:{clientId}`	180s (3 min)	Manual on update

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L30-L36)

README.md30-36 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L30-L36) High-Level Diagrams 1 & 3

Deployment Model

The service is containerized using Docker multi-stage builds:

1. **Base Stage**: Node.js 16 Alpine with system dependencies
2. **Builder Stage**: Full dependency installation and TypeScript compilation
3. **Production Stage**: Minimal runtime with compiled JavaScript and configuration

The `entrypoint.sh` script orchestrates environment setup by sourcing domain-specific configuration files before launching the application via `node build/src/index.js`.

Health Monitoring: The service exposes a `/currentTime` endpoint for container health checks (30-second interval).

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L132-L234)

README.md132-234 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L132-L234)[

dockerfile](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile)) (implied),[

entrypoint.sh](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh (https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh)) (implied), High-Level Diagram 4

Request Processing Model

All API operations are processed asynchronously through Kafka message patterns:

1. **Client** publishes request message to Kafka `user` topic
2. **HandleAAARequest.ts** consumes message and routes based on URI pattern
3. **Service Layer** processes authentication/authorization logic
4. **Data Access Layer** queries MySQL with Redis caching
5. **Response** published back to Kafka for client consumption

This event-driven architecture enables:

- **Horizontal Scaling**: Multiple AAA service instances consume from same topic
- **Fault Tolerance**: Message replay on failures
- **Async Processing**: Non-blocking request handling
- **Service Decoupling**: Clients don't require direct HTTP endpoints

Sources: High-Level Diagrams 2 & 3, (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L38-L51)

README.md38-51 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L38-L51)

Supported Grant Types Reference

The service implements 18 OAuth2 grant types and extensions:

Category	Grant Types
Password-Based	`password`, `password_otp`, `password_faceid`, `password_totp`, `password_hardware_otp`, `password_smart_otp`
Biometric	`biometric`, `otp_faceid`, `otp_hardware_otp_faceid`, `otp_mobile_otp_faceid`
Social OAuth	`access_google`, `access_facebook`, `access_apple`, `access_techx`
Organization	`organization_login`, `domain_login`
Service Auth	`client_credentials`
Special	`demo`, `link_account`

For detailed implementation of each grant type, see Grant Types and Login Methods.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L200-L211)

README.md200-211 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L200-L211)[

GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/GrantType.ts (https://github.com/nhsv-tradex/aaa/blob/59961d60/GrantType.ts)) (implied)

Monitoring and Observability

Aspect	Implementation
Logging	Log4js with file rotation via `conf.ts` configuration
Health Checks	`/currentTime` endpoint returns server timestamp
Metrics	Kafka consumer lag, message processing rates
Error Tracking	Comprehensive error codes with `tradex-common` framework
Audit Logs	Transaction records in MySQL for authentication events

For error handling details, see Error Handling.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L227-L258)

README.md227-258 (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L227-L258)[

conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/conf.ts (https://github.com/nhsv-tradex/aaa/blob/59961d60/conf.ts)) (implied)

System Architecture

Overview

Relevant source files

- [

  README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md (https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md))
- [

  package.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json))
- [

src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts))

This document provides a comprehensive overview of the AAA service's system architecture, including component relationships, data flow patterns, external service integration points, and the microservices ecosystem. The AAA service implements an OAuth2-compliant authentication and authorization server within a distributed microservices architecture using Apache Kafka for inter-service communication.

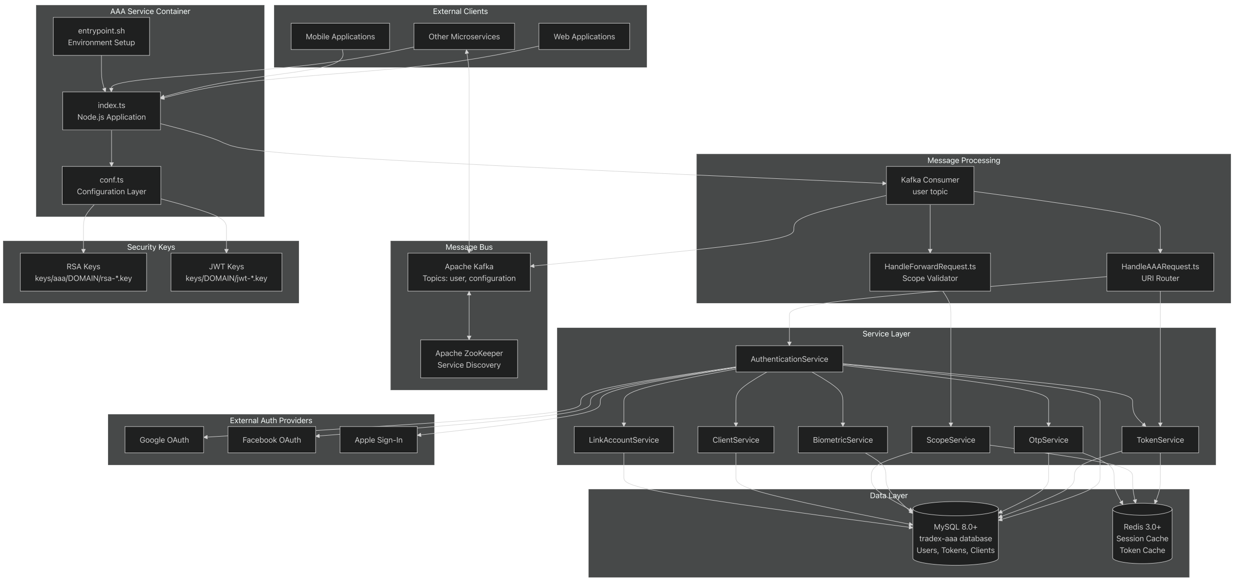
For detailed information about:

- **Deployment and build process**: See Deployment Architecture
- **Configuration management**: See Configuration System
- **Authentication flows**: See Authentication System
- **Database models and data access patterns**: See Data Layer Architecture
- **External service integrations**: See External Integrations

High-Level System Architecture

The AAA service operates as a containerized microservice within a distributed ecosystem, communicating via Apache Kafka message bus and managing authentication/authorization state through MySQL and Redis. The architecture follows a layered pattern with clear separation of concerns.

AAA Service Ecosystem



The AAA service container is initialized via `entrypoint.sh` which sets up environment variables before launching `index.ts`. The application consumes messages from the Kafka `user` topic, routing them through `HandleAAARequest.ts` for authentication flows and `HandleForwardRequest.ts` for scope-validated request forwarding. The service layer coordinates authentication methods, token management, and account operations while maintaining state in MySQL and caching in Redis.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1-L51)

src/index.ts1-51 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1-L51)[

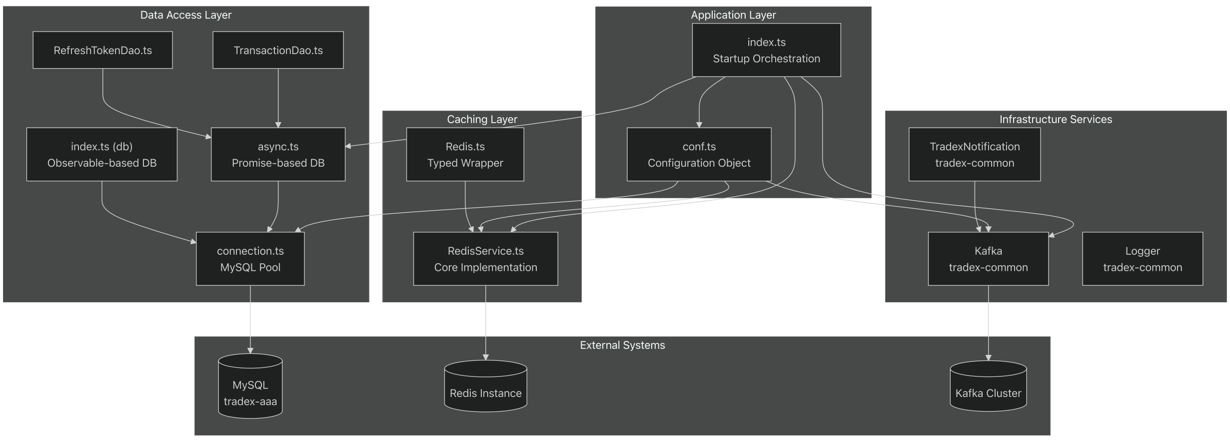
src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L1-L184)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L1-L184)%5B)

entrypoint.sh1-36](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh#L1-L36)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh#L1-L36)%5B)

src/HandleAAARequest.ts1-359](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)%5B)

src/HandleForwardRequest.ts1-87](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleForwardRequest.ts#L1-L87 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleForwardRequest.ts#L1-L87))

Internal Component Architecture



The internal architecture maintains strict layer separation: the application layer (`index.ts` , `conf.ts`) initializes infrastructure, the data access layer (`async.ts` , `index.ts` , DAOs) provides both Promise-based and Observable-based database access with connection pooling, and the caching layer (`RedisService.ts` , `Redis.ts`) provides typed caching with automatic serialization.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1-L51)

src/index.ts1-51 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1-L51)[

src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L1-L184)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L1-L184)%5B)

src/db/async.ts1-260](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts#L1-L260)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts#L1-L260)%5B)

src/db/index.ts1-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/index.ts#L1-L173)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/index.ts#L1-L173)%5B)

src/db/connection.ts1-37](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connection.ts#L1-L37)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connection.ts#L1-L37)%5B)

src/services/redis/RedisService.ts1-233](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/RedisService.ts#L1-L233)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/RedisService.ts#L1-L233)%5B)

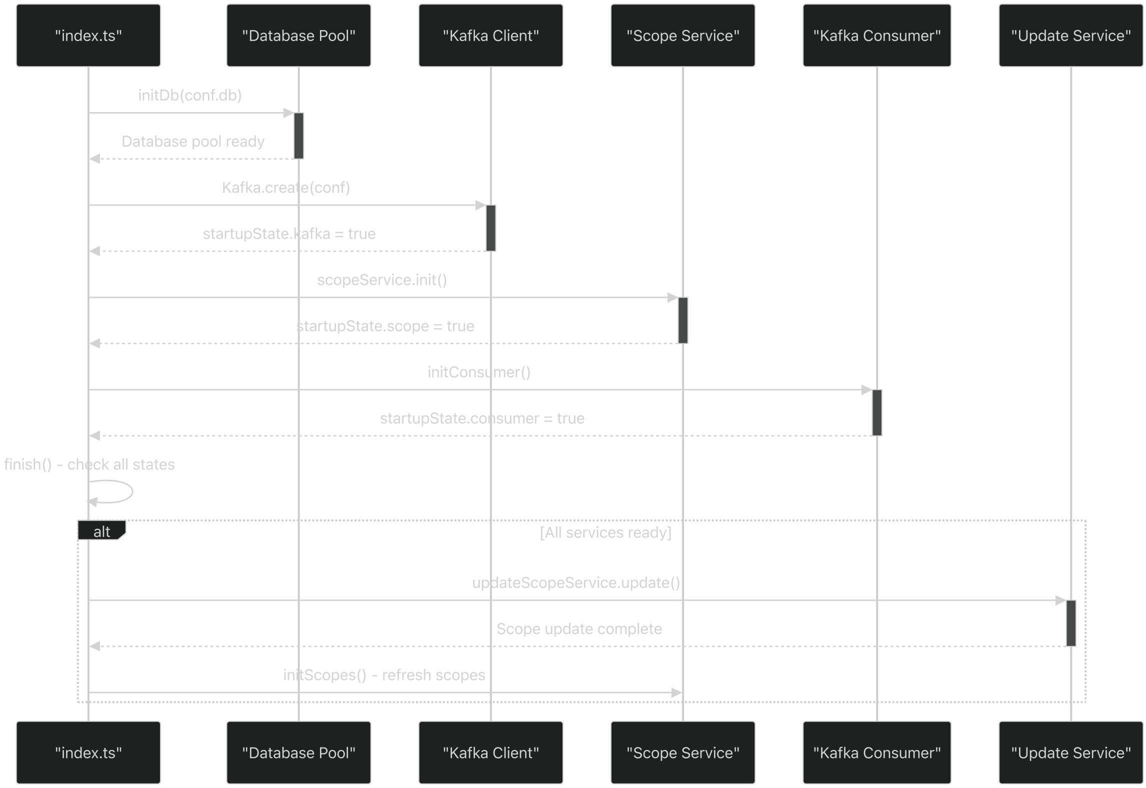
src/services/redis/Redis.ts1-53](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/Redis.ts#L1-L53)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/Redis.ts#L1-L53)%5B)

src/dao/TransactionDao.ts1-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/TransactionDao.ts#L1-L14)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/TransactionDao.ts#L1-L14)%5B)

src/dao/RefreshTokenDao.ts1-93](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/RefreshTokenDao.ts#L1-L93 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/RefreshTokenDao.ts#L1-L93)))

Application Startup Sequence

The AAA service follows a controlled startup sequence that ensures all dependencies are properly initialized before the service becomes available.



The startup process manages three critical state flags: `kafka`, `scope`, and `consumer`. Only when all three are successfully initialized does the service proceed to the final configuration update phase.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L12-L50)

src/index.ts12-50 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L12-L50)[

src/conf.ts12-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184))

Core Infrastructure Components

Database Connection Management

The AAA service provides both synchronous Observable-based and asynchronous Promise-based database access patterns.

Component	Purpose	Key Features
`connection.ts`	MySQL connection pooling	Connection pool management, type casting for BIT fields
`async.ts`	Promise-based DB operations	Async/await support, transaction management, bulk operations
`index.ts`	Observable-based DB operations	RxJS integration, reactive query patterns

The async layer provides the `Connection` class with methods for transaction management:

```
// Key methods from Connection class
beginTransaction(): Promise<any>
commit(): Promise<any>
rollback(): Promise<any>
query(sql: string, params: any[]): Promise<Pair<any, FieldInfo[]>>
insertBulk<T>(insertQuery: Query<T>, records: T[]): Promise<any>
```

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connection.ts#L1-L37)

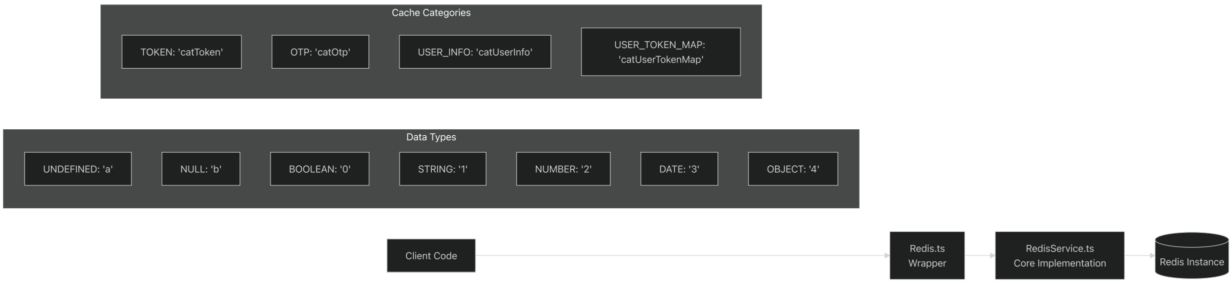
src/db/connection.ts1-37 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connection.ts#L1-L37)[

src/db/async.ts8-260](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts#L8-L260)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts#L8-L260)%5B)

src/db/index.ts15-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/index.ts#L15-L173 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/index.ts#L15-L173))

Redis Caching Service

The Redis service provides typed caching with automatic serialization and clustering support.



The Redis service supports automatic type serialization and deserialization, with cluster-aware key prefixing using `conf.clusterId`.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/RedisService.ts#L5-L233)

src/services/redis/RedisService.ts5-233 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/RedisService.ts#L5-L233)[

src/services/redis/Redis.ts1-53](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/Redis.ts#L1-L53 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/redis/Redis.ts#L1-L53))

Configuration Management

The configuration system loads from multiple sources with environment variable overrides and external configuration file support.

Configuration Area	Key Properties	Environment Variables
Database	`host`, `port`, `user`, `password`, `database`	`TRADEX_ENV_MYSQL_*`
Redis	`host`, `port`, `options`	Default: localhost:6379
Kafka	`kafkaUrls`, `kafkaConsumerOptions`, `kafkaProducerOptions`	`TRADEX_ENV_KAFKA_URLS`
JWT	`privateKeyFile`, `publicKeyFile`, domain-specific keys	Domain-based key paths
Tokens	`accessToken.expiredInSeconds`, `refreshToken.expiredInSeconds`	900s, 86400s defaults

The configuration system supports an external `env.js` file that can override any default values using a VM context.

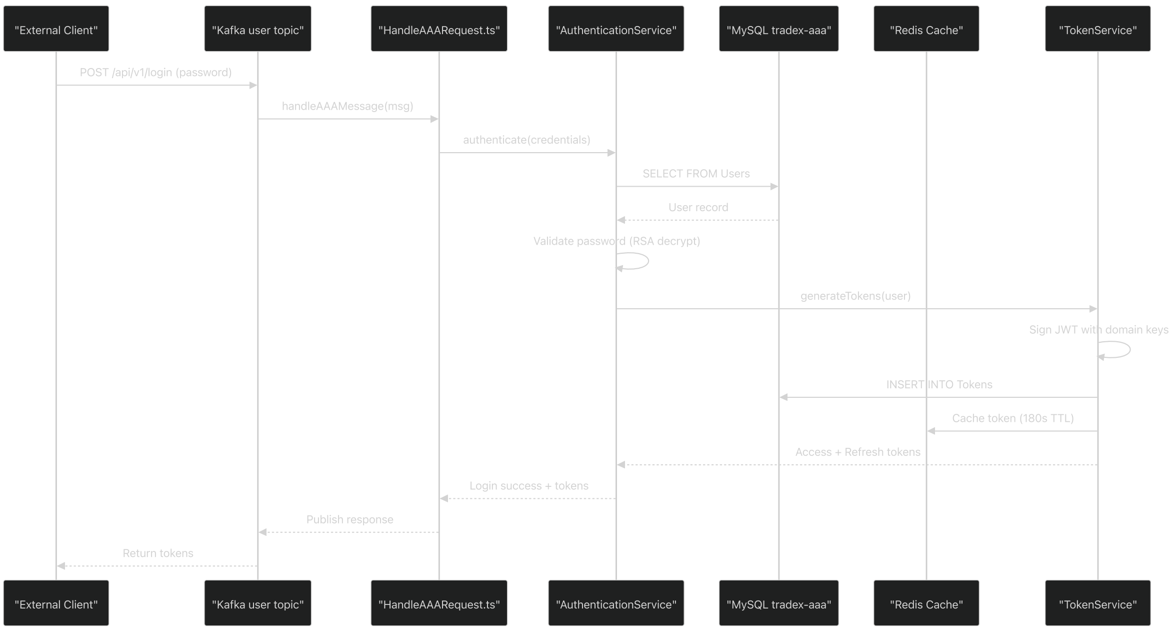
Sources: [(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184)](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184)

[src/conf.ts12-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184) (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L12-L184)

Data Flow Patterns

The AAA service implements multiple data flow patterns to handle different types of operations: authentication flows, token operations, and configuration updates.

Authentication Request Flow



Authentication flows begin when `HandleAAARequest.ts` receives login requests via Kafka. The service validates credentials (decrypting passwords with RSA keys from `keys/aaa/DOMAIN/rsa-private.key`), generates JWT tokens signed with domain-specific keys (`keys/DOMAIN/jwt-private.key`), persists tokens to MySQL, and caches them in Redis with a 180-second TTL (configurable via `conf.cacheTimeout`).

Sources: [(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)

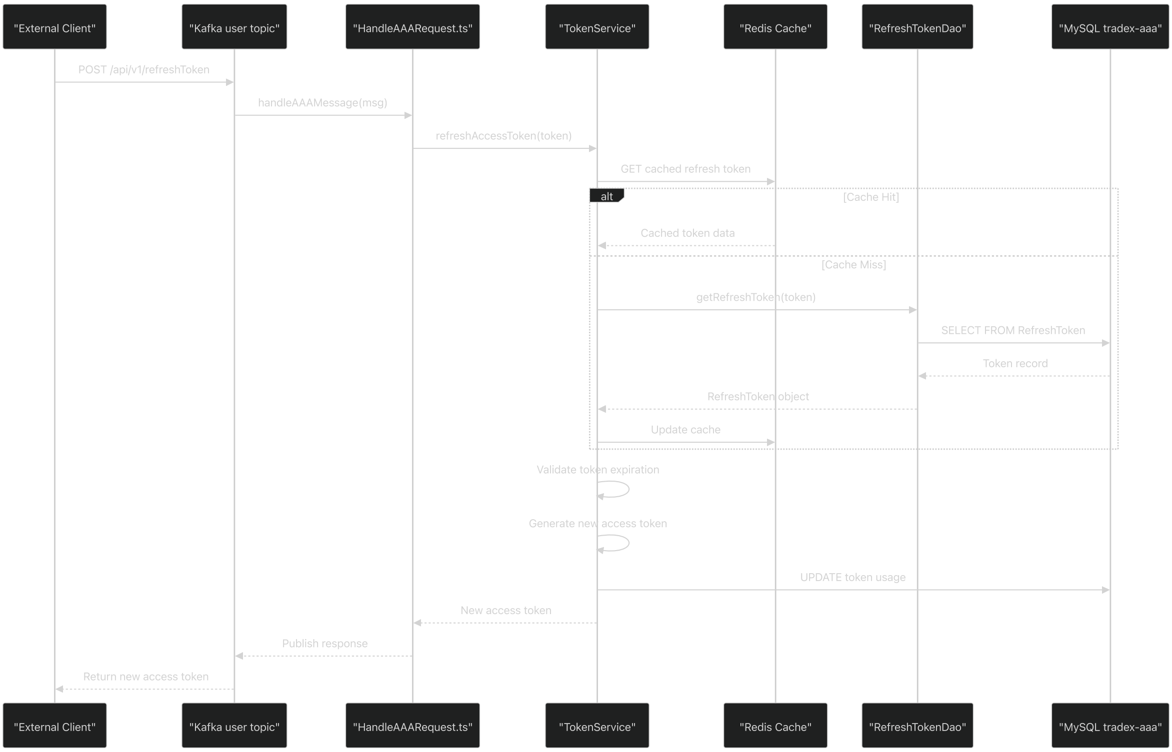
[src/HandleAAARequest.ts1-359](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359) (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)[

src/conf.ts42-46](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L42-L46)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L42-L46)%5B) src/conf.ts59-68](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L59-L68)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L59-L68)%5B)

src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L69-L78)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L69-L78)%5B)

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L127-L127 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L127-L127))

Token Refresh Flow



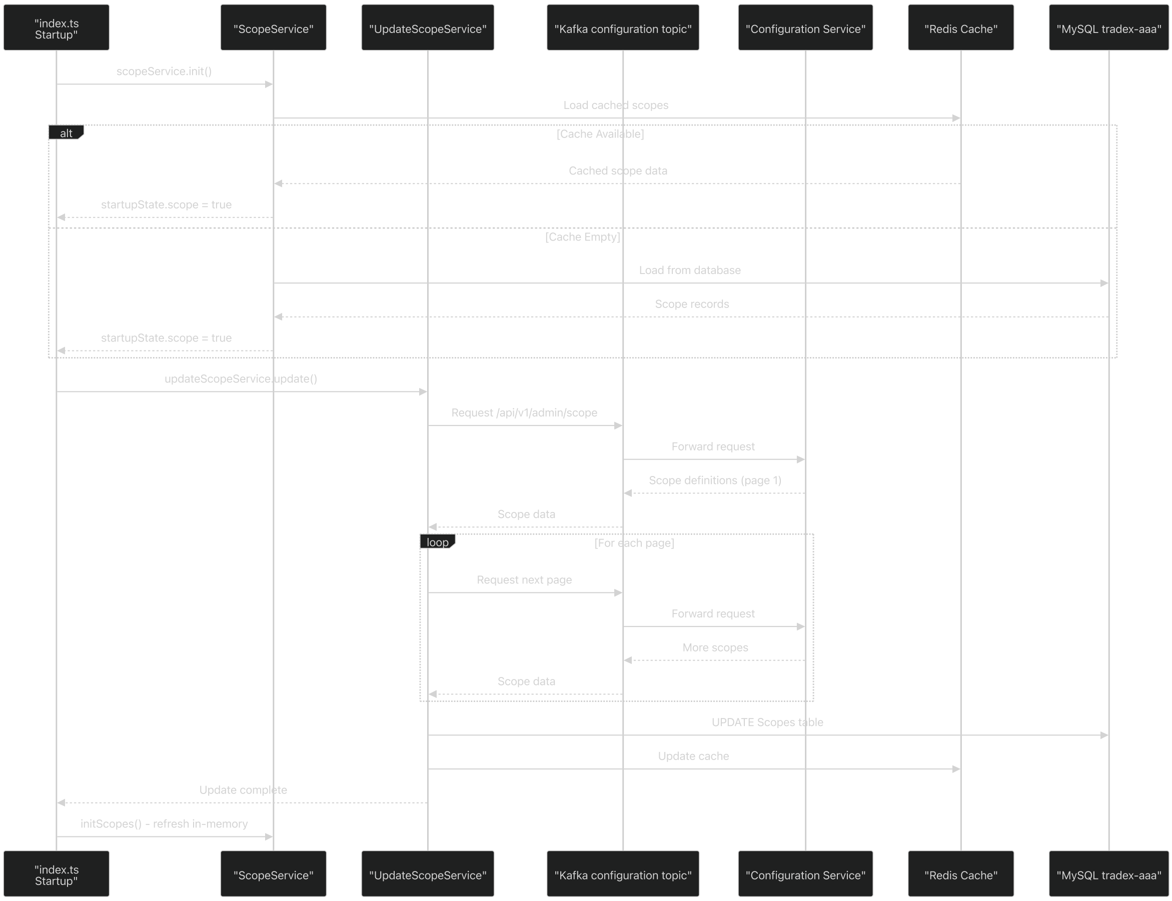
Token refresh operations prioritize Redis cache lookups before falling back to database queries via `RefreshTokenDao`. The service validates token expiration (default 86400s for refresh tokens, 2592000s with remember-me), generates new access tokens (default 900s expiration), and updates usage tracking in MySQL.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)

src/HandleAAARequest.ts1-359 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/HandleAAARequest.ts#L1-L359)[

src/dao/RefreshTokenDao.ts1-93](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/RefreshTokenDao.ts#L1-L93)[(https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/RefreshTokenDao.ts#L1-L93)%5B) src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L69-L78 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L69-L78))

Configuration Loading Flow



Configuration loading occurs during startup: `ScopeService` initializes from cache/database, then `UpdateScopeService` refreshes data from the `configuration` topic. Scopes are loaded via `/api/v1/admin/scope` with pagination (pageSize: 100 configured in `conf.scopes.loadFrom.uris.pageSize`). Similar flows occur for scope groups (`/api/v1/admin/scopeGroup`) and OAuth2 clients (`/api/v1/system/client`).

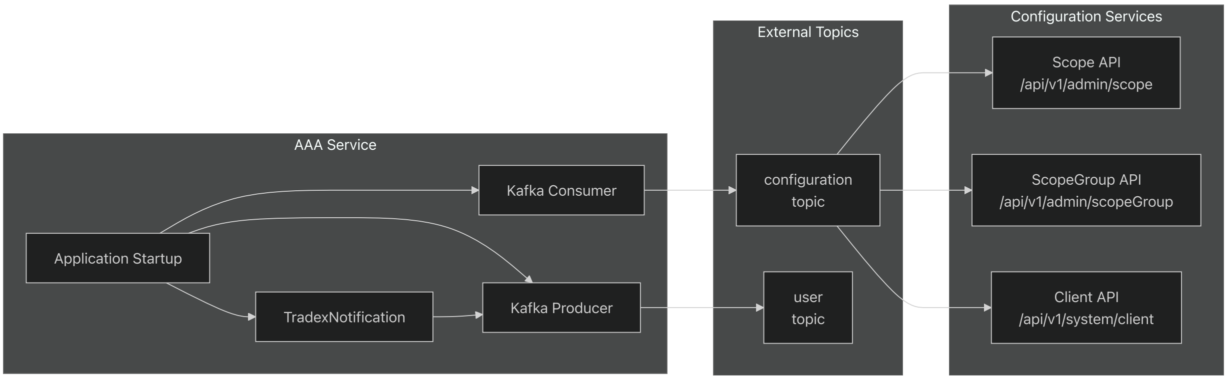
Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L12-L50)

[src/index.ts12-50](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L12-L50) (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L12-L50)[

src/conf.ts95-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L95-L115 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L95-L115))

Inter-Service Communication

The AAA service integrates with the broader microservices ecosystem through Kafka messaging and direct API calls.



The service loads configuration data from external services through Kafka topics, with configurable URIs and page sizes for batch loading operations.

Sources: (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L31-L36)

src/index.ts31-36 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L31-L36)[

src/conf.ts95-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L95-L115 (https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L95-L115))

Deployment Architecture

Relevant source files

- [

  dockerfile](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile))
- [

  entrypoint.sh](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh (https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh))
- [

  package.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json))

Purpose and Scope

This document describes the deployment architecture of the AAA service, covering the Docker containerization process, multi-stage build pipeline, runtime environment setup, and health monitoring. It focuses on how the application is built from source code, packaged into a Docker container, and prepared for execution.

For information about runtime configuration values and domain-specific settings, see Configuration System. For information about the application entry point and service initialization, see Authentication Service Core.

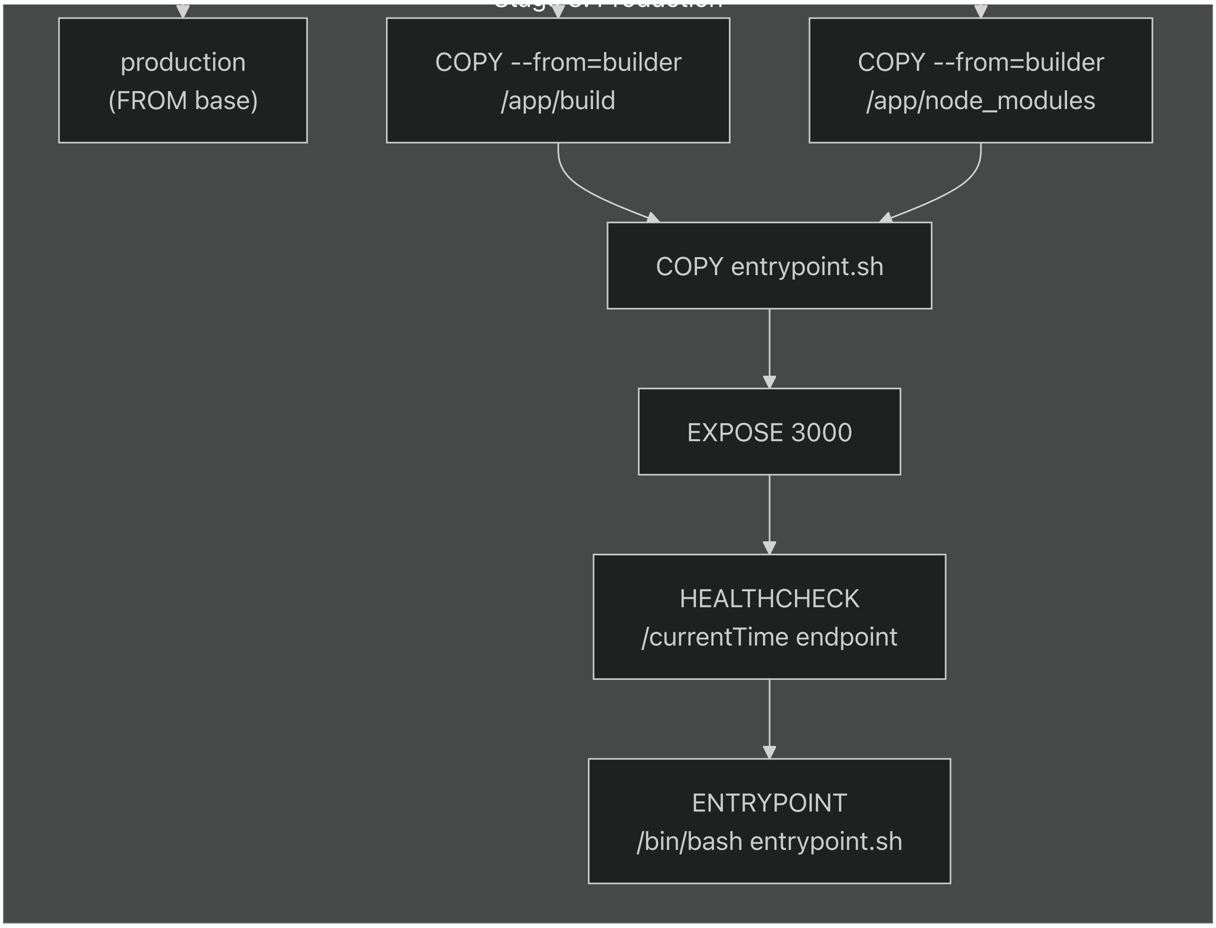
Multi-Stage Docker Build

The AAA service uses a multi-stage Docker build defined in (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#L1-L57)

dockerfile1-57 (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#L1-L57) to optimize image size and separate build-time dependencies from runtime dependencies.

Build Stages


Stage 1: Base
base
(node:16-alpine3.11)
Install system
dependencies
bash, git, python3, make,
g++
Copy package*.json
package-lock.json
npm install
--frozen-lockfile
--production=true
Stage 2: Builder
builder
(FROM base)
Copy source code
COPY . .
npm install
(all dependencies)
npm run build
(compile TypeScript)
Stage 3: Production



Sources: dockerfile

Stage Details

Stage	Base Image	Purpose	Key Operations
`base`	`node:16-alpine3.11`	Foundation layer	Install system dependencies (bash, git, python3, make, g++), install production npm dependencies with `--frozen-lockfile`
`builder`	`base`	Build environment	Copy full source code, install all dependencies (including devDependencies), execute `npm run build`
`production`	`base`	Runtime container	Copy only `/app/build` and `/app/node_modules` from builder, add `entrypoint.sh`, configure health check

Sources: dockerfile:2-57

System Dependencies

The base stage installs native build tools required for compiling native Node.js modules:

```
bash, git, python3, make, g++
```

These dependencies are installed via `apk` (Alpine Package Keeper) and the cache is cleared to minimize image size (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#L5-L11)

dockerfile5-11 (https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#L5-L11)

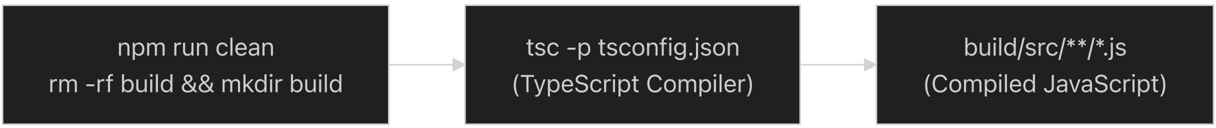
Sources: dockerfile:5-11

Build Process

TypeScript Compilation

The build process is triggered by `npm run build` (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L12-L12)

package.json12 (https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L12-L12) which executes the following steps:



Sources: package.json:7-12

Build Artifacts

The build process generates the following artifact structure:

```
build/
├── src/
│   ├── index.js           # Application entry point
│   ├── conf.js            # Configuration module
│   ├── HandleAAARequest.js
│   ├── services/          # Service layer
│   ├── dao/               # Data access objects
│   └── utils/             # Utility functions
└── node_modules/          # Runtime dependencies
```

Sources: package.json:12, dockerfile:44

Local Development Build

For local development, additional build steps are available:

Script	Command	Purpose
`copyKeys`	`mkdir -p build/src/keys && cp -R ../tradex-key/dev/aaa build/src/keys/aaa`	Copy RSA and JWT keys from external repository
`run-local`	`cp src/env.js build/src/env.js && npm run copyKeys && cd build/src && node index.js`	Copy environment config, keys, and run locally
`build-local`	`npm run build && npm run run-local`	Full local build and execution

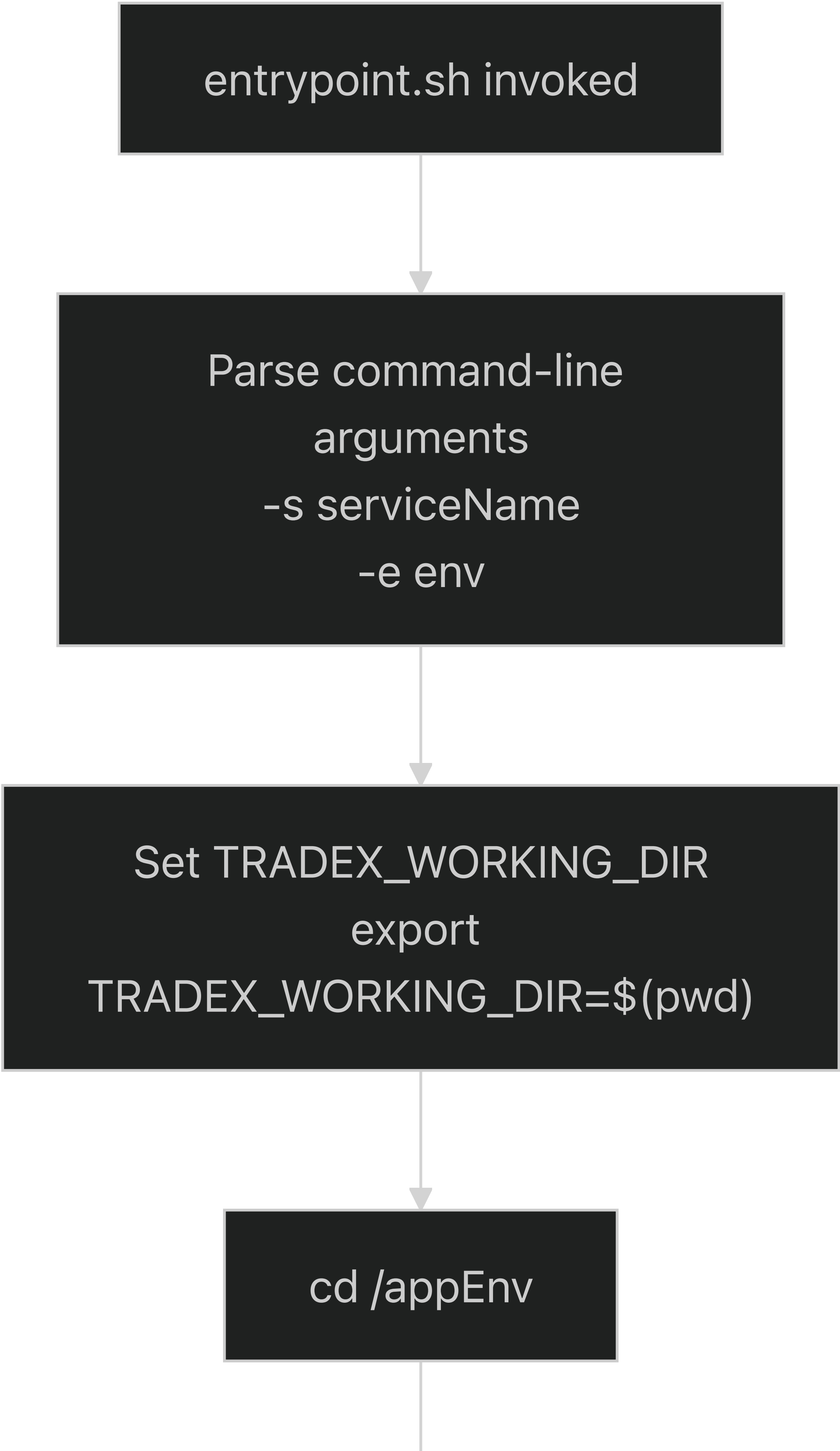
Sources: package.json:10-13

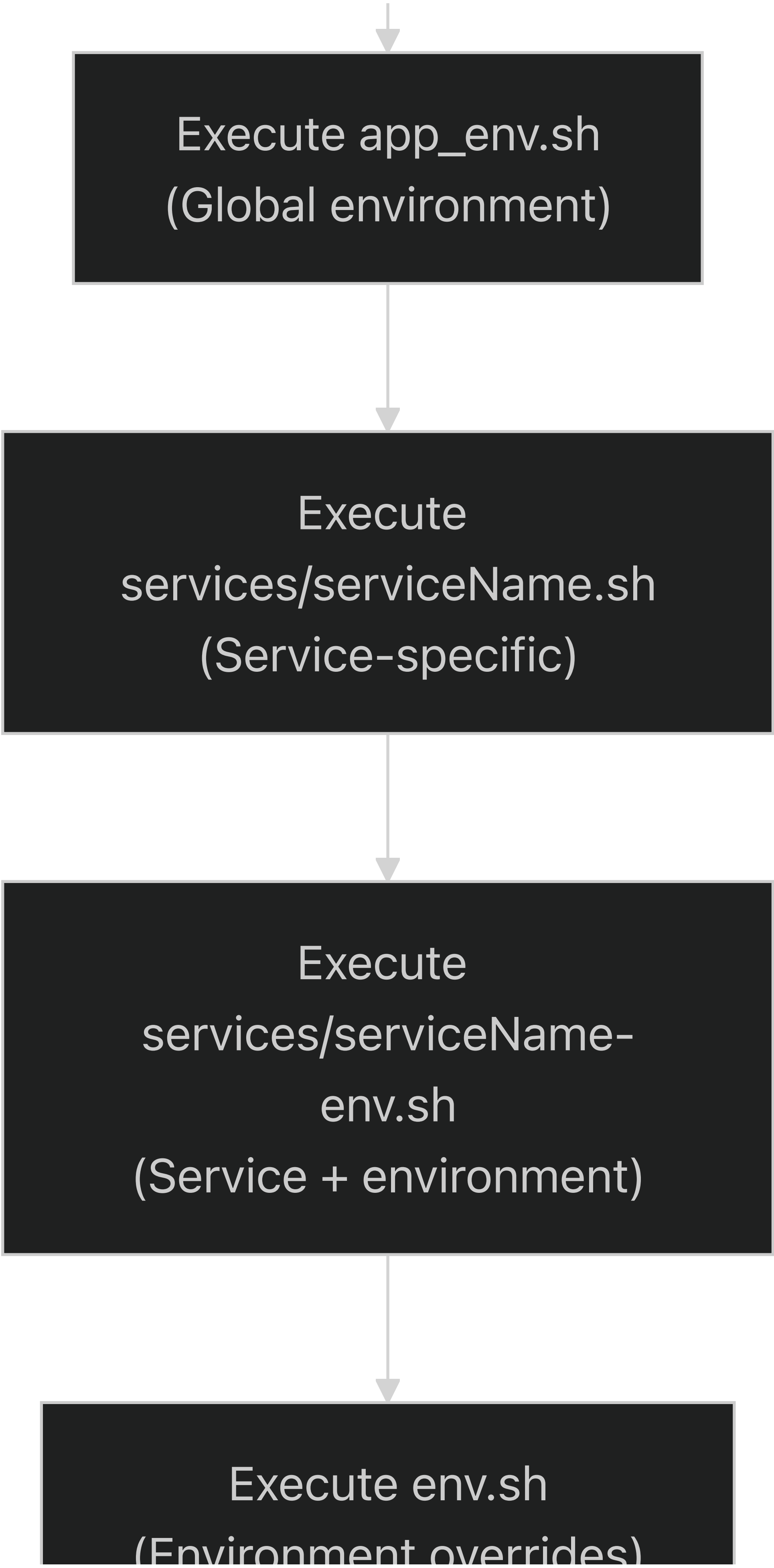
Container Entrypoint

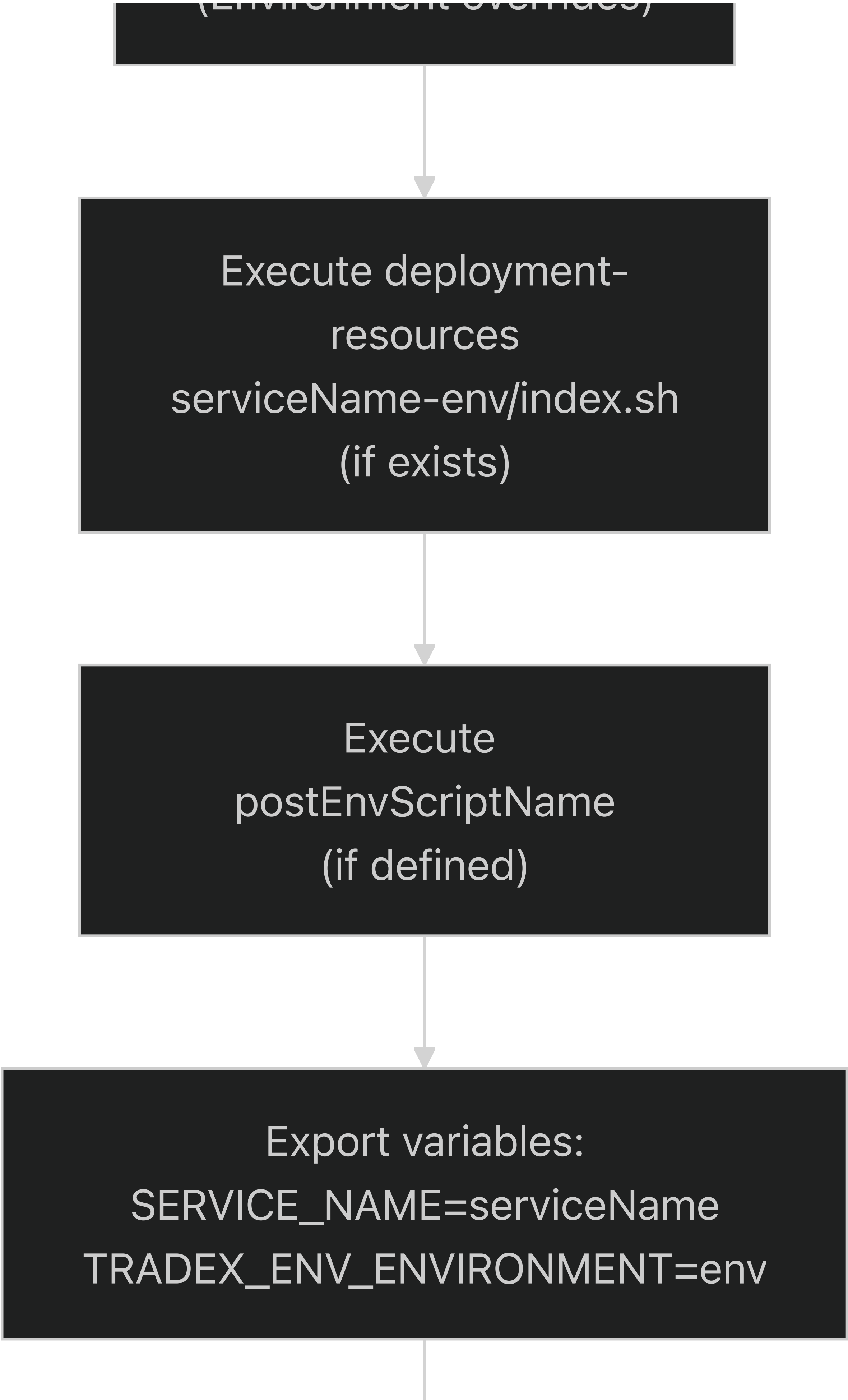
Entrypoint Script Flow

The `entrypoint.sh` script (https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh#L1-L90)

entrypoint.sh1-90 (https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.sh#L1-L90) orchestrates environment setup before launching the Node.js application.


entrypoint.sh invoked
Parse command-line arguments
-s serviceName
-e env
Set TRADEX_WORKING_DIR
export TRADEX_WORKING_DIR=$(pwd)
cd /appEnv


Execute app_env.sh
(Global environment)
Execute
services/serviceName.sh
(Service-specific)
Execute
services/serviceName-
env.sh
(Service + environment)
Execute env.sh


Execute deployment-resources serviceName-env/index.sh (if exists)
Execute postEnvScriptName (if defined)
Export variables: SERVICE_NAME=serviceName TRADEX_ENV_ENVIRONMENT=env

Sources: entrypoint.sh:1-90

Environment Script Hierarchy

The entrypoint sources environment scripts in a specific order, allowing for layered configuration overrides:

Order	Script Pattern	Purpose	Example
1	`/appEnv/app_env.sh`	Global environment variables shared across all services	Database URLs, Kafka brokers
2	`/appEnv/services/{serviceName}.sh`	Service-specific configuration	AAA-specific settings
3	`/appEnv/services/{serviceName}-{env}.sh`	Service and environment specific	AAA development vs production
4	`/appEnv/env.sh`	Environment-wide overrides	Environment-specific credentials
5	`/appEnv/services/deployment-resources/{serviceName}-{env}/index.sh`	Deployment resources (optional)	Kubernetes secrets, config maps
6	`{postEnvScriptName}`	Post-environment script (if defined)	Final initialization steps

Sources: entrypoint.sh:37-82

Command-Line Arguments

The entrypoint script accepts the following arguments:

```
entrypoint.sh -s
 | --serviceName <name> -e | --env <environment> -- <command> |
|--- | --- |

-   `-s|--serviceName`: Name of the service (e.g., "aaa")
-   `-e|--env`: Environment name (e.g., "dev", "staging", "prod")
-   `--`: Delimiter before the actual run command
-   `<command>`: The command to execute after environment setup (e.g., "nod

**Sources**: entrypoint.sh:7-30

##### Environment Variables Exported

The entrypoint script exports the following variables before launching the
```

TRADEX_WORKING_DIR=/app
SERVICE_NAME=
TRADEX_ENV_ENVIRONMENT= # Only if not already set

```
**Sources**: entrypoint.sh:35-87

* * *

#### Docker Container Configuration

##### Port Exposure

The container exposes port 3000 for HTTP traffic[](https://github.com/nhsv-

[dockerfile50](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#
```

EXPOSE 3000

```
**Sources**: dockerfile:50

##### Health Check

The production container includes a health check that probes the `/currentT

![Chart 4](aaa-images/aaa-section2-4.svg)

The health check command[](https://github.com/nhsv-tradex/aaa/blob/59961d60

[dockerfile53-54](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfi
```

node -e "require('http').get('http://localhost:3000/currentTime', (res) => {
  process.exit(res.statusCode === 200 ? 0 : 1)
})"

```
This executes a lightweight HTTP request to verify the service is respondin

-   Runs every 30 seconds after a 5-second startup grace period
-   Times out after 3 seconds
-   Marks the container unhealthy after 3 consecutive failures
-   Expects a 200 status code from `/currentTime`

**Sources**: dockerfile:53-54

* * *

#### Runtime Execution

##### Final Command Execution

After environment setup completes, the entrypoint script executes the run c

[entrypoint.sh89-90](https://github.com/nhsv-tradex/aaa/blob/59961d60/entry
```

echo "***********$(date) starting $runCommand ************"
eval $runCommand

```
In the Docker container, this evaluates to:
```

node build/src/index.js

```
**Sources**: entrypoint.sh:89-90, dockerfile:56

##### Complete Deployment Flow

![Chart 5](aaa-images/aaa-section2-5.svg)

**Sources**: dockerfile, entrypoint.sh, package.json

* * *

#### Dependency Management

##### Production Dependencies

The production container includes only runtime dependencies installed with

[dockerfile21](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#

 | Category | Key Dependencies | Version Constraint |
|--- | --- | --- |
 | Authentication | `jsonwebtoken`, `apache-crypt`, `otp-generator` | Vario
 | Database | `mysql` | ^2.16.0 |
 | Caching | `redis` | ^3.0.2 |
 | Messaging | Via `tradex-common` | GitHub source |
 | Service Discovery | `node-zookeeper-client` | ^0.2.2 |
 | HTTP | `node-fetch` | ^2.7.0 |
 | Utilities | `moment`, `uuid`, `nanomatch` | Various |
 | Dependency Injection | `typedi` | ^0.8.0 |

**Sources**: package.json:18-41

##### Development Dependencies

Development dependencies (TypeScript compiler, ESLint, testing tools) are e
```

@types/, *typescript, eslint, @typescript-eslint/*, mocha, prettier, rimraf, pkg

```
**Sources**: package.json:42-53

* * *

#### Container Runtime Characteristics

##### Working Directory

The container operates with `/app` as the working directory[](https://githu

[dockerfile14](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#

##### Process Management

The container uses bash to execute the entrypoint script, which then spawns
```

ENTRYPOINT ["/bin/bash", "/app/entrypoint.sh", "node build/src/index.js"]

```
The entrypoint receives `"node build/src/index.js"` as positional arguments

[entrypoint.sh33](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoi

**Sources**: dockerfile:56, entrypoint.sh:25-33

##### Signal Handling

Note: The dockerfile includes a commented-out section for `dumb-init`[](htt

[dockerfile38](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfile#

**Sources**: dockerfile:38

* * *

#### Build Optimization

##### Image Size Optimization

The multi-stage build optimizes image size through several techniques:

1.  **Separate build and runtime stages**: Build tools and devDependencies
2.  **Production-only dependencies**: The base stage installs only producti

    dockerfile21](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfi
3.  **Cache cleanup**: APK cache is removed after installing system depende

    dockerfile11](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfi
4.  **Selective copying**: Only `build/` and `node_modules/` are copied fro

    dockerfile44-45](https://github.com/nhsv-tradex/aaa/blob/59961d60/docke

**Sources**: dockerfile:2-45

##### Build Cache Optimization

The Dockerfile is structured to maximize Docker layer caching:

1.  **Package files copied first**[

    dockerfile17-18](https://github.com/nhsv-tradex/aaa/blob/59961d60/docke
2.  **Dependencies installed before source copy**: Only source changes trig
3.  **Frozen lockfile**: `--frozen-lockfile` ensures reproducible dependenc
```

```
    dockerfile21](https://github.com/nhsv-tradex/aaa/blob/59961d60/dockerfi

**Sources**: dockerfile:17-21

* * *

#### Summary

The AAA service deployment architecture implements a three-stage Docker bui

**Sources**: dockerfile, entrypoint.sh, package.json

---

### Configuration System

Relevant source files

-   [

    entrypoint.sh](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypo
-   [

    global.d.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.
-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

#### Purpose and Scope

The Configuration System manages all runtime configuration for the AAA serv

For information about deployment and containerization, see [Deployment Arch

* * *

#### Configuration Architecture Overview

The configuration system operates through three distinct layers that are pr

![Chart 1](aaa-images/aaa-section3-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.
```

```
[entrypoint.sh1-90](https://github.com/nhsv-tradex/aaa/blob/59961d60/entryp

src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

* * *

#### Configuration Layers

##### Layer 1: Hard-coded Defaults

The base configuration layer provides sensible defaults for all settings. T

 | Configuration Key | Default Value | Description |
|--- | --- | --- |
 | `clusterId` | `"aaa"` | Service cluster identifier |
 | `retryTimes` | `3` | Default retry attempts |
 | `defaultLongRedisDuration` | `604800` (7 days) | Default Redis cache TTL
 | `defaultKafkaTimeout` | `10000` | Default Kafka operation timeout in mil
 | `enableHandleOtp` | `true` | Whether to handle OTP at AAA service |
 | `accessToken.expiredInSeconds` | `900` (15 min) | Access token lifetime
 | `refreshToken.expiredInSeconds` | `86400` (1 day) | Refresh token lifeti
 | `refreshToken.expiredInSecondsWithRememberMe` | `2592000` (30 days) | Re
 | `otpToken.expiredInSeconds` | `90` | OTP token lifetime |
 | `cacheTimeout` | `180000` (3 min) | Scope cache timeout in milliseconds

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts12-141](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Layer 2: Environment Variables

Environment variables prefixed with `TRADEX_ENV_` override default values.

 | Utility Function | Purpose | Example Usage |
|--- | --- | --- |
 | `Utils.getEnvStr()` | Read string value | `Utils.getEnvStr("TRADEX_ENV_D
 | `Utils.getEnvBool()` | Read boolean value | `Utils.getEnvBool("TRADEX_EN
 | `Utils.getEnvNum()` | Read numeric value | `Utils.getEnvNum("TRADEX_ENV_
 | `Utils.getEnvArr()` | Read array value | `Utils.getEnvArr("TRADEX_ENV_KA

**Key Environment Variables:**

![Chart 2](aaa-images/aaa-section3-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts
```

```
[src/conf.ts13-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts17](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

src/conf.ts18](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

src/conf.ts48-55](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

##### Layer 3: Runtime Overrides (env.js)

The highest priority configuration layer loads runtime overrides from the `

![Chart 3](aaa-images/aaa-section3-3.svg)

The VM sandbox provides access to:

-   `conf` / `config`: The configuration object (aliased for convenience)
-   `process`: Node.js process object for environment variable access

**Implementation:**

[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L146-L158)

[src/conf.ts146-158](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts146-160](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

* * *

#### Environment Setup Process

The `entrypoint.sh` script orchestrates the environment setup before the No

![Chart 4](aaa-images/aaa-section3-4.svg)

**Script Execution Order:**

1.  **[

    entrypoint.sh37-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
2.  **[
```

```
    entrypoint.sh43-48](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
3.  **[

    entrypoint.sh50-54](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
4.  **[

    entrypoint.sh58-62](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
5.  **[

    entrypoint.sh65-74](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
6.  **[

    entrypoint.sh78-82](https://github.com/nhsv-tradex/aaa/blob/59961d60/en
7.  **[

    entrypoint.sh84-90](https://github.com/nhsv-tradex/aaa/blob/59961d60/en

This layered approach allows environment-specific configuration at multiple

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/entrypoint.

[entrypoint.sh1-90](https://github.com/nhsv-tradex/aaa/blob/59961d60/entryp

* * *

#### Central Configuration Object

The `conf.ts` file defines and exports the central configuration object use

![Chart 5](aaa-images/aaa-section3-5.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts12-141](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Domain-Specific Configuration

The configuration system supports multi-tenant deployments where different

##### Domain Resolution

![Chart 6](aaa-images/aaa-section3-6.svg)
```

```
**Domain-Specific Settings:**

 | Setting | Default Domain | Custom Domain |
|--- | --- | --- |
 | `conf.domain` | Value of `TRADEX_DOMAIN` | Value of `TRADEX_ENV_DOMAIN`
 | `conf.forceSecCode` | `null` | Same as `conf.domain` |
 | RSA key path | `keys/aaa/TRADEX_DOMAIN/rsa-*.key` | `keys/aaa/{domain}/r
 | JWT key path | `keys/jwt-*.key` | `keys/{domain}/jwt-*.key` (if configur

**Implementation:**

-   **[

    src/conf.ts13](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con
-   **[

    src/conf.ts171-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr
-   **[

    src/conf.ts174-176](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts10](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts13](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

src/conf.ts171-176](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Key Management

##### RSA Keys

RSA keys are used for password encryption. Each domain has its own RSA key

**Configuration:**
```

rsa: {
  enableEncryptPassword: true,
  publicKey: `keys/aaa/${domain}/rsa-public.key`,
  privateKey: `keys/aaa/${domain}/rsa-private.key`,
}

```
The keys are located at[](https://github.com/nhsv-tradex/aaa/blob/59961d60/

[src/conf.ts42-46](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

##### JWT Keys

JWT keys are used for signing and verifying access tokens. The system suppo

**JWT Configuration Structure:**

![Chart 7](aaa-images/aaa-section3-7.svg)

**Base Configuration:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/

[src/conf.ts59-68](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

The `Utils.createJwtConfig()` function[](https://github.com/nhsv-tradex/aaa

[src/conf.ts174-175](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

-   `conf` - Configuration object to modify
-   `conf.domain` - Current domain
-   `TRADEX_ENV_DOMAINS` - Array of all supported domains
-   `"keys"` - Base directory for key files
-   `conf.clusterId` - Service identifier (`"aaa"`)
-   Key filenames for default keys

The `Utils.processJwtKey()` function[](https://github.com/nhsv-tradex/aaa/b

[src/conf.ts176](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts59-68](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts174-176](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Infrastructure Configuration

##### Database Configuration

MySQL database settings are configured through environment variables:
```

```
 | Configuration Key | Environment Variable | Default Value |
|--- | --- | --- |
 | `db.host` | `TRADEX_ENV_MYSQL_HOST` | Required |
 | `db.port` | `TRADEX_ENV_MYSQL_PORT` | `3306` |
 | `db.user` | `TRADEX_ENV_MYSQL_USER` | Required |
 | `db.password` | `TRADEX_ENV_MYSQL_PASSWORD` | Required |
 | `db.database` | \- | `"tradex-aaa"` |
 | `db.connectionLimit` | \- | `10` |
 | `db.timezone` | \- | `"UTC"` |

**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts47-55](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

##### Redis Configuration

Redis is used for caching tokens, scopes, and session data:
```

redis: {
  port: 6379,
  host: '127.0.0.1',
  options: {},
}

```
**Default Cache Durations:**

-   **Long duration:**[

    src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con
-   **Scope cache:**[

    src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

##### Kafka Configuration

Kafka settings control message queue integration:

![Chart 8](aaa-images/aaa-section3-8.svg)

**Option Merging:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts162-169](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

**Topics:**

-   **User topic:**[

    src/conf.ts56-58](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/
-   **Configuration topic:**[

    src/conf.ts99](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

    src/conf.ts109](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts18-23](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts56-58](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts162-169](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *
```

```
#### Service-Specific Settings

##### Timeout Configuration

Service timeouts define maximum wait times for external operations:

 | Timeout Key | Default Value (ms) | Description |
|--- | --- | --- |
 | `loginFacebook` | `15000` | Facebook OAuth timeout |
 | `loginGoogle` | `15000` | Google OAuth timeout |
 | `loginApple` | `15000` | Apple Sign-In timeout |
 | `loginTechx` | `15000` | TechX integration timeout |
 | `loginPasswordOtp` | `45000` | Password + OTP flow timeout |
 | `otpService` | `10000` (default) | OTP service timeout |
 | `loginDirectToService` | `10000` | Direct service login timeout |
 | `loginDomain` | `10000` (default) | Domain login timeout |
 | `loadScope` | `30000` | Scope loading timeout |

Timeouts set to `null` are replaced with `defaultKafkaTimeout`[](https://gi

[src/conf.ts178-182](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts31-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts178-182](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### OTP Configuration

OTP (One-Time Password) settings control authentication behavior:
```

enableHandleOtp: true, // Handle OTP at AAA service vs forwarding to core
otp_type_uri_map: OTP_TYPE_URI_MAP, // Maps OTP types to URIs
nhsv: {
  OTP_SENT_KEY_MAX_LIFE_TIME: 86400, // 24 hours
  MAX_DAILY_SENT: 5,   // Maximum OTPs per day
  TRIGGER_TIME_INTERVAL: 30   // Seconds between sends
}

```
**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts30](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts128](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Biometric Configuration

Biometric authentication can be enabled per deployment:

 | Configuration Key | Environment Variable | Default |
|--- | --- | --- |
 | `isEnableBiometric` | `TRADEX_ENV_ENABLE_BIOMETRIC` | `false` |
 | `isRegisBiometricWithPassword` | `TRADEX_ENV_BIOMETRIC_VALIDATE_PASSWORD

**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts14-15](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

##### Scope and Client Loading

Scopes and clients are loaded from Kafka configuration topics:

**Scope Configuration:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60

[src/conf.ts95-106](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
```

```
scopes: {
  verifyOtpScope: "VERIFY_OTP",
  unAuthenticated: "PUBLIC",
  loadFrom: {
    topic: "configuration",
    uris: {
      scope: "/api/v1/admin/scope",
      scopeGroup: "/api/v1/admin/scopeGroup",
      pageSize: 100,
    },
  },
}
```

```
**Client Configuration:**[](https://github.com/nhsv-tradex/aaa/blob/59961d6

[src/conf.ts107-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c
```

clients: {
  loadFrom: {
    topic: "configuration",
    uris: {
      client: "/api/v1/system/client",
      pageSize: 100,
    },
  },
}

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts31-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts95-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Global Configuration Access

The configuration object is made available to the entire application throug

##### Module Export

The configuration is exported as the default export from `conf.ts`:
```

export default conf;

```
This allows services to import the configuration:
```

import conf from './conf';

```
**Implementation:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

[src/conf.ts183](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

##### Global Object

The configuration is also assigned to a global property for legacy access p
```

global.systemConf = conf;

```
This is defined at[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

[src/conf.ts144](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

**Type Definition:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/glo

[global.d.ts1-7](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.

![Chart 9](aaa-images/aaa-section3-9.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts144](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

src/conf.ts183](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

global.d.ts1-7](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.t

* * *

#### Configuration Validation

After all configuration layers are processed, the system logs the final con
```

Logger.info('configuration after injecting:', conf);

```
This occurs at[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

[src/conf.ts160](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts160](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

---

## Authentication System

### Overview

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/constants/GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

This page documents the authentication flows and mechanisms supported by th

For token lifecycle management including JWT generation and refresh token h

#### Authentication Flow Overview

The authentication system follows a centralized delegation pattern where al

![Chart 1](aaa-images/aaa-section4-1.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts52-173](https://github.com/nhsv-trade

#### Core Authentication Components

The authentication system is built around several key components that work
```

```
| Component | File Location | Responsibility |
|--- | --- | --- |
| `authenticate` | `authenticationService.ts:52-68` | Main entry point wit
| `authenticateTx` | `authenticationService.ts:70-173` | Transaction orche
| `findOrGet` | `ClientService.ts` | Client validation and retrieval |
| `findLoginMethods` | `LoginMethodService.ts` | Login method resolution b
| `loginMethodCache` | `authenticationService.ts:42` | Caching layer for l

##### Client Validation Process

Before any authentication can proceed, the system validates the OAuth2 clie

![Chart 2](aaa-images/aaa-section4-2.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts75-98](https://github.com/nhsv-tradex

##### Login Method Resolution

The system uses a caching mechanism to efficiently resolve login methods ba

![Chart 3](aaa-images/aaa-section4-3.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts102-118](https://github.com/nhsv-trad

#### Grant Type Delegation System

The authentication system supports multiple OAuth2 grant types and custom a

![Chart 4](aaa-images/aaa-section4-4.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts121-172](https://github.com/nhsv-trad

#### Third-Party Authentication Flow

Social and third-party authentication follows a common pattern where creden

![Chart 5](aaa-images/aaa-section4-5.svg)

##### Supported Social Providers
```

```
 | Provider | Grant Type | Handler Function | Required Parameters |
|--- | --- | --- | --- |
 | Facebook | `ACCESS_FACEBOOK` | `loginFacebook()` | `access_token` |
 | Google | `ACCESS_GOOGLE` | `loginGoogle()` | `access_token`, `id_token`
 | Apple | `ACCESS_APPLE` | `loginApple()` | Device information only |
 | Generic Social | `SOCIAL_LOGIN` | `loginSocialPaave()` | `login_social_t

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts236-359](https://github.com/nhsv-trad

src/services/authen/commonLoginThirdParty.ts13-43](https://github.com/nhsv-

#### Client Credentials Grant

The `CLIENT_CREDENTIALS` grant type provides machine-to-machine authenticat

![Chart 6](aaa-images/aaa-section4-6.svg)

The client credentials flow skips user authentication and directly creates

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authen/loginClientCredential.ts10-30](https://github.com/nhsv

#### Demo Account Authentication

Demo accounts provide a way to authenticate with pre-configured test creden

![Chart 7](aaa-images/aaa-section4-7.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authen/loginAccountDemoService.ts19-85](https://github.com/nh

#### Direct Service Login

The `loginDirectToService` function provides a common pattern for authentic

![Chart 8](aaa-images/aaa-section4-8.svg)

This pattern is used by several authentication methods including CA login a

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au
```

```
[src/services/authen/loginDirectToService.ts23-82](https://github.com/nhsv-

#### Error Handling

The authentication system implements comprehensive error handling with spec

 | Error Type | Trigger Condition | File Location |
|--- | --- | --- |
 | `InvalidParameterError` | Missing required parameters | Throughout valid
 | `InvalidIdSecretError` | Invalid client credentials | `authenticationSer
 | `NoLoginMethodError` | No matching login method found | `authenticationS
 | `MultipleLoginMethodError` | Multiple login methods found | `authenticat
 | `NoGrantTypeError` | Unsupported grant type | `authenticationService.ts:
 | `ForwardError` | External service errors | Third-party auth flows |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts38-172](https://github.com/nhsv-trade

src/services/authen/loginDirectToService.ts40](https://github.com/nhsv-trad

src/services/authen/commonLoginThirdParty.ts1-43](https://github.com/nhsv-t

### Authentication Service Core

Relevant source files

-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
-   [

    src/consumer/index.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60

#### Purpose and Scope

This document describes the core authentication service implementation that

-   Kafka consumer initialization and message consumption
-   URI-based request routing to service handlers
-   Service orchestration across authentication, token management, OTP, bio

For detailed information about specific authentication methods and grant ty
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts1-131](https://github.com/nhsv-tradex/aaa/

src/consumer/index.ts1-11](https://github.com/nhsv-tradex/aaa/blob/59961d60

* * *

#### Architecture Overview

The Authentication Service Core implements a message-driven architecture wh

![Chart 1](aaa-images/aaa-section5-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts1-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/consumer/HandleAAARequest.ts1-131](https://github.com/nhsv-tradex/aaa/b

* * *

#### Kafka Consumer Infrastructure

The consumer infrastructure is initialized in[](https://github.com/nhsv-tra

[src/consumer/index.ts6-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

##### Consumer Initialization

 | Component | Class | Configuration Source | Purpose |
|--- | --- | --- | --- |
 | Stream Handler | `Kafka.StreamHandler` | `conf.kafkaConsumerOptions` | E
 | Message Handler | `Kafka.MessageHandler` | N/A | Wraps business logic ha
 | Cluster Subscription | Array | `[conf.clusterId]` | Subscribes to specif
 | Processing Function | `handleAAAMessage` | Imported from HandleAAAReques

The initialization flow:

![Chart 2](aaa-images/aaa-section5-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts6-11](https://github.com/nhsv-tradex/aaa/blob/59961d6
```

```
* * *

#### Message Routing Pipeline

The `handleAAAMessage()` function at[](https://github.com/nhsv-tradex/aaa/b

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

##### URI Pattern Matching Strategy

The routing function uses a series of conditional statements to match exact
```

function handleAAAMessage(msg: Kafka.IMessage): Promise | boolean

```
##### Authentication Endpoints

These URIs route to the core `authenticate()` function:

 | URI Pattern | Grant Type | Description |
|--- | --- | --- |
 | `post:/api/v1/login` | password | Basic username/password authentication
 | `post:/api/v1/login/sec` | password\_otp | Password with OTP challenge |
 | `post:/api/v1/login/domain` | domain | Domain-based login |
 | `post:/api/v1/login/social` | social | Social OAuth login |
 | `post:/api/v1/loginCA` | client\_credentials | Client authentication |
 | `post:/api/v1/login/biometric` | biometric | Biometric authentication |
 | `post:/api/v1/user/SocialLogin` | social | Alternative social login endp
 | `post:/api/v1/login/organization` | organization | Organization-based lo
 | `post:/api/v1/login/social/organization` | social+organization | Combine

All these URIs are handled by[](https://github.com/nhsv-tradex/aaa/blob/599

[src/consumer/HandleAAARequest.ts35-45](https://github.com/nhsv-tradex/aaa/

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-45](https://github.com/nhsv-tradex/aaa/

* * *

#### Service Orchestration

The routing function delegates to specialized service modules for different

![Chart 3](aaa-images/aaa-section5-3.svg)

##### Service Import Mapping

The following table maps URI patterns to their corresponding service functi

 | Service Domain | Function | Import Statement | Lines |
|--- | --- | --- | --- |
 | Authentication | `authenticate()` | `import { authenticate } from "../se
 | 1](https://github.com/nhsv-tradex/aaa/blob/59961d60/1) |
 | Token Management | `refreshAccessToken()`, `revokeToken()` | `import { r

4](https://github.com/nhsv-tradex/aaa/blob/59961d60/4) |
| OTP Verification | `verifyOtp()` | `import { verifyOtp } from "../service
```

```
5](https://github.com/nhsv-tradex/aaa/blob/59961d60/5) |
| Account Linking | `linkAccountService.*` | `import * as linkAccountServic

6](https://github.com/nhsv-tradex/aaa/blob/59961d60/6) |
| Scope Management | `scopeService` | `import { scopeService } from "../ser

7](https://github.com/nhsv-tradex/aaa/blob/59961d60/7) |
| OTP Operations | `generateSaveOtpToken()`, `notifyMobileOtp()`, etc. | `i

8](https://github.com/nhsv-tradex/aaa/blob/59961d60/8) |
| User Management | `registerMobileOtp()`, `unregisterMobileOtp()`, `update

10-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/10-14) |
| Biometrics | `registerBiometric()`, `verifyBiometricOTP()`, etc. | `impor

18-25](https://github.com/nhsv-tradex/aaa/blob/59961d60/18-25) |
| Client Management | `changeClientSecret()`, `updateAppVersion()` | `impor

27-28](https://github.com/nhsv-tradex/aaa/blob/59961d60/27-28) |
| Partner Credentials | `loginPartnerCredential()` | `import { loginPartner

29](https://github.com/nhsv-tradex/aaa/blob/59961d60/29) |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts1-31](https://github.com/nhsv-tradex/aaa/b

* * *

#### Request Flow

##### Message Structure

All incoming messages conform to the `Kafka.IMessage` interface from the `t

-   `msg.uri` - The URI pattern used for routing (e.g., `"post:/api/v1/logi
-   `msg.data` - The request payload containing credentials, tokens, or oth
-   `msg.transactionId` - Unique identifier for request tracking and loggin

##### Flow Diagram: Authentication Request

![Chart 4](aaa-images/aaa-section5-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume
```

```
[src/consumer/index.ts6-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa/

* * *

#### Complete URI Routing Table

The following table documents all URI patterns handled by the core authenti

 | URI Pattern | Method | Lines | Handler Function | Service Module | Descr
|--- | --- | --- | --- | --- | --- |
 | `post:/api/v1/login` | POST | [ |
 | 35](https://github.com/nhsv-tradex/aaa/blob/59961d60/35) | `authenticate
 | `post:/api/v1/login/sec` | POST | [ |

36](https://github.com/nhsv-tradex/aaa/blob/59961d60/36)
 | `authenticate()` | authenticationService | Secure login with OTP |
|--- | --- | --- |

37](https://github.com/nhsv-tradex/aaa/blob/59961d60/37)
 | `authenticate()` | authenticationService | Domain-based login |
|--- | --- | --- |

38](https://github.com/nhsv-tradex/aaa/blob/59961d60/38)
 | `authenticate()` | authenticationService | Social OAuth login |
|--- | --- | --- |

39](https://github.com/nhsv-tradex/aaa/blob/59961d60/39)
 | `authenticate()` | authenticationService | Client authentication |
|--- | --- | --- |

40](https://github.com/nhsv-tradex/aaa/blob/59961d60/40)
 | `authenticate()` | authenticationService | Biometric login |
|--- | --- | --- |

41](https://github.com/nhsv-tradex/aaa/blob/59961d60/41)
 | `authenticate()` | authenticationService | Alternative social endpoint |
|--- | --- | --- |

42](https://github.com/nhsv-tradex/aaa/blob/59961d60/42)
 | `authenticate()` | authenticationService | Organization login |
|--- | --- | --- |
```

```
43](https://github.com/nhsv-tradex/aaa/blob/59961d60/43)
 | `authenticate()` | authenticationService | Combined social+org |
|--- | --- | --- |

46](https://github.com/nhsv-tradex/aaa/blob/59961d60/46)
 | `loginPartnerCredential()` | loginPartnerCredentialService | Partner cre
|--- | --- | --- |

48](https://github.com/nhsv-tradex/aaa/blob/59961d60/48)
 | `refreshAccessToken()` | TokenService | Refresh access token |
|--- | --- | --- |

50](https://github.com/nhsv-tradex/aaa/blob/59961d60/50)
 | `revokeToken()` | TokenService | Revoke token |
|--- | --- | --- |

52](https://github.com/nhsv-tradex/aaa/blob/59961d60/52)
 | `changeClientSecret()` | ClientService | Change client secret |
|--- | --- | --- |

54-58](https://github.com/nhsv-tradex/aaa/blob/59961d60/54-58)
 | `verifyOtp()` or `loginVerifyOtp()` | VerifyOtp / authen/loginOtp | OTP
|--- | --- | --- |

59](https://github.com/nhsv-tradex/aaa/blob/59961d60/59)
 | `findAllPartners()` | linkAccountService | List partners |
|--- | --- | --- |

61](https://github.com/nhsv-tradex/aaa/blob/59961d60/61)
 | `findAllLinkAccount()` | linkAccountService | List linked accounts |
|--- | --- | --- |

63](https://github.com/nhsv-tradex/aaa/blob/59961d60/63)
 | `createLinkAccount()` | linkAccountService | Create account link |
|--- | --- | --- |

65](https://github.com/nhsv-tradex/aaa/blob/59961d60/65)
 | `deleteLinkAccount()` | linkAccountService | Delete account link |
|--- | --- | --- |

67](https://github.com/nhsv-tradex/aaa/blob/59961d60/67)
 | `loginLinkAccount()` | linkAccountService | Login with linked account |
|--- | --- | --- |

69](https://github.com/nhsv-tradex/aaa/blob/59961d60/69)
```

```
 | `confirmLinkAccount()` | linkAccountService | Confirm account link |
|--- | --- | --- |

71](https://github.com/nhsv-tradex/aaa/blob/59961d60/71)
 | `unlink()` | linkAccountService | Unlink accounts |
|--- | --- | --- |

73](https://github.com/nhsv-tradex/aaa/blob/59961d60/73)
 | `initLinkAccount()` | linkAccountService | Initialize account link |
|--- | --- | --- |

75](https://github.com/nhsv-tradex/aaa/blob/59961d60/75)
 | `notifyOtpPartner()` | linkAccountService | Notify OTP to partner |
|--- | --- | --- |

77](https://github.com/nhsv-tradex/aaa/blob/59961d60/77)
 | `notifyOtpFromPartner()` | linkAccountService | Receive OTP from partner
|--- | --- | --- |

79](https://github.com/nhsv-tradex/aaa/blob/59961d60/79)
 | `changeUsername()` | linkAccountService | Change linked username |
|--- | --- | --- |

81](https://github.com/nhsv-tradex/aaa/blob/59961d60/81)
 | `createLinkAccountApprove()` | linkAccountService | Approve account link
|--- | --- | --- |

83](https://github.com/nhsv-tradex/aaa/blob/59961d60/83)
 | `getUserIdByPartnerName()` | linkAccountService | Get user by partner |
|--- | --- | --- |

85-89](https://github.com/nhsv-tradex/aaa/blob/59961d60/85-89)
 | `scopeService.getScopesByScopeGroups()` | ScopeService | Search scopes |
|--- | --- | --- |

90](https://github.com/nhsv-tradex/aaa/blob/59961d60/90)
 | `registerMobileOtp()` | UserService | Register mobile OTP |
|--- | --- | --- |

92](https://github.com/nhsv-tradex/aaa/blob/59961d60/92)
 | `registerBiometric()` | BiometricService | Register biometric |
|--- | --- | --- |

94](https://github.com/nhsv-tradex/aaa/blob/59961d60/94)
 | `registerBiometricKis()` | BiometricService | Register biometric (KIS) |
```

```
|--- | --- | --- |

96](https://github.com/nhsv-tradex/aaa/blob/59961d60/96)
 | `verifyBiometricOTP()` | BiometricService | Verify biometric OTP |
|--- | --- | --- |

98](https://github.com/nhsv-tradex/aaa/blob/59961d60/98)
 | `cancelBiometricRegister()` | BiometricService | Unregister biometric |
|--- | --- | --- |

100](https://github.com/nhsv-tradex/aaa/blob/59961d60/100)
 | `unregisterMobileOtp()` | UserService | Unregister mobile OTP |
|--- | --- | --- |

102](https://github.com/nhsv-tradex/aaa/blob/59961d60/102)
 | `queryBiometricStatus()` | BiometricService | Query biometric status |
|--- | --- | --- |

104](https://github.com/nhsv-tradex/aaa/blob/59961d60/104)
 | `notifyMobileOtp()` | otpService | Notify mobile OTP |
|--- | --- | --- |

106](https://github.com/nhsv-tradex/aaa/blob/59961d60/106)
 | `notifyMobileOtpNhsv()` | otpService | Notify mobile OTP (NHSV) |
|--- | --- | --- |

108](https://github.com/nhsv-tradex/aaa/blob/59961d60/108)
 | `notifyMobileOtpKisTtl()` | otpService | Notify mobile OTP (KIS TTL) |
|--- | --- | --- |

110](https://github.com/nhsv-tradex/aaa/blob/59961d60/110)
 | `notifyMobileBiometricOtp()` | otpService | Notify biometric OTP |
|--- | --- | --- |

112](https://github.com/nhsv-tradex/aaa/blob/59961d60/112)
 | `jsonDbImport()` | ImportDbJsonService | Import database data |
|--- | --- | --- |

114](https://github.com/nhsv-tradex/aaa/blob/59961d60/114)
 | `updateProfile()` | UserService | Update user profile |
|--- | --- | --- |

116](https://github.com/nhsv-tradex/aaa/blob/59961d60/116)
 | `generateSaveOtpToken()` | otpService | Verify and save OTP token |
|--- | --- | --- |
```

```
118](https://github.com/nhsv-tradex/aaa/blob/59961d60/118)
 | `verifyPwdBiometric()` | BiometricService | Verify password biometric |
|--- | --- | --- |

120](https://github.com/nhsv-tradex/aaa/blob/59961d60/120)
 | `putLeaderboardSetting()` | linkAccountService | Leaderboard settings |
|--- | --- | --- |

122](https://github.com/nhsv-tradex/aaa/blob/59961d60/122) | `updateAppVers

If no URI pattern matches, the request is forwarded to `handleMessage()` at

[src/consumer/HandleAAARequest.ts125](https://github.com/nhsv-tradex/aaa/bl

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

* * *

#### Configuration Dependencies

The core authentication service relies on configuration from `conf.ts` for:

-   **Kafka Consumer Options**: Connection settings, topic names, consumer

    src/consumer/index.ts8](https://github.com/nhsv-tradex/aaa/blob/59961d6
-   **Cluster ID**: Service cluster identifier for topic subscription ([

    src/consumer/index.ts9](https://github.com/nhsv-tradex/aaa/blob/59961d6
-   **OTP Handling Mode**: The `conf.enableHandleOtp` flag determines wheth

    src/consumer/HandleAAARequest.ts55-58](https://github.com/nhsv-tradex/a

For detailed configuration management, see [Configuration System](#configur

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts3](https://github.com/nhsv-tradex/aaa/blob/59961d60/s

src/consumer/HandleAAARequest.ts30](https://github.com/nhsv-tradex/aaa/blob

src/consumer/HandleAAARequest.ts54-58](https://github.com/nhsv-tradex/aaa/b
```

```
---

### Grant Types and Login Methods

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/constants/GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

This document covers the configuration and management of login methods and

#### Login Method Configuration Architecture

Login methods are configured through a database-driven architecture that de

![Chart 1](aaa-images/aaa-section6-1.svg)

The system uses `findLoginMethods()` to resolve which `LoginMethod` configu

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LoginMethod.ts4-31](https://github.com/nhsv-tradex/aaa/blob/

src/models/db/ClientLoginMethodMap.ts4-21](https://github.com/nhsv-tradex/a

src/models/db/LoginMethodScopeGroupMap.ts4-21](https://github.com/nhsv-trad

src/services/LoginMethodService.ts8-31](https://github.com/nhsv-tradex/aaa/

#### Request Parameter Mapping

The `ILoginReq` interface defines all possible parameters that can be inclu

![Chart 2](aaa-images/aaa-section6-2.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/requ

[src/models/request/ILoginReq.ts3-29](https://github.com/nhsv-tradex/aaa/bl
```

```
#### LoginMethod Database Model

The `LoginMethod` model defines the core configuration for each authenticat

##### LoginMethod Configuration Fields

 | Field | Type | Description | Usage |
|--- | --- | --- | --- |
 | `id` | `number` | Primary key identifier | Foreign key reference |
 | `service_code` | `string` | Optional service identifier | Used with `sec
 | `grant_type` | `string` | OAuth2 grant type | Route determination |
 | `access_token_ttl` | `number` | Access token TTL (seconds) | Token expir
 | `refresh_token_ttl` | `number` | Standard refresh token TTL | Standard s
 | `refresh_token_long_ttl` | `number` | Extended refresh token TTL | "Reme
 | `multi_factor_ttl` | `number` | Multi-factor auth TTL | OTP/biometric fl
 | `public_scopes` | `string` | Default scope assignment | Permission defau
 | `ms_name` | `string` | Target microservice name | External service routi
 | `ms_uri` | `string` | Target microservice URI | External service routing
 | `is_default` | `number` | Default method flag | Selection priority |
 | `extra_data` | `string` | Additional configuration | Custom parameters |

##### Client-LoginMethod Mapping

The `findLoginMethods()` function resolves valid login methods by joining t

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LoginMethod.ts4-31](https://github.com/nhsv-tradex/aaa/blob/

src/models/db/ClientLoginMethodMap.ts4-21](https://github.com/nhsv-tradex/a

src/services/LoginMethodService.ts8-31](https://github.com/nhsv-tradex/aaa/

#### Grant Types and Login Methods

The system supports multiple OAuth2 grant types, each mapped to specific au

#### Supported Grant Types and Parameters

The system supports multiple OAuth2 grant types, each with specific paramet

 | Grant Type | Handler Function | Required Parameters | Optional Parameter
|--- | --- | --- | --- | --- |
 | `CLIENT_CREDENTIALS` | `loginClientCredentials()` | `client_id`, `client
```

```
| `PASSWORD` | `loginPassword()` | `username`, `password` | `remember_me`
| `KB_FINA` | `loginPassword()` | `username`, `password` | None | All TTL
| `PASSWORD_OTP` | `loginPasswordWithOtp()` | `username`, `password`, `sec
| `PASSWORD_TRADEX` | `loginPasswordTechx()` | `username`, `password` | `r
| `ACCESS_FACEBOOK` | `loginFacebook()` | `access_token` | None | Standard
| `ACCESS_GOOGLE` | `loginGoogle()` | `access_token`, `id_token` | None |
| `ACCESS_APPLE` | `loginApple()` | Platform tokens | None | Standard TTL
| `LOGIN_BIOMETRIC` | `loginBiometric()` | Biometric data | `device_id` |
| `DEMO` | `loginAccountDemo()` | None | None | All TTL fields |
| `LINK_ACCOUNT` | `loginLinkAccount()` | Partner tokens | None | Standard
| `SOCIAL_LOGIN` | `loginSocialPaave()` | `login_social_token`, `login_soc
| `ORGANIZATION_LOGIN` | `loginOrganization()` | `orgLoginToken` | `organi
| `BIOMETRIC_OTP` | `loginBiometricOtp()` | Biometric + OTP data | None |

##### Grant Type Processing Flow

![Chart 3](aaa-images/aaa-section6-3.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts121-172](https://github.com/nhsv-trad

#### Multi-Step Authentication Configuration

Some login methods support multi-step authentication flows configured throu

##### LoginMethodStep Configuration

The `LoginMethodStep` model defines individual steps in multi-factor authen

| Field | Description | Usage |
|--- | --- | --- |
| `login_method_id` | Foreign key to `LoginMethod` | Links step to parent
| `step` | Step sequence number | Execution order |
| `name` | Step display name | UI/logging purposes |
| `desc` | Step description | Documentation |

##### Scope Group Assignment

Login methods can be assigned specific scope groups through `LoginMethodSco

![Chart 4](aaa-images/aaa-section6-4.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L
```

```
[src/models/db/LoginMethodStep.ts7-27](https://github.com/nhsv-tradex/aaa/b

src/models/db/LoginMethodScopeGroupMap.ts4-21](https://github.com/nhsv-trad

#### Request Validation and Method Resolution

The authentication process validates requests and resolves the appropriate

![Chart 5](aaa-images/aaa-section6-5.svg)

##### Validation Rules

The system enforces several validation rules during method resolution:

-   **Single Method Rule**: Exactly one `LoginMethod` must match the `clien
-   **Service Code Matching**: When `sec_code` is provided, it must match `
-   **Client Authorization**: Client must have explicit permission via `Cli
-   **Version Constraints**: App version must meet minimum requirements if

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts52-118](https://github.com/nhsv-trade

src/services/LoginMethodService.ts8-31](https://github.com/nhsv-tradex/aaa/

#### Password-Based Authentication

Password authentication supports multiple variants depending on the target

##### Standard Password Authentication

The `PASSWORD` and `KB_FINA` grant types use the standard password flow:

![Chart 6](aaa-images/aaa-section6-6.svg)

##### TradEX Password Authentication

The `PASSWORD_TRADEX` grant type handles TradEX platform-specific login:

![Chart 7](aaa-images/aaa-section6-7.svg)

##### Certificate Authority Authentication

The `PASSWORD_CA` grant type processes certificate-based authentication dat
```

```
![Chart 8](aaa-images/aaa-section6-8.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts126-129](https://github.com/nhsv-trad

src/services/authenticationService.ts146-147](https://github.com/nhsv-trade

src/services/authenticationService.ts154-155](https://github.com/nhsv-trade

src/models/ILoginTechxReq.ts1-10](https://github.com/nhsv-tradex/aaa/blob/5

src/models/ILoginCAReq.ts1-9](https://github.com/nhsv-tradex/aaa/blob/59961

#### Social Authentication Methods

Social login methods integrate with external OAuth providers and follow a c

##### Facebook Authentication

![Chart 9](aaa-images/aaa-section6-9.svg)

##### Google Authentication

![Chart 10](aaa-images/aaa-section6-10.svg)

##### Generic Social Login

The `SOCIAL_LOGIN` grant type supports various social providers through a u

![Chart 11](aaa-images/aaa-section6-11.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts236-277](https://github.com/nhsv-trad

src/services/authenticationService.ts279-324](https://github.com/nhsv-trade

src/services/authenticationService.ts175-220](https://github.com/nhsv-trade

src/models/ILoginFacebookReq.ts1-14](https://github.com/nhsv-tradex/aaa/blo

src/models/ILoginGoogleReq.ts1-15](https://github.com/nhsv-tradex/aaa/blob/

src/models/ILoginSocialPaaveReq.ts1-14](https://github.com/nhsv-tradex/aaa/
```

```
#### Client Credentials and Demo Authentication

##### Client Credentials Grant

The `CLIENT_CREDENTIALS` grant type provides service-to-service authenticat

![Chart 12](aaa-images/aaa-section6-12.svg)

##### Demo Account Login

The `DEMO` grant type allows access to pre-configured demo accounts:

![Chart 13](aaa-images/aaa-section6-13.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authen/loginClientCredential.ts10-30](https://github.com/nhsv

src/services/authen/loginAccountDemoService.ts19-85](https://github.com/nhs

#### Common Authentication Patterns

##### Third-Party Login Processing

All social and external authentication methods converge through the `loginV

![Chart 14](aaa-images/aaa-section6-14.svg)

##### Direct Service Integration

Many login methods use the `loginDirectToService` pattern for external syst

![Chart 15](aaa-images/aaa-section6-15.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authen/commonLoginThirdParty.ts13-43](https://github.com/nhsv

src/services/authen/loginDirectToService.ts23-84](https://github.com/nhsv-t

#### Error Handling and Validation

The authentication system implements comprehensive error handling for vario
```

```
 | Error Type | Description | Trigger Conditions |
|--- | --- | --- |
 | `InvalidParameterError` | Missing or invalid request parameters | Missin
 | `InvalidIdSecretError` | Invalid client credentials | Client not found o
 | `NoLoginMethodError` | No valid login method found | No `LoginMethod` en
 | `MultipleLoginMethodError` | Ambiguous login method configuration | Mult
 | `NoGrantTypeError` | Unsupported grant type | Grant type not implemented
 | `ForwardError` | External service error | Error response from external a
 | `GeneralError` | Application version validation | App version below mini

The system also implements caching mechanisms to improve performance:

-   **Login Method Cache**: Caches `LoginMethod` objects by `client_id` and
-   **Scope Groups Cache**: Caches scope group mappings for login methods
-   **Client Cache**: Caches client configurations to reduce database queri

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/au

[src/services/authenticationService.ts56-66](https://github.com/nhsv-tradex

src/services/authenticationService.ts75-78](https://github.com/nhsv-tradex/

src/services/authenticationService.ts102-118](https://github.com/nhsv-trade

src/services/authenticationService.ts172-173](https://github.com/nhsv-trade

---

### OTP Authentication

Relevant source files

-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/constants/OtpTypeUriMap.ts](https://github.com/nhsv-tradex/aaa/blob
-   [

    src/constants/messages.ts](https://github.com/nhsv-tradex/aaa/blob/5996
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
```

```
#### Purpose and Scope

This document describes the One-Time Password (OTP) authentication subsyste

For information about the overall authentication architecture and how OTP f

#### OTP Types and Supported Methods

The AAA service supports five distinct OTP types, each with partner-specifi

 | OTP Type | Description | Partner Support | Typical Use Case |
|--- | --- | --- | --- |
 | `SMS_OTP` | One-time code sent via SMS | mas, kis | Standard 2FA for web
 | `MOBILE_OTP` | Push notification to registered mobile app | mas, kis | P
 | `HARDWARE_OTP` | Physical token generating time-based codes | mas | High
 | `SMART_OTP` | Software-based OTP generator (e.g., Google Authenticator)
 | `MATRIX_CARD` | Physical card with grid of numbers | mas, kis | Legacy a

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

#### OTP Configuration

The OTP subsystem is configured through several parameters in the configura

 | Configuration Key | Default Value | Purpose |
|--- | --- | --- |
 | `enableHandleOtp` | `true` | Controls whether OTP verification is handle
 | `otpToken.expiredInSeconds` | `90` | OTP token validity period (90 secon
 | `timeouts.loginPasswordOtp` | `45000` ms | Timeout for password+OTP logi
 | `timeouts.otpService` | `10000` ms (default) | Timeout for OTP service c
 | `nhsv.MAX_DAILY_SENT` | `5` | Maximum OTP requests per user per day |
 | `nhsv.TRIGGER_TIME_INTERVAL` | `30` seconds | Minimum interval between O
 | `nhsv.OTP_SENT_KEY_MAX_LIFE_TIME` | `86400` seconds | Cache lifetime for

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts30](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts76-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts36-37](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
```

```
#### OTP Type Routing Architecture

The system uses a routing table to map OTP types to partner-specific endpoi

##### OTP Type URI Mapping

![Chart 1](aaa-images/aaa-section7-1.svg)

**Diagram: Partner-Specific OTP Routing Configuration**

The routing table is stored in `OTP_TYPE_URI_MAP` and includes both the RES

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

src/conf.ts128](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

#### OTP API Endpoints

The AAA service exposes multiple endpoints for OTP-related operations, all

##### OTP Verification and Authentication

![Chart 2](aaa-images/aaa-section7-2.svg)

**Diagram: OTP Endpoint Routing in HandleAAARequest**

 | Endpoint | Method | Handler Function | Purpose |
|--- | --- | --- | --- |
 | `/api/v1/login/sec/verifyOTP` | POST | `verifyOtp()` or `loginVerifyOtp(
 | `/api/v1/verifyAndSaveOTP` | POST | `generateSaveOtpToken()` | Generate
 | `/api/v1/registerMobileOtp` | POST | `registerMobileOtp()` | Register de
 | `/api/v1/unregisterMobileOtp` | POST | `unregisterMobileOtp()` | Remove
 | `/api/v1/notifyMobileOtp` | POST | `notifyMobileOtp()` | Send mobile pus
 | `/api/v1/notifyMobileOtpNhsv` | POST | `notifyMobileOtpNhsv()` | Send NH
 | `/api/v1/notifyMobileOtpKisTtl` | POST | `notifyMobileOtpKisTtl()` | Sen
 | `/api/v1/notifyBiometricMobileOtp` | POST | `notifyMobileBiometricOtp()`

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts54-111](https://github.com/nhsv-tradex/aaa

##### Conditional OTP Handling
```

```
The system supports two modes of OTP verification controlled by the `enable

![Chart 3](aaa-images/aaa-section7-3.svg)

**Diagram: Conditional OTP Verification Routing**

When `enableHandleOtp` is `true` (default), the AAA service handles OTP ver

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts54-58](https://github.com/nhsv-tradex/aaa/

src/conf.ts27-30](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

#### OTP Token Lifecycle

##### Token Generation and Expiration

OTP tokens have a short lifecycle to maintain security:

**Diagram: OTP Request and Verification Flow with Rate Limiting**

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts76-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

src/constants/messages.ts1-5](https://github.com/nhsv-tradex/aaa/blob/59961

#### Rate Limiting and Security Controls

##### Daily Rate Limiting

The system implements rate limiting to prevent OTP abuse and protect agains

 | Rate Limit Type | Configuration | Purpose |
|--- | --- | --- |
 | Maximum daily OTP requests | `nhsv.MAX_DAILY_SENT = 5` | Prevents excess
 | Minimum interval between requests | `nhsv.TRIGGER_TIME_INTERVAL = 30` se
 | Rate limit cache lifetime | `nhsv.OTP_SENT_KEY_MAX_LIFE_TIME = 86400` se

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts
```

[src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

Error Messages

The system returns specific error messages for rate limit violations:

- `OTP_GENERATE_TO_FAST`: Returned when user requests OTP within 30 secon
- `OTP_LIMIT_GENERATE`: Returned when user exceeds 5 OTP requests per day
- `SUCCESS`: Returned on successful OTP generation

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/messages.ts1-5](https://github.com/nhsv-tradex/aaa/blob/5996

Mobile OTP Variants

The service supports multiple mobile OTP notification mechanisms for differ

Standard Mobile OTP

Endpoint: `/api/v1/notifyMobileOtp`
Handler: `notifyMobileOtp(msg.data, msg.transactionId)`

Standard mobile push notification for OTP delivery. Used for general-purpos

NHSV-Specific Mobile OTP

Endpoint: `/api/v1/notifyMobileOtpNhsv`
Handler: `notifyMobileOtpNhsv(msg.data, msg.transactionId)`

Specialized mobile OTP notification for NHSV partner with custom rate limit

KIS TTL-Specific Mobile OTP

Endpoint: `/api/v1/notifyMobileOtpKisTtl`
Handler: `notifyMobileOtpKisTtl(msg.data, msg.transactionId)`

KIS partner-specific mobile OTP with custom TTL (Time-To-Live) handling.

Biometric Mobile OTP

Endpoint: `/api/v1/notifyBiometricMobileOtp`
Handler: `notifyMobileBiometricOtp(msg.data, msg.transactionId)`

Mobile OTP notification used during biometric registration flows. Combines

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts104-111](https://github.com/nhsv-tradex/aa

#### Integration with Other Systems

##### Kafka-Based OTP Routing

![Chart 4](aaa-images/aaa-section7-4.svg)

**Diagram: OTP Verification Routing via Kafka Topics**

OTP verification requests are routed to partner services via Kafka topics.

1.  **Partner identifier** (mas, kis, etc.)
2.  **OTP type** (MATRIX\_CARD, SMART\_OTP, HARDWARE\_OTP, SMS\_OTP)
3.  **Communication protocol** (REST vs WebSocket bridge)

This architecture decouples the AAA service from partner-specific OTP imple

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

##### Transaction Tracking

All OTP-related operations accept a transaction ID parameter for end-to-end
```

// Example from HandleAAARequest.ts
notifyMobileOtp(msg.data, `${msg.transactionId}`)
notifyMobileOtpNhsv(msg.data, `${msg.transactionId}`)
notifyMobileOtpKisTtl(msg.data, `${msg.transactionId}`)

```
The transaction ID flows through the entire request chain, enabling correla

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts105-109](https://github.com/nhsv-tradex/aa

#### Security Considerations

##### OTP Token Security

-   **Short expiration**: OTP tokens expire after 90 seconds to minimize th
-   **Single-use**: OTP codes are invalidated after successful verification
-   **Rate limiting**: Maximum 5 OTP requests per user per day prevents enu
-   **Minimum interval**: 30-second cooldown between requests prevents rapi

##### Cache-Based Rate Limiting

Rate limiting counters are stored in Redis with configurable TTLs:

-   Daily request counter: 86400 seconds (24 hours)
-   Interval enforcement: 30 seconds between requests

This approach provides fast rate limit checks without database queries whil

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts76-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts135-139](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

---

### Biometric Authentication

Relevant source files

-   [

    src/constants/Constants.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/constants/GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [
```

```
    src/constants/biometricKeys.ts](https://github.com/nhsv-tradex/aaa/blob
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

This document covers the biometric authentication system within the AAA ser

For general authentication flows and grant types, see [Authentication Servi

#### System Overview

The biometric authentication system enables secure login using biometric si

##### Core Components

 | Component | Purpose | Key Features |
|--- | --- | --- |
 | `BiometricService` | Main service logic | Registration, login, verificat
 | `Biometric` model | Database representation | Stores public keys, user a
 | Request/Response models | API contracts | Validation, type safety for di
 | Cryptographic verification | Security layer | RSA-SHA256 signature verif

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts1-734](https://github.com/nhsv-tradex/aaa/

src/models/db/Biometric.ts1-41](https://github.com/nhsv-tradex/aaa/blob/599

#### Registration Flow

##### Standard Registration Process

![Chart 1](aaa-images/aaa-section8-1.svg)

**Standard Registration Flow (`registerBiometric`)** Sources:[](https://git

[src/services/BiometricService.ts150-242](https://github.com/nhsv-tradex/aa

##### KIS-Specific Registration Process

![Chart 2](aaa-images/aaa-section8-2.svg)

**KIS Registration Flow (`registerBiometricKis`)** Sources:[](https://githu

[src/services/BiometricService.ts87-148](https://github.com/nhsv-tradex/aaa
```

```
##### Registration Data Model

The `Biometric` model stores comprehensive device and authentication inform

 | Field | Type | Purpose |
|--- | --- | --- |
 | `publicKey` | string | RSA public key for signature verification |
 | `username` | string | Associated user account |
 | `password` | string | Encrypted password for service login |
 | `deviceId` | string | Device identifier for device-specific bindings |
 | `clientId` | number | OAuth client identifier |
 | `status` | string | ACTIVE/INACTIVE status |
 | `biometricType` | string | Type of biometric authentication |
 | `otpIndex/otpValue` | string | OTP information when applicable |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/Biometric.ts4-26](https://github.com/nhsv-tradex/aaa/blob/59

src/services/BiometricService.ts117-140](https://github.com/nhsv-tradex/aaa

#### Login Flow

##### Biometric Login Process

![Chart 3](aaa-images/aaa-section8-3.svg)

**Standard Biometric Login (`loginBiometric`)** Sources:[](https://github.c

[src/services/BiometricService.ts496-590](https://github.com/nhsv-tradex/aa

##### OTP-Based Biometric Login

![Chart 4](aaa-images/aaa-section8-4.svg)

**OTP Biometric Login (`loginBiometricOtp`)** Sources:[](https://github.com

[src/services/BiometricService.ts592-689](https://github.com/nhsv-tradex/aa

src/services/BiometricService.ts691-734](https://github.com/nhsv-tradex/aaa

#### Cryptographic Verification

##### RSA Signature Verification Process
```

```
The system uses RSA-SHA256 for signature verification:
```

// Public key format reconstruction
let publicKey = `-----BEGIN PUBLIC KEY-----\n{key}\n-----END PUBLIC KEY-----`;
const verify = crypto.createVerify("RSA-SHA256");
publicKey = publicKey.replace(/{key}/g, queryResponse.public_key);

// Signature verification against username
verify.update(request.username.toUpperCase());
const isValid = verify.verify(publicKey, signature, "base64");

```
**Verification Components**:

-   **Algorithm**: RSA-SHA256
-   **Data Signed**: Username (uppercase)
-   **Signature Format**: Base64-encoded
-   **Public Key Storage**: Database field `public_key`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts502-503](https://github.com/nhsv-tradex/aa

src/services/BiometricService.ts517-523](https://github.com/nhsv-tradex/aaa

src/services/BiometricService.ts613-619](https://github.com/nhsv-tradex/aaa

##### Password Verification Integration

For certain scenarios, biometric verification integrates with external pass

![Chart 5](aaa-images/aaa-section8-5.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts386-402](https://github.com/nhsv-tradex/aa

#### Status and Lifecycle Management

##### Status Management

The biometric system maintains several status states:

 | Status | Description | Transitions |
|--- | --- | --- |
 | `INACTIVE` | Newly registered, awaiting OTP verification | → ACTIVE (aft
 | `ACTIVE` | Verified and ready for login | → INACTIVE (password change, d
 | Deleted (`is_deleted: true`) | Cancelled or replaced | Final state |

**Status Constants**:

-   `BIOMETRIC_STATUS.ACTIVE = "ACTIVE"`
-   `BIOMETRIC_STATUS.INACTIVE = "INACTIVE"`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constants/b
```

```
[src/constants/biometricKeys.ts13-16](https://github.com/nhsv-tradex/aaa/bl

src/services/BiometricService.ts233-235](https://github.com/nhsv-tradex/aaa

##### Cancellation and Cleanup

![Chart 6](aaa-images/aaa-section8-6.svg)

**Cancellation Reasons**:

-   `CHANGE_DEVICE`: User switched devices
-   `CHANGE_PASS`: User changed password
-   `CANCEL`: User manually cancelled
-   `OTP_VERIFY`: Failed OTP verification

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts408-429](https://github.com/nhsv-tradex/aa

src/services/BiometricService.ts267-276](https://github.com/nhsv-tradex/aaa

src/constants/biometricKeys.ts18-23](https://github.com/nhsv-tradex/aaa/blo

#### OTP Verification for Biometrics

##### OTP Verification Process

![Chart 7](aaa-images/aaa-section8-7.svg)

The OTP verification uses Apache crypt3 for secure comparison:

-   Input OTP is encrypted using stored checksum as salt
-   Comparison ensures timing attack resistance
-   Successful verification activates the biometric record

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts307-353](https://github.com/nhsv-tradex/aa

#### Error Handling

##### Biometric-Specific Error Codes

 | Error Code | Trigger Condition | Resolution |
|--- | --- | --- |
```

```
 | `LOGIN_BIOMETRIC_NOT_FOUND` | No biometric record exists | User must reg
 | `LOGIN_BIOMETRIC_SIGNATURE_VERIFICATION_FAILED` | Invalid signature | Ch
 | `LOGIN_BIOMETRIC_PASSWORD_NOT_MATCH` | Backend password verification fai
 | `LOGIN_BIOMETRIC_OTP_DOES_NOT_VERIFIED` | Biometric not OTP-activated |
 | `BIOMETRIC_ACTIVE_ON_ANOTHER_DEVICE` | Device mismatch detected | Re-reg

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constants/b

[src/constants/biometricKeys.ts1-11](https://github.com/nhsv-tradex/aaa/blo

##### Domain-Specific Login Errors

The system handles different error codes from various partner domains:
```

const BIOMETRIC_LOGIN_ERROR = {
  kbsv: "INVALID_CLIENT_CREDENTIAL",
  vcsc: "LOGIN_WRONG_PASSWORD",
  kbfinance: "INVALID_CLIENT_CREDENTIAL",
  kis: "LOGIN_WRONG_PASSWORD"
};

```
Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constants/b

[src/constants/biometricKeys.ts25-30](https://github.com/nhsv-tradex/aaa/bl

src/services/BiometricService.ts584-587](https://github.com/nhsv-tradex/aaa

#### Integration Points

##### Database Schema

The biometric system uses the `t_biometric` table with the following key re

![Chart 8](aaa-images/aaa-section8-8.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/Biometric.ts35-37](https://github.com/nhsv-tradex/aaa/blob/5

##### Service Integration

The biometric service integrates with several other system components:

-   **LoginMethodService**: For finding appropriate login methods
-   **ClientService**: For client validation
-   **TokenService**: For generating authentication tokens
-   **Kafka messaging**: For external service communication
-   **Redis caching**: For login method caching

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/services/Bi

[src/services/BiometricService.ts24-25](https://github.com/nhsv-tradex/aaa/

src/services/BiometricService.ts635-651](https://github.com/nhsv-tradex/aaa

---

### Social and Organization Login

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [
```

```
    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/constants/GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

#### Purpose and Scope

This document describes the social OAuth authentication and organization-ba

#### Overview

The AAA service supports two primary categories of alternative authenticati

1.  **Social OAuth Authentication** - Integration with third-party OAuth pr
2.  **Organization Login** - Domain-based and partner credential authentica

These methods enable users to authenticate without managing separate creden

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md11-19](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md

src/constants/GrantType.ts1-20](https://github.com/nhsv-tradex/aaa/blob/599

#### Social OAuth Authentication

##### Supported Providers and Grant Types

The service integrates with four external OAuth providers, each mapped to a

 | Provider | Grant Type | Configuration Key |
|--- | --- | --- |
 | Google OAuth | `access_google` | `timeouts.loginGoogle` |
 | Facebook OAuth | `access_facebook` | `timeouts.loginFacebook` |
 | Apple Sign-In | `access_apple` | `timeouts.loginApple` |
 | TechX | N/A | `timeouts.loginTechx` |
 | Generic Social | `social_login` | N/A |
 | Organization Social | `organization_social_login` | N/A |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan
```

```
[src/constants/GrantType.ts8-14](https://github.com/nhsv-tradex/aaa/blob/59

src/conf.ts31-35](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

##### Social Login Flow

![Chart 1](aaa-images/aaa-section9-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts35-45](https://github.com/nhsv-tradex/aaa/

##### Request URI Patterns

Social login requests are routed through the following URI patterns:
```

post:/api/v1/login/social post:/api/v1/user/SocialLogin post:/api/v1/login/social/organization

```
All three URIs route to the `authenticate` function, which determines the s

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts35-45](https://github.com/nhsv-tradex/aaa/

##### Configuration and Timeouts

Social OAuth provider timeouts are configured in the `timeouts` section of
```

```
timeouts: {
  loginFacebook: 15000,   // 15 seconds
  loginGoogle: 15000,     // 15 seconds
  loginApple: 15000,      // 15 seconds
  loginTechx: 15000,      // 15 seconds
}
```

```
If a timeout value is `null`, it defaults to `conf.defaultKafkaTimeout` (10

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts31-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts178-182](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

#### Organization Login

##### Domain-Based Authentication

Organization login enables enterprise users to authenticate using domain-sp

 | Grant Type | URI Pattern | Description |
|--- | --- | --- |
 | `access_domain` | `post:/api/v1/login/domain` | Domain-based authenticat
 | `organization_login` | `post:/api/v1/login/organization` | Organization
 | `organization_social_login` | `post:/api/v1/login/social/organization` |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/GrantType.ts11-18](https://github.com/nhsv-tradex/aaa/blob/5

src/consumer/HandleAAARequest.ts35-43](https://github.com/nhsv-tradex/aaa/b

##### Organization Login Flow

![Chart 2](aaa-images/aaa-section9-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts35-45](https://github.com/nhsv-tradex/aaa/

src/conf.ts171-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Domain-Specific Configuration

When the service domain differs from the default `TRADEX_DOMAIN`, the `forc
```

if (conf.domain !== Utils.TRADEX_DOMAIN) {
  conf.forceSecCode = conf.domain;
}

```
This enables domain-specific security validation during the authentication

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts171-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/conf.ts10-13](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

##### Organization Login Timeout

The organization login timeout is configured via `timeouts.loginDomain`:
```

timeouts: {
  loginDomain: null, // Defaults to defaultKafkaTimeout (10000ms)
}

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts39](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

#### Partner Credential Authentication

##### Overview

Partner credential authentication enables trusted partners to authenticate

##### Request Routing

![Chart 3](aaa-images/aaa-section9-3.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts46-47](https://github.com/nhsv-tradex/aaa/

##### Service Integration

The `loginPartnerCredential` function is defined in a dedicated service mod
```

if (msg.uri === "post:/api/v1/login/partnerCredential") {
  return loginPartnerCredential(msg.data, msg);
}

```
This service handles partner-specific authentication logic, validating part

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts46-47](https://github.com/nhsv-tradex/aaa/

src/consumer/HandleAAARequest.ts29](https://github.com/nhsv-tradex/aaa/blob

#### Authentication Message Routing

##### URI-to-Handler Mapping

The following table maps all social and organization login URIs to their ha

 | URI Pattern | Grant Type(s) | Handler Function | Module |
|--- | --- | --- | --- |
 | `post:/api/v1/login` | Multiple | `authenticate` | authenticationService
 | `post:/api/v1/login/social` | `access_google`, `access_facebook`, `acces
 | `post:/api/v1/user/SocialLogin` | Social grant types | `authenticate` |
 | `post:/api/v1/login/domain` | `access_domain` | `authenticate` | authent
 | `post:/api/v1/login/organization` | `organization_login` | `authenticate
 | `post:/api/v1/login/social/organization` | `organization_social_login` |
 | `post:/api/v1/login/partnerCredential` | N/A | `loginPartnerCredential`

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts35-47](https://github.com/nhsv-tradex/aaa/

##### Message Handler Implementation

![Chart 4](aaa-images/aaa-section9-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-47](https://github.com/nhsv-tradex/aaa/

#### Configuration Reference

##### Environment Variables

The following environment variables affect social and organization login:

 | Variable | Purpose | Default |
|--- | --- | --- |
```

```
 | `TRADEX_ENV_DOMAIN` | Service domain for organization login | `TRADEX_DO
 | `TRADEX_ENV_ENABLE_BIOMETRIC` | Enable biometric auth (affects org+biome

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts13-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

README.md166-172](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

##### Timeout Configuration

Default timeout values for external provider integrations:
```

defaultKafkaTimeout: 10000 // 10 seconds (fallback)

timeouts: {
  loginFacebook: 15000, // 15 seconds
  loginGoogle: 15000, // 15 seconds
  loginApple: 15000, // 15 seconds
  loginTechx: 15000, // 15 seconds
  loginDomain: null, // Uses defaultKafkaTimeout
}

```
Any timeout set to `null` is automatically set to `defaultKafkaTimeout` dur

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts26](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts31-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts178-182](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Domain-Specific RSA Keys

Organization login may use domain-specific RSA keys for password encryption
```

rsa: {
  enableEncryptPassword: true,
  publicKey: `keys/aaa/${domain}/rsa-public.key`,
  privateKey: `keys/aaa/${domain}/rsa-private.key`,
}

```
The domain is determined by `TRADEX_ENV_DOMAIN` environment variable.

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts42-46](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

#### Integration with Other Subsystems

##### Account Linking

Social and organization login integrates with the Account Linking System ([

[#5](https://github.com/nhsv-tradex/aaa/blob/59961d60/#5)) to enable users

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts1-6](https://github.com/nhsv-tradex/aaa/bl

##### Token Management

All successful social and organization authentications generate JWT tokens

[#4](https://github.com/nhsv-tradex/aaa/blob/59961d60/#4)). The token expir

-   Access Token: 900 seconds (15 minutes)
-   Refresh Token: 86400 seconds (24 hours)
-   Refresh Token with Remember Me: 2592000 seconds (30 days)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts69-75](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

README.md158-160](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

##### Scope Management

Social and organization login requests are subject to the same scope-based

[#6](https://github.com/nhsv-tradex/aaa/blob/59961d60/#6)) validates that t

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts1-9](https://github.com/nhsv-tradex/aaa/bl
```

```
---

## Token Management

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
-   [

    src/dao/RefreshTokenDao.ts](https://github.com/nhsv-tradex/aaa/blob/599

### Purpose and Scope

This document describes the token management subsystem responsible for JWT

For information about the initial token generation during authentication fl

**Sources:** README.md, conf.ts, HandleAAARequest.ts

* * *

### Token Types and Expiration Configuration

The AAA service manages four distinct token types, each with specific lifes

 | Token Type | Configuration Key | Default Expiration | Use Case |
|--- | --- | --- | --- |
 | Access Token | `accessToken.expiredInSeconds` | 900 seconds (15 minutes)
 | Refresh Token | `refreshToken.expiredInSeconds` | 86400 seconds (24 hour
 | Refresh Token (Remember Me) | `refreshToken.expiredInSecondsWithRemember
 | OTP Token | `otpToken.expiredInSeconds` | 90 seconds (1.5 minutes) | One

These expiration times are configured in[](https://github.com/nhsv-tradex/a

[src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

![Chart 1](aaa-images/aaa-section10-1.svg)
```

```
**Token Lifecycle Diagram**

**Sources:** src/conf.ts:69-78, README.md:158-160

* * *

### JWT Configuration and Domain-Specific Keys

The token management system uses JWT (JSON Web Tokens) with RSA encryption.

#### Key Path Resolution

The system resolves JWT key paths based on the configured domain:

-   **Default keys:** `keys/jwt-private.key` and `keys/jwt-public.key`
-   **Domain-specific keys:** `keys/{DOMAIN}/jwt-private.key` and `keys/{DO
-   **Fallback logic:** If domain-specific keys don't exist, the system fal

The configuration structure in[](https://github.com/nhsv-tradex/aaa/blob/59

[src/conf.ts59-68](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con
```

```
jwt: {
  privateKeyFile: "keys/jwt-private.key",
  publicKeyFile: "keys/jwt-public.key",
  domains: {
    vcsc: {
      privateKeyFile: "keys/vcsc/jwt-private.key",
      publicKeyFile: "keys/vcsc/jwt-public.key",
    },
  },
}
```

```
The key resolution is processed by `Utils.createJwtConfig()` at[](https://g

[src/conf.ts174-176](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

![Chart 2](aaa-images/aaa-section10-2.svg)

**JWT Key Configuration Architecture**

**Sources:** src/conf.ts:59-68, src/conf.ts:174-176

* * *

### Token Request Handling

Token-related operations are routed through the Kafka message handler at[](

[src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/blob/

#### Refresh Token Endpoint

**URI:** `post:/api/v1/refreshToken`
**Handler:** `refreshAccessToken(msg.data)` from `TokenService`
**Location:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consum

[src/consumer/HandleAAARequest.ts48-49](https://github.com/nhsv-tradex/aaa/

This endpoint allows clients to obtain a new access token using a valid ref

#### Revoke Token Endpoint

**URI:** `post:/api/v1/revokeToken`
**Handler:** `revokeToken(msg.data)` from `TokenService`
**Location:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consum

[src/consumer/HandleAAARequest.ts50-51](https://github.com/nhsv-tradex/aaa/

This endpoint invalidates both access and refresh tokens, forcing the user

![Chart 3](aaa-images/aaa-section10-3.svg)

**Token Request Processing Sequence**

**Sources:** src/consumer/HandleAAARequest.ts:48-51, README.md:183-185
```

```
* * *

### Refresh Token Data Access Layer

The `RefreshTokenDao` module at[](https://github.com/nhsv-tradex/aaa/blob/5

[src/dao/RefreshTokenDao.ts](https://github.com/nhsv-tradex/aaa/blob/59961d

#### Core DAO Operations

 | Function | Purpose | Location |
|--- | --- | --- |
 | `getRefreshToken(token, con)` | Retrieve token by token string | [ |
 | src/dao/RefreshTokenDao.ts34-42](https://github.com/nhsv-tradex/aaa/blob
 | `getRefreshTokenById(id, con)` | Retrieve token by numeric ID | [ |

src/dao/RefreshTokenDao.ts44-52](https://github.com/nhsv-tradex/aaa/blob/59
| `deleteRefreshToken(token, con)` | Delete token and parent cascade | [

src/dao/RefreshTokenDao.ts12-21](https://github.com/nhsv-tradex/aaa/blob/59
| `deleteRefreshTokenById(token, con)` | Delete token by ID with cascade |

src/dao/RefreshTokenDao.ts23-32](https://github.com/nhsv-tradex/aaa/blob/59

#### Cascading Delete Logic

Refresh tokens support a parent-child relationship through the `parentId` f

1.  Query the token record from the database
2.  Delete the token itself
3.  If `parentId` exists, delete the parent token as well

This cascading behavior is implemented in both delete methods at[](https://

[src/dao/RefreshTokenDao.ts12-21](https://github.com/nhsv-tradex/aaa/blob/5

src/dao/RefreshTokenDao.ts23-32](https://github.com/nhsv-tradex/aaa/blob/59

#### Query Construction

The DAO uses the `Query<RefreshToken>` builder pattern from `BaseModel` to
```

// Token lookup queries
const queryByToken: Query = new Query(new RefreshToken())

```
  .where((model: RefreshToken) => model.token, "=?");

const queryById: Query = new Query(new RefreshToken())
  .where((model: RefreshToken) => model.id, "=?");
```

```
These queries are defined at[](https://github.com/nhsv-tradex/aaa/blob/5996

[src/dao/RefreshTokenDao.ts6-10](https://github.com/nhsv-tradex/aaa/blob/59

#### Error Handling

When a token is not found during retrieval operations, the DAO throws `Erro

[src/dao/RefreshTokenDao.ts39](https://github.com/nhsv-tradex/aaa/blob/5996

src/dao/RefreshTokenDao.ts49](https://github.com/nhsv-tradex/aaa/blob/59961

![Chart 4](aaa-images/aaa-section10-4.svg)

**RefreshTokenDao Architecture**

**Sources:** src/dao/RefreshTokenDao.ts:1-52

* * *

### Token Storage Architecture

Tokens are persisted using a dual-storage strategy combining MySQL for dura

#### MySQL Storage

The MySQL database contains the `RefreshToken` table which stores:

-   Token string (unique identifier)
-   User ID and client ID associations
-   Expiration timestamps
-   Parent token relationships (for token chains)
-   Creation and update timestamps

This is the authoritative source of truth for token data, referenced in[](h

[README.md222](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

#### Redis Caching

Redis caches token data with configurable TTL (Time To Live) settings:

-   **Default cache duration:** 7 days (`defaultLongRedisDuration`) at[
```

```
    src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con
-   **Scope cache timeout:** 180 seconds (`cacheTimeout`) at[

    src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
-   **Redis configuration:** Host, port, and options at[

    src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr

The caching strategy prioritizes cache hits to minimize database queries du

#### Cache Invalidation

Token revocation triggers cache invalidation to ensure deleted tokens are i

![Chart 5](aaa-images/aaa-section10-5.svg)

**Token Storage and Caching Strategy**

**Sources:** src/conf.ts:24, src/conf.ts:127, src/conf.ts:130-134, README.m

* * *

### Token Lifecycle Summary

The complete token lifecycle progresses through the following stages:

1.  **Generation:** Tokens are created after successful authentication (see
2.  **Signing:** JWT tokens are signed with domain-specific RSA keys
3.  **Storage:** Tokens are persisted to MySQL and cached in Redis
4.  **Validation:** During API requests, tokens are validated against cache
5.  **Refresh:** Access tokens are renewed using refresh tokens via `/api/v
6.  **Expiration:** Tokens automatically expire based on configured timeout
7.  **Revocation:** Tokens can be explicitly revoked via `/api/v1/revokeTok

The system maintains different expiration windows for different token types

**Sources:** README.md:22-28, src/conf.ts:69-78

---

## Account Linking System

Relevant source files

-   [
```

```
    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

### Purpose and Scope

The Account Linking System enables users to associate multiple authenticati

This document covers the link account creation/deletion flows, partner mana

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md19](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L1

README.md62](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L62

README.md193-195](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

### System Architecture

The Account Linking System operates as a specialized subsystem within the A

#### Component Overview

![Chart 1](aaa-images/aaa-section11-1.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts1-131](https://github.com/nhsv-tradex/aaa/

README.md62](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L62

### Account Linking Flow

The account linking process involves multiple steps including partner disco

#### Link Account Creation Sequence

![Chart 2](aaa-images/aaa-section11-2.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts63-76](https://github.com/nhsv-tradex/aaa/
```

```
#### Link Account Login Flow

![Chart 3](aaa-images/aaa-section11-3.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts67-68](https://github.com/nhsv-tradex/aaa/

### API Endpoints

The Account Linking System exposes 14 API endpoints for managing account as

 | Endpoint | Method | Handler Function | Description |
|--- | --- | --- | --- |
 | `/api/v1/partners` | GET | `findAllPartners()` | Retrieve list of all av
 | `/api/v1/linkAccounts` | GET | `findAllLinkAccount(data)` | Get all link
 | `/api/v1/linkAccounts` | POST | `createLinkAccount(data, txId)` | Create
 | `/api/v1/linkAccounts/{partnerId}` | DELETE | `deleteLinkAccount(data, t
 | `/api/v1/linkAccounts/login` | POST | `loginLinkAccount(data, txId)` | A
 | `/api/v1/linkAccounts/confirm` | POST | `confirmLinkAccount(data, txId)`
 | `/api/v1/linkAccounts/unlink` | POST | `unlink(data, txId)` | Unlink acc
 | `/api/v1/linkAccounts/init` | POST | `initLinkAccount(data)` | Initializ
 | `/api/v1/linkAccounts/notifyOtpPartner` | \- | `notifyOtpPartner(data, t
 | `/api/v1/linkAccounts/notifyOtpFromPartner` | \- | `notifyOtpFromPartner
 | `/api/v1/linkAccounts/changeUsername` | \- | `changeUsername(data)` | Up
 | `/api/v1/linkAccounts/approve` | POST | `createLinkAccountApprove(data,
 | `/api/v1/linkAccounts/getUserIdByPartnerName` | \- | `getUserIdByPartner
 | `/api/v1/linkAccounts/leaderboard/settings` | \- | `putLeaderboardSettin

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts59-122](https://github.com/nhsv-tradex/aaa

### Message Routing and Handler Integration

All account linking requests flow through the unified message handler, whic

![Chart 4](aaa-images/aaa-section11-4.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-125](https://github.com/nhsv-tradex/aaa

src/consumer/HandleAAARequest.ts6](https://github.com/nhsv-tradex/aaa/blob/
```

```
### Partner Management

Partners represent external services or platforms that can be linked to use

#### Partner Discovery

The `findAllPartners()` function retrieves all registered partner platforms

![Chart 5](aaa-images/aaa-section11-5.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts59-60](https://github.com/nhsv-tradex/aaa/

#### Partner Data Model

The Partners table stores information about each linkable service:

-   `partnerId`: Unique identifier for the partner
-   `name`: Display name of the partner service
-   `config`: Partner-specific configuration (JSON)
-   `status`: Partner availability status

### OTP Notification System

The Account Linking System implements a bidirectional OTP notification mech

#### Cross-Partner OTP Flow

![Chart 6](aaa-images/aaa-section11-6.svg)

#### OTP Notification Functions

 | Function | Direction | Purpose |
|--- | --- | --- |
 | `notifyOtpPartner(data, transactionId)` | AAA → Partner | Request partne
 | `notifyOtpFromPartner(data, transactionId)` | Partner → AAA | Receive OT

Both functions use transaction IDs for request-response correlation and aud

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts75-78](https://github.com/nhsv-tradex/aaa/
```

```
### Link Account Data Model

The `LinkAccounts` table maintains the associations between users and their

#### Link Account Record Structure

![Chart 7](aaa-images/aaa-section11-7.svg)

#### Link Account States

The linking process progresses through multiple states:

 | State | Description |
|--- | --- |
 | `PENDING` | Link request initiated, awaiting OTP generation |
 | `OTP_SENT` | OTP has been sent to user via partner |
 | `ACTIVE` | Link confirmed and active |
 | `INACTIVE` | Link deactivated by user |
 | `REJECTED` | Link request denied |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md225](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

### Transaction Tracking

All account linking operations are tracked using transaction IDs for auditi

#### Transaction Flow

![Chart 8](aaa-images/aaa-section11-8.svg)

Each `linkAccountService` function accepts a `transactionId` parameter (oft

-   Correlating multi-step operations across services
-   Auditing and debugging
-   Tracking request flow through distributed systems

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts64-82](https://github.com/nhsv-tradex/aaa/

### Security Considerations

The Account Linking System implements multiple security measures to protect
```

```
#### Security Controls

 | Control | Implementation |
|--- | --- |
 | **Identity Verification** | OTP verification required before link activa
 | **Partner Validation** | Partner must be registered in Partners table |
 | **User Authorization** | User must be authenticated to create/delete lin
 | **Transaction Integrity** | All operations tracked with unique transacti
 | **Status Management** | Links progress through defined states preventing
 | **Partner Isolation** | Each partner has isolated configuration and cred

#### OTP Security

-   OTPs are generated and sent by the partner service, not stored in AAA
-   Only OTP tokens (references) are stored in LinkAccounts
-   OTP verification happens during confirmation step
-   Failed OTP attempts tracked for rate limiting

#### Link Deletion Security

The `deleteLinkAccount()` and `unlink()` functions ensure that only the acc

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts65-72](https://github.com/nhsv-tradex/aaa/

README.md21-28](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#

### Additional Operations

#### Username Change

The `changeUsername()` function allows users to update the username associa

#### User ID Resolution

The `getUserIdByPartnerName()` function enables partner services to resolve

#### Leaderboard Settings

The `putLeaderboardSetting()` function manages configuration for linked acc

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume
```

```
[src/consumer/HandleAAARequest.ts79-122](https://github.com/nhsv-tradex/aaa

### Integration with Authentication System

Account linking integrates with the broader authentication system:

-   **Grant Type**: The `link_account` grant type enables authentication us
-   **Token Issuance**: Successful link account login issues standard AAA J
-   **OTP Integration**: Link creation may require OTP verification using t

The `loginLinkAccount()` function retrieves the linked account mapping and

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts67-68](https://github.com/nhsv-tradex/aaa/

README.md211](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L2

---

## Authorization and Scope Management

Relevant source files

-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
-   [

    src/consumer/HandleForwardRequest.ts](https://github.com/nhsv-tradex/aa

### Purpose and Scope

This document describes the authorization and scope management system withi

For information about authentication and token generation, see [Authenticat

### Core Concepts

#### Scopes

A **scope** represents a specific permission that grants access to a partic
```

```
-   A unique identifier (`scope_id`)
-   A URI pattern that defines which requests the scope applies to
-   Metadata returned via `getScopeResponse()` method

#### Scope Groups

**Scope groups** are collections of scopes that are assigned together. Toke

#### URI Patterns

URI patterns use glob-style matching (via the `nanomatch` library) to deter

### Scope Management Architecture

![Chart 1](aaa-images/aaa-section12-1.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts95-106](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

src/consumer/HandleForwardRequest.ts1-41](https://github.com/nhsv-tradex/aa

### Configuration

Scope management configuration is centralized in[](https://github.com/nhsv-

[src/conf.ts95-106](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

 | Configuration Key | Value | Description |
|--- | --- | --- |
 | `scopes.verifyOtpScope` | `"VERIFY_OTP"` | Special scope for OTP verific
 | `scopes.unAuthenticated` | `"PUBLIC"` | Scope for public endpoints |
 | `scopes.loadFrom.topic` | `"configuration"` | Kafka topic for loading sc
 | `scopes.loadFrom.uris.scope` | `"/api/v1/admin/scope"` | URI for fetchin
 | `scopes.loadFrom.uris.scopeGroup` | `"/api/v1/admin/scopeGroup"` | URI f
 | `scopes.loadFrom.uris.pageSize` | `100` | Pagination size for loading |
 | `cacheTimeout` | `180000` | Cache timeout in milliseconds (3 minutes) |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts95-128](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

### Authorization Flow
```

```
The following diagram illustrates the complete authorization flow from requ

![Chart 2](aaa-images/aaa-section12-2.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/a

### Request Routing and Scope Checking

#### URI-Based Routing

The `handleAAAMessage` function in[](https://github.com/nhsv-tradex/aaa/blo

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

**AAA-specific URIs** (no scope check required):

-   `post:/api/v1/login*` - Various login endpoints
-   `post:/api/v1/refreshToken` - Token refresh
-   `post:/api/v1/revokeToken` - Token revocation
-   `get:/api/v1/scopes/search` - Scope search
-   Link account endpoints (`/api/v1/linkAccounts/*`)
-   OTP and biometric endpoints

**Forwarded URIs** (require scope check):

-   All other URIs that don't match AAA-specific patterns

#### Scope Search Endpoint

The scope search endpoint at[](https://github.com/nhsv-tradex/aaa/blob/5996

[src/consumer/HandleAAARequest.ts86-89](https://github.com/nhsv-tradex/aaa/
```

// Endpoint: get:/api/v1/scopes/search
// Input: msg.data.scopeGroupIds (array of scope group IDs)
// Output: IScopeSearchRes with array of scope responses

```
The implementation retrieves scopes using `scopeService.getScopesByScopeGro

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts86-89](https://github.com/nhsv-tradex/aaa/

### Token-Based Authorization

#### Token Structure

Tokens contain authorization information including:

 | Token Field | Type | Description |
|--- | --- | --- |
 | `userId` | number/string | User identifier |
 | `serviceUserId` | number/string | Service-specific user ID |
 | `clientId` | string | OAuth2 client identifier |
 | `refreshTokenId` | string | Associated refresh token ID |
 | `scopeGroupIds` | number\[\] | Array of scope group IDs |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts14-26](https://github.com/nhsv-tradex/

#### Transaction Tracking

When a token is present, the system creates an `ITransaction` record at[](h

[src/consumer/HandleForwardRequest.ts15-25](https://github.com/nhsv-tradex/

-   User and client identifiers from the token
-   Transaction ID for tracing
-   Full message data as JSON
-   Scope ID (set during forwarding)
-   Destination topic and URI (set during forwarding)

This transaction record enables auditing and troubleshooting of authorizati

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts15-25](https://github.com/nhsv-tradex/

### URI Pattern Matching
```

```
#### Matching Algorithm

The authorization system uses the `nanomatch` library at[](https://github.c

[src/consumer/HandleForwardRequest.ts1-30](https://github.com/nhsv-tradex/a

![Chart 3](aaa-images/aaa-section12-3.svg)

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts28-35](https://github.com/nhsv-tradex/

#### First-Match Strategy

The authorization system uses a **first-match strategy**. Once a scope's UR

1.  Logs the matched scope at[

    src/consumer/HandleForwardRequest.ts32](https://github.com/nhsv-tradex/
2.  Calls `forwardMessageWithScope()` to forward the request
3.  Exits the loop without checking remaining scopes

This ensures efficient processing and prevents duplicate forwarding.

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts28-35](https://github.com/nhsv-tradex/

### Public Access

Requests without tokens (unauthenticated requests) can still access public

1.  Token extraction returns `null` at[

    src/consumer/HandleForwardRequest.ts14](https://github.com/nhsv-tradex/
2.  No transaction record is created (line 15 checks `!token`)
3.  `scopeGroupIds` defaults to empty array at[

    src/consumer/HandleForwardRequest.ts26](https://github.com/nhsv-tradex/
4.  `getScopesByScopeGroups([])` returns public scopes only
5.  Pattern matching proceeds normally

The configuration defines `scopes.unAuthenticated = "PUBLIC"` at[](https://

[src/conf.ts97](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t
```

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts13-27](https://github.com/nhsv-tradex/

src/conf.ts97](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

### Scope Loading from Kafka

#### Configuration Topic Integration

Scopes and scope groups are loaded from the Kafka `configuration` topic rat

-   Dynamic permission updates without service restart
-   Centralized permission management across microservices
-   Consistency across the distributed system

The loading configuration at[](https://github.com/nhsv-tradex/aaa/blob/5996

[src/conf.ts98-105](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

-   **Topic**: `configuration`
-   **Scope URI**: `/api/v1/admin/scope`
-   **Scope Group URI**: `/api/v1/admin/scopeGroup`
-   **Page Size**: 100 records per request

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts98-105](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

#### Caching Strategy

The system caches scope data with a 180-second (3-minute) TTL configured at

[src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

-   **Performance**: Avoids repeated database/Kafka lookups
-   **Consistency**: Updates propagate within 3 minutes
-   **Memory**: Reasonable cache size for scope metadata

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

### Integration with Message Forwarding
```

```
When a scope match is found, the `forwardMessageWithScope()` function (refe

[src/consumer/HandleForwardRequest.ts33](https://github.com/nhsv-tradex/aaa

1.  Receives the original message
2.  Receives the matched `Scope` object
3.  Receives the request headers with token
4.  Receives the transaction record for auditing
5.  Forwards the message to the destination service based on scope configur

The scope object contains the destination topic and other routing informati

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts33](https://github.com/nhsv-tradex/aaa

### Authorization Code Flow

#### HandleForwardRequest.ts Implementation

The complete authorization logic in[](https://github.com/nhsv-tradex/aaa/bl

[src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/
```

Function: handleMessage(msg: Kafka.IMessage) ├── Extract headers and token from msg.data ├── Create ITransaction record if token exists │ ├── user_id from token │ ├── client_id from token
│ ├── refresh_token_id from token │ ├── transaction_id from msg │ └── serialized message data ├── Extract scopeGroupIds from token (or [] if no token) ├── Call scopeService.getScopesByScopeGroups(sgIds) ├── Loop through returned scopes: │ ├── Call nanomatch(msg.uri, scope.uriPattern) │ └── If match found: │ ├── Log matched scope │ ├── Call forwardMessageWithScope() │ └── Return (exit loop) └── Return false if no matches (unauthorized)

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/

#### HandleAAARequest.ts Integration

The main message handler at[](https://github.com/nhsv-tradex/aaa/blob/59961

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa
```

Function: handleAAAMessage(msg: Kafka.IMessage) ├── Check msg.uri against AAA-specific patterns │ ├── Login endpoints → authenticate() │ ├── Token operations → refreshAccessToken() / revokeToken() │ ├── Scope search → scopeService.getScopesByScopeGroups() │ ├── Link account operations → linkAccountService methods │ ├── OTP/Biometric operations → service methods │ └── Other AAA operations → direct service calls └── Fall through to handleMessage() for scope-based forwarding

```
This architecture ensures AAA-internal operations bypass scope checking whi

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

### Special Scopes

#### VERIFY\_OTP Scope

The `VERIFY_OTP` scope at[](https://github.com/nhsv-tradex/aaa/blob/59961d6

[src/conf.ts96](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

-   Secondary authentication challenges
-   Transaction authorization
-   Sensitive operation confirmation

#### PUBLIC Scope

The `PUBLIC` scope at[](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr

[src/conf.ts97](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

-   Health check endpoints
-   Public information endpoints
-   Initial authentication endpoints

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts96-97](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

### Error Handling

The authorization system uses implicit error handling:

1.  **No matching scope**: Returns `false` from `handleMessage()` at[

    src/consumer/HandleForwardRequest.ts36](https://github.com/nhsv-tradex/
2.  **Missing token**: Proceeds with empty scope groups, allowing only PUBL
3.  **Invalid token**: Token validation occurs before reaching scope checki

The boolean return value (`false`) indicates authorization failure, allowin
```

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts36](https://github.com/nhsv-tradex/aaa

---

## User and Client Management

Relevant source files

-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/constants/Constants.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/constants/UpdateObjectKeys.ts](https://github.com/nhsv-tradex/aaa/b
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl

### Purpose and Scope

This page documents the user profile management and OAuth2 client managemen

For authentication methods and login flows, see [Authentication System](#au

* * *

### OAuth2 Client Management

The AAA service implements OAuth2 client management to validate and control

#### Client Configuration and Loading

Clients are loaded from a centralized configuration service via Kafka messa

**Configuration Structure:**

 | Configuration Key | Value | Description |
|--- | --- | --- |
 | `clients.loadFrom.topic` | `"configuration"` | Kafka topic for client da
 | `clients.loadFrom.uris.client` | `"/api/v1/system/client"` | URI for cli
```

```
| `clients.loadFrom.uris.pageSize` | `100` | Page size for batch loading |
| `enableForcePartnerClientId` | `true` | Enforces partner client validati

The client loading process supports pagination for efficient data retrieval

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts107-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/conf.ts140](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

#### Client Validation Flow

![Chart 1](aaa-images/aaa-section13-1.svg)

**Diagram: Client Validation and Lifecycle Management**

The diagram illustrates how clients are loaded from the configuration servi

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts107-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/consumer/HandleAAARequest.ts52-53](https://github.com/nhsv-tradex/aaa/b

src/consumer/HandleAAARequest.ts122-123](https://github.com/nhsv-tradex/aaa

#### Client Secret Management

Client secrets authenticate OAuth2 clients to prevent unauthorized applicat

**Endpoint:** `PUT /api/v1/client/{id}/changeSecret`

**Handler:** `changeClientSecret(msg.data)`

**Request Flow:**

![Chart 2](aaa-images/aaa-section13-2.svg)

**Diagram: Client Secret Change Flow**

When a client secret change is requested, the system validates the request,

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume
```

```
[src/consumer/HandleAAARequest.ts52-53](https://github.com/nhsv-tradex/aaa/

src/constants/UpdateObjectKeys.ts1-3](https://github.com/nhsv-tradex/aaa/bl

#### Mobile Application Version Validation

The AAA service enforces minimum mobile application versions to ensure clie

**Configuration:**
```

checkMobileAppVersion: {
  android: {
    version: "1.0.17",
    url: "https://play.google.com/store/apps/details?id=com.tradex.vcsc"
  },
  ios: {
    version: "1.0.17",
    url: "https://apps.apple.com/us/app/v-mobile-new/id1492065191"
  }
}

```
**Endpoint:** `POST /api/v1/client/updateAppVersion`

**Handler:** `updateAppVersion(msg.data)`

 | Platform | Minimum Version | Update URL |
|--- | --- | --- |
 | Android | 1.0.17 | Google Play Store |
 | iOS | 1.0.17 | Apple App Store |

The configuration can be set to `null` to disable version checking[](https:

[src/conf.ts142](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts117-126](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/conf.ts142](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/consumer/HandleAAARequest.ts122-123](https://github.com/nhsv-tradex/aaa

* * *

### User Profile Management

The AAA service maintains user profiles and provides operations for profile

#### Profile Update Operations

Users can update their profile information through the `updateProfile` endp

**Endpoint:** `POST /api/v1/updateProfile`

**Handler:** `updateProfile(msg.data)`

![Chart 3](aaa-images/aaa-section13-3.svg)

**Diagram: User Profile Update Flow**

Profile updates are routed through the Kafka message handler to the `UserSe

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts114-115](https://github.com/nhsv-tradex/aa
```

```
#### Mobile OTP Device Management

Users can register and unregister mobile devices for receiving one-time pas

**Mobile OTP Registration:**

**Endpoint:** `POST /api/v1/registerMobileOtp`

**Handler:** `registerMobileOtp(msg.data)`

This operation associates a mobile device with the user's account, enabling

**Mobile OTP Unregistration:**

**Endpoint:** `POST /api/v1/unregisterMobileOtp`

**Handler:** `unregisterMobileOtp(msg.data)`

This operation removes a mobile device from the user's registered OTP devic

![Chart 4](aaa-images/aaa-section13-4.svg)

**Diagram: Mobile OTP Device Management**

The diagram shows the complete mobile OTP device lifecycle. Registration st

For details on OTP notification services and rate limiting, see [OTP Authen

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts90-91](https://github.com/nhsv-tradex/aaa/

src/consumer/HandleAAARequest.ts100-101](https://github.com/nhsv-tradex/aaa

src/consumer/HandleAAARequest.ts104-109](https://github.com/nhsv-tradex/aaa

* * *

### User Account Data Management

#### Account Number Validation

The system includes validation for account numbers to ensure required field
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/Constants.ts2](https://github.com/nhsv-tradex/aaa/blob/59961

#### User Data Model

User data is persisted in the MySQL `tradex-aaa` database. The Users table

-   User credentials (username, encrypted password)
-   Profile information (name, email, phone, avatar)
-   Account status and verification flags
-   Device registration data for mobile OTP
-   Biometric registration data
-   Organization and partner associations

The default avatar URL is configured at `https://s3-ap-southeast-1.amazonaw

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts25](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts47-55](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

* * *

### Integration with Other Subsystems

#### Client Validation in Authentication Flow

During authentication, the system validates the requesting client before pr

When enabled, authentication requests must include a valid `client_id` that

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts140](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.

#### Update Event Publishing

When client data changes (secret rotation, version updates), the system pub

**Update Event Structure:**
```

```
{
  objectType: UPDATE_OBJECT_KEYS.CLIENT,
  operation: "UPDATE",
  data: {
    clientId: "...",
    // Updated fields
  }
}
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/UpdateObjectKeys.ts1-3](https://github.com/nhsv-tradex/aaa/b

* * *

### Security Considerations

#### Client Secret Storage

Client secrets are stored securely in the database and never exposed in log

#### Profile Update Authorization

Profile update operations require valid authentication tokens with appropri

For scope validation details, see [Authorization and Scope Management](#aut

#### Mobile App Version Enforcement

Version checking prevents security vulnerabilities by blocking outdated mob

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts117-126](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

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## Data Layer Architecture

### Overview

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/db/async.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/d
-   [

    src/db/connection.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/
```

```
This page documents the data layer architecture of the AAA service, includi

For information about API request and response models, see [Request and Res

#### Data Layer Architecture Overview

The AAA service implements a comprehensive data layer with multiple abstrac

##### Architecture Components

![Chart 1](aaa-images/aaa-section14-1.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/BaseModel.ts1-331](https://github.com/nhsv-tradex/aaa/blob/5

src/db/async.ts1-260](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

src/db/index.ts1-173](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

src/dao/TransactionDao.ts1-14](https://github.com/nhsv-tradex/aaa/blob/5996

#### ORM-like Abstraction Layer

##### BaseModel and FieldModel Pattern

The system uses a custom ORM-like abstraction built around `BaseModel<T>` a

![Chart 2](aaa-images/aaa-section14-2.svg)

##### Query Builder System

The `Query<T>` class provides a fluent interface for building SQL queries w
```

// Example usage pattern from codebase
const transaction = new Transaction(data);
const query = new Query(transaction);
const insertSql = query.insert();
const results = await connection.queryResult(insertSql, [data]);

```
**Key Query Builder Features:**

-   **Type-safe field selection**: `query.selectFields(model => [model.fiel
-   **Conditional queries**: `query.where(model => model.id, "= 123")`
-   **Join support**: `query.join(otherModel, fromField, toField)`
-   **Bulk operations**: `insertByFields()`, `insertBulkIterator()`
-   **Update/Insert**: `insertUpdateOnDuplicate()` for upsert operations

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/BaseModel.ts41-202](https://github.com/nhsv-tradex/aaa/blob/

src/dao/TransactionDao.ts5-13](https://github.com/nhsv-tradex/aaa/blob/5996

##### Condition System

The query builder supports complex conditional logic through the `ICondition

 | Condition Class | Purpose | Usage |
|--- | --- | --- |
 | `FieldCondition` | Single field condition | `WHERE field = value` |
 | `AndCondition` | Multiple AND conditions | `WHERE (cond1 AND cond2)` |
 | `OrCondition` | Multiple OR conditions | `WHERE (cond1 OR cond2)` |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/BaseModel.ts275-309](https://github.com/nhsv-tradex/aaa/blob

#### Database Connection Management

##### Connection Pool Architecture

The system provides connection pooling through the `mysql` package with cus

![Chart 3](aaa-images/aaa-section14-3.svg)

##### Connection Configuration

The connection pool supports custom type casting for MySQL-specific data ty
```

```
// BIT(1) fields are automatically cast to boolean
typeCast: (field, useDefaultTypeCasting) => {
  if ((field.type === "BIT") && (field.length === 1)) {
```

```
    const bytes = field.buffer();
    return (bytes[0] === 1);
  }
  return (useDefaultTypeCasting());
}
```

```
Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connecti

[src/db/connection.ts6-36](https://github.com/nhsv-tradex/aaa/blob/59961d60

src/db/async.ts195-207](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr

src/db/index.ts65-80](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/

##### Transaction Management

Both connection APIs support transaction management with automatic rollback

**Promise-based Transactions:**
```

```
export async function doJobInTransaction(
  func: (connection: Connection) => Promise,
  msgId?: string
): Promise
```

```
**Observable-based Transactions:**
```

```
function doJobInTransaction(
  subj: Subject,
  func: (connection: Connection) => Observable
): Observable
```

```
The transaction system provides:

-   Automatic connection acquisition from pool
-   Transaction begin/commit/rollback lifecycle
-   Error handling with automatic rollback
-   Connection release on completion
-   Transaction count tracking for debugging

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts

[src/db/async.ts225-249](https://github.com/nhsv-tradex/aaa/blob/59961d60/s

src/db/index.ts108-153](https://github.com/nhsv-tradex/aaa/blob/59961d60/sr

#### Database Query APIs

##### Dual API Architecture

The system provides both Observable-based and Promise-based APIs for databa

 | API Type | Module | Use Case | Key Methods |
|--- | --- | --- | --- |
 | Promise-based | `src/db/async.ts` | Modern async/await patterns | `conne
 | Observable-based | `src/db/index.ts` | Reactive programming patterns | `

##### Promise-based API Example
```

// From TransactionDao implementation
export async function createTransaction(data: ITransaction, connection?: Connection): Promise {
  return connectAndDo(async (con: Connection) => {
    const transaction: Transaction = new Transaction(data);
    const query: Query = new Query(transaction);
    const results: any = await con.queryResult(query.insert(), [data]);
    transaction.id.set(results.insertId);
    return transaction;
  }, connection);
}

```
##### Bulk Operation Support

The async Connection class provides specialized bulk operation methods:

-   `insertBulk<T>()`: Batch insert with configurable batch size
-   `insertBulkIterator<T>()`: Iterator-based bulk insert for memory effici
-   `insertBulkTransform<T>()`: Transform-based bulk insert with state mana

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/async.ts

[src/db/async.ts105-162](https://github.com/nhsv-tradex/aaa/blob/59961d60/s

src/dao/TransactionDao.ts5-13](https://github.com/nhsv-tradex/aaa/blob/5996

#### Data Access Patterns

##### DAO Pattern Implementation

The system uses Data Access Object (DAO) pattern for encapsulating database

![Chart 4](aaa-images/aaa-section14-4.svg)

##### DAO Implementation Example

The `TransactionDao` demonstrates the standard DAO pattern:
```

```
export async function createTransaction(data: ITransaction, connection?: Connection): Promise {
  return connectAndDo(async (con: Connection) => {
    const transaction: Transaction = new Transaction(data);
    const query: Query = new Query(transaction);
    const results: any = await con.queryResult(query.insert(), [data]);
    transaction.id.set(results.insertId);
    return transaction;
  }, connection);
}
```

```
**DAO Pattern Benefits:**

-   **Connection reuse**: Optional connection parameter for transaction con
-   **Type safety**: Generic model types ensure compile-time safety
-   **Error handling**: Automatic connection cleanup on errors
-   **Transaction support**: Can participate in larger transactions

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/dao/Transac

[src/dao/TransactionDao.ts1-14](https://github.com/nhsv-tradex/aaa/blob/599

#### Core Data Models

##### Model Inheritance Hierarchy

![Chart 5](aaa-images/aaa-section14-5.svg)

##### Key Model Categories

 | Category | Models | Purpose |
|--- | --- | --- |
 | **Authorization** | `Scope`, `ScopeGroup` | Permission and access contro
 | **Audit** | `Transaction` | Transaction logging and audit trail |
 | **Authentication** | `Biometric`, `Client` | Authentication credential s
 | **Integration** | `Partner`, `LinkAccount`, `LinkAccountDraft`, `LinkAcc
 | **Testing** | `AccountDemo` | Demo and testing accounts |

##### Scope Model

The `Scope` model defines access permissions and request forwarding rules w

 | Field | Type | Purpose |
|--- | --- | --- |
 | `name` | `string` | Scope identifier name |
 | `uriPattern` | `string` | URI pattern for matching requests |
 | `forwardType` | `string` | Type of forwarding (kafka, http, etc.) |
 | `forwardDataJson` | `string` | JSON configuration for forwarding |

**Key Methods:**

-   `getForwardData()`: Parses JSON configuration with caching
-   `getScopeResponse()`: Converts to API response format

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/S
```

```
[src/models/db/Scope.ts1-61](https://github.com/nhsv-tradex/aaa/blob/59961d

##### Transaction Model

The `Transaction` model provides comprehensive audit logging for all system
```

```
interface ITransaction {
  user_id: number;
  scope_id: number;
  service_user_id: number;
  client_id: number;
  refresh_token_id: number;
  transaction_id: string;
  data: string;
  to_topic: string;
  to_uri: string;
}
```

```
Each transaction record captures the complete context for audit and debuggi

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/T

[src/models/db/Transaction.ts4-46](https://github.com/nhsv-tradex/aaa/blob/

#### Core Database Models

##### Client Model

The `Client` class represents OAuth2 client applications that can authentic

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | `number` | `id` | Primary key |
 | `userId` | `number` | `user_id` | Associated user identifier |
 | `clientId` | `string` | `client_id` | OAuth2 client identifier |
 | `clientSecret` | `string` | `client_secret` | OAuth2 client secret |
 | `desciption` | `string` | `desciption` | Client description |
 | `status` | `number` | `status` | Client status flag |
 | `appVersion` | `string` | `app_version` | Application version |

The `Client` class extends `DefaultBaseModel<Client>` and maps to the `t_cl

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/C

[src/models/db/Client.ts4-25](https://github.com/nhsv-tradex/aaa/blob/59961

##### Biometric Model

The `Biometric` class stores biometric authentication data including RSA pu

Key fields include:

-   `publicKey`: RSA public key for signature verification
-   `biometricType`: Type of biometric authentication (fingerprint, face, e
-   `deviceId`: Unique device identifier for device binding
-   `otpIndex` and `otpValue`: OTP integration for two-factor authenticatio
-   `isDeleted` and `deleteReason`: Soft deletion support

The model tracks device metadata (`platform`, `osVersion`, `sourceIp`) for

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B
```

```
[src/models/db/Biometric.ts4-40](https://github.com/nhsv-tradex/aaa/blob/59

##### Partner Integration Models

###### Partner Model

The `Partner` class defines external partner systems for account linking an

 | Field | Type | Purpose |
|--- | --- | --- |
 | `targetPartnerId` | `string` | External partner system identifier |
 | `name` | `string` | Partner display name |
 | `loginUrl` | `string` | Partner authentication endpoint |
 | `loginClientId` | `string` | OAuth credentials for partner |
 | `confirmLinkUrl` | `string` | Account linking confirmation endpoint |
 | `unlinkUrl` | `string` | Account unlinking endpoint |
 | `initLinkUrl` | `string` | Account linking initiation endpoint |
 | `integrationType` | `string` | Integration method type |
 | `privateKey` | `string` | RSA private key for secure communication |
 | `publicKey` | `string` | RSA public key for verification |
 | `notyOtpUrl` | `string` | OTP notification endpoint |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/P

[src/models/db/Partner.ts4-35](https://github.com/nhsv-tradex/aaa/blob/5996

###### LinkAccount Model

The `LinkAccount` class manages active account linking relationships betwee

Critical fields:

-   `partnerId`: References the `Partner.id` for the external system
-   `partnerUserId` and `partnerUsername`: User identification in partner s
-   `userId` and `username`: User identification in AAA system
-   `joinLeaderboard`: Permission flag for leaderboard participation
-   `subAccount`: Sub-account identifier for multi-account support
-   `initRequest` and `confirmRequest`: Stores linking transaction data
-   `infoAccessGranted`: Permission flag for information sharing

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccount.ts4-32](https://github.com/nhsv-tradex/aaa/blob/

###### LinkAccountHistory Model
```

```
The `LinkAccountHistory` class maintains audit records of deleted account l

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccountHistory.ts4-27](https://github.com/nhsv-tradex/aa

###### LinkAccountDraft Model

The `LinkAccountDraft` class stores temporary account linking requests duri

-   `authCode`: Temporary authorization code for linking verification
-   `expirationAt`: Code expiration timestamp
-   `request`: Serialized linking request data

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccountDraft.ts4-27](https://github.com/nhsv-tradex/aaa/

##### Demo Account Model

The `AccountDemo` class provides pre-configured demonstration accounts for

 | Field | Type | Purpose |
|--- | --- | --- |
 | `username` | `string` | Demo account username |
 | `password` | `string` | Demo account password |
 | `realUsername` | `string` | Actual backend username |
 | `realPassword` | `string` | Actual backend password |
 | `domain` | `string` | Account domain/environment |
 | `loginResponse` | `string` | Cached login response data |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/A

[src/models/db/AccountDemo.ts4-29](https://github.com/nhsv-tradex/aaa/blob/

#### Schema Conventions and Design Patterns

##### Table and Field Naming Conventions

The database follows consistent naming patterns:

 | Convention | Pattern | Examples |
|--- | --- | --- |
 | **Table Names** | `t_` prefix with underscores | `t_client`, `t_biometri
```

```
 | **Primary Keys** | `id` (auto-increment integer) | Most tables use `id`
 | **Foreign Keys** | `{table}_id` format | `client_id`, `partner_id`, `use
 | **Timestamps** | `created_at`, `updated_at` | Standardized across all `D
 | **Soft Delete** | `is_deleted`, `deleted_at` | `t_biometric`, `t_link_ac

##### Field Type Mapping

The ORM layer handles MySQL-to-TypeScript type mapping:

 | MySQL Type | TypeScript Type | Special Handling |
|--- | --- | --- |
 | `BIT(1)` | `boolean` | Automatic byte-to-boolean conversion |
 | `DATETIME` | `Date` | Automatic parsing in `FieldModel` |
 | `JSON` | `string` | Stored as string, parsed in model methods |
 | `TEXT` | `string` | Standard string mapping |
 | `INT` | `number` | Standard numeric mapping |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/db/connecti

[src/db/connection.ts18-24](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/models/db/BaseModel.ts312-320](https://github.com/nhsv-tradex/aaa/blob/

#### Schema Conventions and Patterns

##### Field Mapping Pattern

All models use the `FieldModel<T>` pattern for type-safe database field acc
```

public fieldName: FieldModel = this.field("database_column", this.getRow);

```
##### Table Naming Convention

Database tables follow the `t_` prefix convention:

-   `t_client` - OAuth clients
-   `t_biometric` - Biometric authentication data
-   `t_partner` - External partner definitions
-   `t_link_account` - Active account links
-   `t_link_account_history` - Deleted account links
-   `t_link_account_draft` - Temporary account links
-   `t_account_demo` - Demo accounts

##### Timestamp Management

Models extending `DefaultBaseModel` automatically include:

-   `created_at DATETIME` - Record creation timestamp
-   `updated_at DATETIME` - Last modification timestamp

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/C

[src/models/db/Client.ts20-22](https://github.com/nhsv-tradex/aaa/blob/5996

src/models/db/Biometric.ts35-37](https://github.com/nhsv-tradex/aaa/blob/59

src/models/db/AccountDemo.ts24-26](https://github.com/nhsv-tradex/aaa/blob/

#### Primary Key and Relationship Patterns

##### Primary Key Strategy

-   Most models use auto-incrementing integer primary keys (`id`)
-   `Partner` model uses string primary key for external system identifiers
-   Foreign key relationships use descriptive names (e.g., `client_id`, `pa

##### Soft Deletion Support

The `Biometric` model implements soft deletion with:

-   `is_deleted` boolean flag
-   `delete_reason` for audit trail

The `LinkAccountHistory` model uses `deleted_at` timestamp for historical a
```

```
Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/Biometric.ts17-18](https://github.com/nhsv-tradex/aaa/blob/5

src/models/db/LinkAccountHistory.ts13](https://github.com/nhsv-tradex/aaa/b

### Database Connection and Query Management

Relevant source files

-   [

    src/db/async.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/d
-   [

    src/db/connection.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/

This document provides a comprehensive reference for all database model cla

For API request/response data structures, see [Request and Response Models]

#### Model Architecture

All database models in the AAA service follow a consistent inheritance patt

![Chart 1](aaa-images/aaa-section15-1.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/BaseModel.ts](https://github.com/nhsv-tradex/aaa/blob/59961d

src/models/db/FieldModel.ts](https://github.com/nhsv-tradex/aaa/blob/59961d

#### Core Entity Models

##### Client Model

The `Client` model represents OAuth2 client applications that can authentic

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
 | `userId` | number | `user_id` | Associated user ID |
 | `clientId` | string | `client_id` | OAuth2 client identifier |
 | `clientSecret` | string | `client_secret` | OAuth2 client secret |
```

```
 | `desciption` | string | `desciption` | Client description |
 | `status` | number | `status` | Client status code |
 | `appVersion` | string | `app_version` | Application version |

**Table:** `t_client`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/C

[src/models/db/Client.ts1-26](https://github.com/nhsv-tradex/aaa/blob/59961

##### Partner Model

The `Partner` model defines external partner systems that can integrate wit

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | string | `id` | Primary key |
 | `targetPartnerId` | string | `partner_id` | External partner identifier
 | `name` | string | `name` | Partner display name |
 | `loginUrl` | string | `login_url` | Partner login endpoint |
 | `loginClientId` | string | `login_client_id` | Partner OAuth client ID |
 | `loginClientSecret` | string | `login_client_secret` | Partner OAuth sec
 | `confirmLinkUrl` | string | `confirm_link_url` | Link confirmation endpo
 | `unlinkUrl` | string | `unlink_url` | Account unlink endpoint |
 | `initLinkUrl` | string | `init_link_url` | Link initialization endpoint
 | `integrationType` | string | `type` | Integration type classifier |
 | `privateKey` | string | `private_key` | RSA private key for signing |
 | `publicKey` | string | `public_key` | RSA public key for verification |
 | `iconUrl` | string | `icon_url` | Partner icon URL |
 | `notyOtpUrl` | string | `noty_otp_url` | OTP notification endpoint |

**Table:** `t_partner`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/P

[src/models/db/Partner.ts1-36](https://github.com/nhsv-tradex/aaa/blob/5996

##### Biometric Model

The `Biometric` model stores biometric authentication registrations includi

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
 | `userId` | number | `user_id` | Associated user ID |
```

```
 | `password` | string | `password` | Encrypted password hash |
 | `publicKey` | string | `public_key` | RSA public key for verification |
 | `username` | string | `username` | Username for biometric auth |
 | `grantType` | string | `grant_type` | OAuth2 grant type |
 | `clientId` | number | `client_id` | Associated client ID |
 | `platform` | string | `platform` | Device platform (iOS/Android) |
 | `osVersion` | string | `os_version` | Operating system version |
 | `appVersion` | string | `app_version` | Application version |
 | `deviceId` | string | `device_id` | Unique device identifier |
 | `sourceIp` | string | `source_ip` | Registration IP address |
 | `isDeleted` | boolean | `is_deleted` | Soft delete flag |
 | `deleteReason` | string | `delete_reason` | Deletion reason |
 | `biometricType` | string | `biometric_type` | Type of biometric (fingerp
 | `otpIndex` | number/string | `otp_index` | OTP sequence index |
 | `otpValue` | string | `otp_value` | Current OTP value |
 | `status` | string | `status` | Registration status |

**Table:** `t_biometric`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/B

[src/models/db/Biometric.ts1-41](https://github.com/nhsv-tradex/aaa/blob/59

##### AccountDemo Model

The `AccountDemo` model manages demo account credentials for testing and de

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
 | `username` | string | `username` | Demo username |
 | `password` | string | `password` | Demo password |
 | `realUsername` | string | `real_username` | Actual system username |
 | `realPassword` | string | `real_password` | Actual system password |
 | `domain` | string | `domain` | Account domain |
 | `description` | string | `description` | Demo account description |
 | `loginResponse` | string | `login_response` | Predefined login response

**Table:** `t_account_demo`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/A

[src/models/db/AccountDemo.ts1-30](https://github.com/nhsv-tradex/aaa/blob/

#### Account Linking Models
```

```
##### LinkAccount Model

The `LinkAccount` model represents established connections between AAA serv

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
 | `partnerId` | string | `partner_id` | Partner system identifier |
 | `partnerUserId` | number | `partner_user_id` | Partner's user ID |
 | `partnerUsername` | string | `partner_username` | Partner's username |
 | `userId` | number | `user_id` | AAA service user ID |
 | `username` | string | `username` | AAA service username |
 | `joinLeaderboard` | boolean | `join_leaderboard` | Leaderboard participa
 | `subAccount` | string | `sub_account` | Sub-account identifier |
 | `initRequest` | string | `init_request` | Initial link request data |
 | `confirmRequest` | string | `confirm_request` | Confirmation request dat
 | `infoAccessGranted` | boolean | `info_access_granted` | Information acce
 | `partnerFullname` | string | `partner_fullname` | Partner's full name |

**Table:** `t_link_account`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccount.ts1-33](https://github.com/nhsv-tradex/aaa/blob/

##### LinkAccountHistory Model

The `LinkAccountHistory` model maintains historical records of deleted acco

**Table:** `t_link_account_history`

Contains the same fields as `LinkAccount` plus a `deletedAt` timestamp fiel

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccountHistory.ts1-28](https://github.com/nhsv-tradex/aa

##### LinkAccountDraft Model

The `LinkAccountDraft` model stores temporary account linking requests duri

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
```

```
| `partnerId` | string | `partner_id` | Target partner system |
| `partnerUsername` | string | `partner_username` | Partner username |
| `partnerUserId` | number | `partner_user_id` | Partner user ID |
| `username` | string | `username` | AAA username |
| `authCode` | string | `auth_code` | Authorization code |
| `expirationAt` | Date | `expiration_at` | Draft expiration time |
| `request` | string | `request` | Original request data |

**Table:** `t_link_account_draft`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LinkAccountDraft.ts1-29](https://github.com/nhsv-tradex/aaa/

#### Authorization Mapping Models

##### LoginMethodScopeGroupMap Model

Maps login methods to their associated scope groups for authorization contr

| Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
| `loginMethodId` | number | `login_method_id` | Login method identifier |
| `groupId` | number | `scope_group_id` | Scope group identifier |

**Table:** `t_login_method_scope_group_map`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LoginMethodScopeGroupMap.ts1-22](https://github.com/nhsv-tra

##### LoginMethodStepScopeGroupMap Model

Maps individual login method steps to scope groups for fine-grained authori

| Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
| `stepId` | number | `step_id` | Login method step ID |
| `groupId` | number | `scope_group_id` | Scope group identifier |

**Table:** `t_login_method_step_scope_group_map`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LoginMethodStepScopeGroupMap.ts1-26](https://github.com/nhsv
```

```
##### ClientLoginMethodMap Model

Defines which login methods are available for each OAuth client.

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `clientId` | number | `client_id` | OAuth client identifier |
 | `loginMethodId` | number | `login_method_id` | Available login method |

**Table:** `t_client_login_method_map`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/C

[src/models/db/ClientLoginMethodMap.ts1-22](https://github.com/nhsv-tradex/

##### LoginMethodStep Model

Defines individual steps within multi-step login methods.

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | `id` | Primary key |
 | `loginMethodId` | number | `login_method_id` | Parent login method |
 | `step` | number | `step` | Step sequence number |
 | `name` | string | `name` | Step display name |
 | `desc` | string | `desc` | Step description |

**Table:** `t_login_method_step`

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/L

[src/models/db/LoginMethodStep.ts1-28](https://github.com/nhsv-tradex/aaa/b

#### Database Schema Relationships

![Chart 2](aaa-images/aaa-section15-2.svg)

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/db/C

[src/models/db/Client.ts20-22](https://github.com/nhsv-tradex/aaa/blob/5996

src/models/db/Partner.ts30-32](https://github.com/nhsv-tradex/aaa/blob/5996

src/models/db/Biometric.ts35-37](https://github.com/nhsv-tradex/aaa/blob/59
```

```
src/models/db/LinkAccount.ts27-29](https://github.com/nhsv-tradex/aaa/blob/

src/models/db/LoginMethodScopeGroupMap.ts16-18](https://github.com/nhsv-tra

src/models/db/ClientLoginMethodMap.ts16-18](https://github.com/nhsv-tradex/

src/models/db/LoginMethodStep.ts22-24](https://github.com/nhsv-tradex/aaa/b

src/models/db/LoginMethodStepScopeGroupMap.ts20-22](https://github.com/nhsv

src/models/db/AccountDemo.ts24-26](https://github.com/nhsv-tradex/aaa/blob/

#### Table Naming Conventions

All database tables follow the prefix convention `t_` followed by the entit

-   Core entities: `t_client`, `t_partner`, `t_biometric`, `t_account_demo`
-   Account linking: `t_link_account`, `t_link_account_history`, `t_link_ac
-   Authorization mappings: `t_login_method_scope_group_map`, `t_client_log

The models provide a `getTableName()` method that returns the corresponding

Sources: All model files provide table name constants via `getTableName()`

---

### Data Access Objects

Relevant source files

-   [

    src/dao/RefreshTokenDao.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/dao/TransactionDao.ts](https://github.com/nhsv-tradex/aaa/blob/5996
-   [

    src/db/async.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/d

This document covers the primary database models that represent the core bu

For information about database connection management and query execution pa
```

```
#### BaseModel Architecture

The data layer is built upon a sophisticated ORM-like abstraction centered

##### Core Model Hierarchy

![Chart 1](aaa-images/aaa-section16-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/BaseModel.ts4-33](https://github.com/nhsv-tradex/aaa/blob/59

src/models/db/BaseModel.ts311-320](https://github.com/nhsv-tradex/aaa/blob/

##### Query Builder System

The `BaseModel` includes an integrated query builder that generates SQL dyn

![Chart 2](aaa-images/aaa-section16-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/BaseModel.ts41-202](https://github.com/nhsv-tradex/aaa/blob/

src/models/db/BaseModel.ts275-309](https://github.com/nhsv-tradex/aaa/blob/

#### Account and Demo Management Models

##### AccountDemo Model

The `AccountDemo` model manages demonstration accounts used for testing and

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | id | Primary key |
 | `username` | string | username | Display username |
 | `password` | string | password | Display password |
 | `realUsername` | string | real\_username | Actual system username |
 | `realPassword` | string | real\_password | Actual system password |
 | `domain` | string | domain | Account domain |
 | `description` | string | description | Account description |
 | `loginResponse` | string | login\_response | Cached login response |

![Chart 3](aaa-images/aaa-section16-3.svg)
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/AccountDemo.ts4-26](https://github.com/nhsv-tradex/aaa/blob/

#### Link Account System Models

The link account system maintains relationships between local users and par

##### LinkAccount Model

The primary model for active account links between the AAA service and part

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `partnerId` | string | partner\_id | Partner system identifier |
 | `partnerUserId` | number | partner\_user\_id | User ID in partner system
 | `partnerUsername` | string | partner\_username | Username in partner sys
 | `userId` | number | user\_id | Local user ID |
 | `username` | string | username | Local username |
 | `joinLeaderboard` | boolean | join\_leaderboard | Leaderboard participat
 | `subAccount` | string | sub\_account | Sub-account identifier |
 | `initRequest` | string | init\_request | Initial linking request data |
 | `confirmRequest` | string | confirm\_request | Confirmation request data
 | `infoAccessGranted` | boolean | info\_access\_granted | Information acce
 | `partnerFullname` | string | partner\_fullname | Partner's full name |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/LinkAccount.ts4-29](https://github.com/nhsv-tradex/aaa/blob/

##### LinkAccountDraft Model

Manages pending account link requests before they are confirmed:

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `partnerId` | string | partner\_id | Target partner system |
 | `partnerUsername` | string | partner\_username | Partner username |
 | `partnerUserId` | number | partner\_user\_id | Partner user ID |
 | `username` | string | username | Local username |
 | `authCode` | string | auth\_code | Authorization code |
 | `expirationAt` | Date | expiration\_at | Code expiration time |
 | `request` | string | request | Request payload |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/
```

```
[src/models/db/LinkAccountDraft.ts4-25](https://github.com/nhsv-tradex/aaa/

##### LinkAccountHistory Model

Maintains historical records of deleted or modified account links:

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `partnerId` | string | partner\_id | Partner system identifier |
 | `partnerUserId` | number | partner\_user\_id | Partner user ID |
 | `userId` | number | user\_id | Local user ID |
 | `deletedAt` | Date | deleted\_at | Deletion timestamp |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/LinkAccountHistory.ts4-25](https://github.com/nhsv-tradex/aa

##### Link Account Relationships

![Chart 4](aaa-images/aaa-section16-4.svg)

#### Authorization Models

##### Scope Model

The `Scope` model defines authorization scopes that control access to speci

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | id | Primary key |
 | `name` | string | name | Scope name |
 | `uriPattern` | string | uri\_pattern | URI pattern for matching |
 | `forwardType` | string | forward\_type | Request forwarding type |
 | `forwardDataJson` | string | forward\_data | JSON forwarding configurati

The model includes specialized methods for handling forwarding configuratio

-   `hasForwardData()`: Checks if forwarding is configured
-   `getForwardData()`: Parses and returns forwarding configuration
-   `setForwardData()`: Sets forwarding configuration
-   `getScopeResponse()`: Returns API-formatted scope data

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/
```

```
[src/models/db/Scope.ts5-58](https://github.com/nhsv-tradex/aaa/blob/59961d

##### ScopeGroup Model

Organizes scopes into hierarchical groups for easier management:

 | Field | Type | Database Column | Purpose |
|--- | --- | --- | --- |
 | `id` | number | id | Primary key |
 | `parentId` | number | parent\_id | Parent group ID |
 | `name` | string | scope\_group\_name | Group name |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/ScopeGroup.ts4-19](https://github.com/nhsv-tradex/aaa/blob/5

##### Authorization Model Relationships

![Chart 5](aaa-images/aaa-section16-5.svg)

#### Model Instantiation and Usage Patterns

All models follow consistent instantiation patterns through the `BaseModel`
```

// Creating new model instances
const accountDemo = new AccountDemo();
const linkAccount = new LinkAccount();

// Creating from database row data
const accountDemo = new AccountDemo(rowData);
const linkAccount = new LinkAccount(rowData);

// Field access and manipulation
linkAccount.partnerId.set("partner_123");
linkAccount.userId.set(456);
const partnerId = linkAccount.partnerId.get();

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/models/

[src/models/db/BaseModel.ts7-11](https://github.com/nhsv-tradex/aaa/blob/59

src/models/db/AccountDemo.ts16-18](https://github.com/nhsv-tradex/aaa/blob/

src/models/db/LinkAccount.ts19-21](https://github.com/nhsv-tradex/aaa/blob/

---

## Infrastructure Services

### Overview

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/consumer/index.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60

This page documents the core infrastructure components that provide message

The infrastructure layer consists of:

-   **Message Processing**: Kafka-based asynchronous communication (detaile
-   **Caching**: Redis-based high-performance caching (detailed in [Redis C
-   **Configuration Management**: Centralized configuration with environmen
-   **Service Coordination**: Startup orchestration and dependency manageme

For database operations, see [Database Connection and Query Management](#da

#### System Startup and Initialization

The AAA service follows a coordinated startup sequence that initializes cri

![Chart 1](aaa-images/aaa-section17-1.svg)

The startup process uses a state tracking mechanism to ensure all component
```

```
Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1

[src/index.ts1-51](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/ind

#### Configuration Management System

The configuration system provides centralized management of application set

![Chart 2](aaa-images/aaa-section17-2.svg)

The configuration system supports multiple sources and processing stages:

 | Configuration Source | Purpose | Example |
|--- | --- | --- |
 | Environment Variables | Runtime configuration | `TRADEX_ENV_MYSQL_HOST`
 | Default Values | Fallback configuration | `accessToken.expiredInSeconds:
 | External Files | Environment-specific overrides | `env.js` script execut
 | Domain-specific | Multi-tenant configuration | JWT keys per domain |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L1-

[src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

#### Redis Caching Service Overview

The AAA service uses Redis as its primary caching layer to optimize perform

##### Cache Configuration

Redis is configured through the `conf.redis` object with the following defa

 | Setting | Default | Purpose |
|--- | --- | --- |
 | `port` | `6379` | Redis server port |
 | `host` | `127.0.0.1` | Redis server host |
 | `defaultLongRedisDuration` | `604800` (7 days) | Default TTL for cached
 | `cacheTimeout` | `180000` (3 minutes) | Cache timeout for scope data |

##### Primary Use Cases

The caching service is utilized for:

-   **Token Storage**: Access tokens, refresh tokens with configurable TTL
-   **Scope Data**: Permission and scope group information (180s cache time
```

```
-   **Session Data**: User session state and authentication context
-   **OTP Data**: One-time password verification state
-   **Rate Limiting**: OTP request rate limiting (5 requests per day)

The Redis service implementation provides type-safe serialization, hash ope

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L24

[src/conf.ts24-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

README.md30-36](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#

#### Kafka Message Processing Overview

The AAA service uses Apache Kafka for asynchronous message-based communicat

##### Consumer Infrastructure

The Kafka consumer infrastructure is initialized through the `init()` funct

[src/consumer/index.ts6-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

![Chart 3](aaa-images/aaa-section17-3.svg)

##### Key Topics and Configuration

The service consumes messages from the following Kafka topics:

 | Topic | Purpose | Configuration Source |
|--- | --- | --- |
 | `user` | Authentication requests and user operations | `conf.topic.user`
 | `configuration` | Scope and client configuration updates | `conf.scopes.
 | `aaa` (clusterId) | Service-specific messages | `conf.clusterId` |

##### Message Handler Integration

The `MessageHandler` routes incoming Kafka messages to the `handleAAAMessag

For detailed message routing, request handling, and topic subscription patt

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consumer/in

[src/consumer/index.ts1-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/conf.ts16-26](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf
```

```
src/conf.ts56-106](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

#### Client Configuration Synchronization

The `UpdateClientService` synchronizes OAuth2 client configurations from th

##### Configuration Loading Process

Client configurations are loaded from the `configuration` Kafka topic using

 | Configuration Parameter | Value | Purpose |
|--- | --- | --- |
 | `topic` | `"configuration"` | Kafka topic for configuration service |
 | `uris.client` | `"/api/v1/system/client"` | Endpoint for client data |
 | `uris.pageSize` | `100` | Pagination size for loading |

##### Transactional Update Flow

![Chart 4](aaa-images/aaa-section17-4.svg)

The update process maintains referential integrity by deleting existing rec

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts#L10

[src/conf.ts107-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

README.md217-226](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

#### Infrastructure Service Integration

The core infrastructure services integrate with the broader AAA system thro

##### Service Dependencies

![Chart 5](aaa-images/aaa-section17-5.svg)

##### Configuration Access Patterns

The configuration system provides centralized access to settings across all

 | Service Component | Configuration Usage | Key Settings |
|--- | --- | --- |
 | `RedisService` | `conf.redis.*` | Host, port, connection options |
 | `UpdateClientService` | `conf.clients.loadFrom.*` | Kafka topics, URIs,
```

```
 | Authentication Services | `conf.accessToken.*`, `conf.jwt.*` | Token TTL
 | Database Services | `conf.db.*` | Connection parameters, pool size |

Sources:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/index.ts#L1

[src/index.ts1-51](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/ind

src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/services/redis/RedisService.ts35-51](https://github.com/nhsv-tradex/aaa

src/services/UpdateClientService.ts108-112](https://github.com/nhsv-tradex/

### Redis Caching Service

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/db/async.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/d

#### Purpose and Scope

This document describes the Redis caching infrastructure used by the AAA se

For database connectivity and query execution, see [Database Connection and

**Sources**: README.md, conf.ts, Diagram 1, Diagram 3

* * *

#### Redis Configuration

The Redis connection is configured through the `conf.ts` configuration file

##### Configuration Structure

 | Configuration Key | Default Value | Description |
|--- | --- | --- |
```

```
 | `redis.host` | `127.0.0.1` | Redis server hostname or IP address |
 | `redis.port` | `6379` | Redis server port |
 | `redis.options` | `{}` | Additional Redis client options |
 | `defaultLongRedisDuration` | `604800` (7 days) | Default TTL for long-li
 | `cacheTimeout` | `180000` (3 minutes) | Short-lived cache timeout in mil

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

#### Cache Duration Strategies

The AAA service implements two distinct TTL strategies for different data t

![Chart 1](aaa-images/aaa-section18-1.svg)

##### TTL Configuration Summary

 | Cache Type | TTL Value | Configuration Key | Use Case |
|--- | --- | --- | --- |
 | Default Long | 7 days (604800s) | `defaultLongRedisDuration` | Refresh t
 | Short Cache | 3 minutes (180000ms) | `cacheTimeout` | Scopes, client val
 | Access Token | 15 minutes (900s) | `accessToken.expiredInSeconds` | JWT
 | Refresh Token | 1 day (86400s) | `refreshToken.expiredInSeconds` | Stand
 | Refresh + Remember | 30 days (2592000s) | `refreshToken.expiredInSeconds
 | OTP Token | 90 seconds | `otpToken.expiredInSeconds` | OTP verification

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

#### Cache-Aside Pattern Implementation
```

```
The AAA service implements a cache-aside pattern where the application code

![Chart 2](aaa-images/aaa-section18-2.svg)

##### Cache Access Flow Characteristics

1.  **Read-Through Pattern**: All reads check cache first before database
2.  **Lazy Loading**: Cache entries populated on cache miss
3.  **TTL-Based Expiration**: No explicit cache invalidation required for m
4.  **Write-Through for Tokens**: New tokens written to both database and c

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Cached Data Types

##### Token Caching

Refresh tokens are cached with the long TTL strategy (7 days) to minimize d

**Cache Key Pattern**: `refresh_token:{tokenValue}` **TTL**: `defaultLongRe

##### Scope Caching

Authorization scopes are cached with the short TTL strategy (3 minutes) to

**Cache Key Pattern**: `scope:{scopeId}` or `scope_group:{groupId}` **TTL**

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts95-106](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Session Data Caching

Session state and user authentication context are cached with the long TTL

**Cache Key Pattern**: `session:{sessionId}` **TTL**: `defaultLongRedisDura
```

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

#### Redis Configuration Management

The Redis configuration follows the same environment-based override pattern

![Chart 3](aaa-images/aaa-section18-3.svg)

##### Environment Variable Override

The Redis configuration can be overridden at runtime using environment vari
```

```
// Example env.js override
conf.redis.host = "redis.production.example.com";
conf.redis.port = 6380;
conf.redis.options = {
  password: "secret",
  db: 0
};
```

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c

src/conf.ts146-158](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Cache Invalidation Strategies

The AAA service primarily relies on TTL-based cache expiration rather than

##### TTL-Based Expiration

 | Strategy | Implementation | Rationale |
|--- | --- | --- |
 | **Passive Expiration** | Redis automatically removes entries after TTL e
 | **Differentiated TTL** | Long TTL (7 days) for tokens, short TTL (3 min)
 | **No Explicit Invalidation** | Cache entries expire naturally | Reduces

##### Implicit Invalidation Scenarios

1.  **Token Revocation**: Relies on TTL expiration; revoked tokens remain i
2.  **Scope Updates**: 3-minute TTL ensures permission changes propagate re
3.  **User Updates**: Session cache expires after 7 days, forcing re-authen

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

#### Integration with Authentication Flow

Redis caching is tightly integrated with the authentication and token manag

![Chart 4](aaa-images/aaa-section18-4.svg)

##### Performance Optimization Points

1.  **Client Validation Cache**: Reduces database queries for repeated clie
2.  **Scope Lookup Cache**: Accelerates authorization checks without databa
```

```
3.  **Token Storage Cache**: Enables fast token refresh without database re
4.  **Session State Cache**: Maintains user context across requests without

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

#### Cache Configuration Best Practices

##### Deployment Considerations

 | Environment | Recommended Configuration | Rationale |
|--- | --- | --- |
 | **Development** | `redis.host = "127.0.0.1"` | Local Redis instance for
 | **Staging** | `redis.host = "redis-staging.internal"` | Dedicated stagin
 | **Production** | `redis.host = "redis-cluster.internal"` |
 | Connection pooling enabled | High availability, connection pool for conc

##### TTL Tuning Guidelines

1.  **Long TTL (7 days)**: Appropriate for data that rarely changes (refres
2.  **Short TTL (3 minutes)**: Appropriate for data requiring quick updates
3.  **Token-Specific TTL**: Match cache TTL to token expiration for consist

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

#### Configuration Reference

##### Redis Configuration Object
```

```
The complete Redis configuration structure in `conf.ts`:
```

conf.redis = { port: 6379, host: '127.0.0.1', options: {} }

```
##### Related Timeout Configurations

 | Configuration Key | Default Value | Related Cache Strategy |
|--- | --- | --- |
 | `defaultLongRedisDuration` | 604800 seconds | Token and session caching
 | `cacheTimeout` | 180000 ms | Scope and client caching |
 | `accessToken.expiredInSeconds` | 900 seconds | Access token validation |
 | `refreshToken.expiredInSeconds` | 86400 seconds | Refresh token lifecycl
 | `otpToken.expiredInSeconds` | 90 seconds | OTP token validation |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts69-78](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

---

### Kafka Message Consumers

Relevant source files

-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
-   [

    src/consumer/HandleForwardRequest.ts](https://github.com/nhsv-tradex/aa
-   [

    src/consumer/index.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60

This document covers the Kafka consumer infrastructure within the AAA servi

For information about Redis caching operations, see page 9.1. For details a

#### Consumer Infrastructure Overview
```

```
The AAA service implements a Kafka consumer that processes authentication a

-   **`StreamHandler`**: Manages Kafka topic subscriptions and message cons
-   **`MessageHandler`**: Processes individual messages and delegates to ha

##### Consumer Architecture

**Diagram: Kafka Consumer Component Structure**

![Chart 1](aaa-images/aaa-section19-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts1-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/consumer/HandleAAARequest.ts1-131](https://github.com/nhsv-tradex/aaa/b

src/consumer/HandleForwardRequest.ts1-42](https://github.com/nhsv-tradex/aa

#### Consumer Initialization

The consumer is initialized in the `init` function which sets up the Kafka

##### Initialization Process

**Diagram: Consumer Initialization Flow**

![Chart 2](aaa-images/aaa-section19-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts6-11](https://github.com/nhsv-tradex/aaa/blob/59961d6

##### Topic Subscription

The consumer subscribes to Kafka topics using the cluster ID from configura

[src/conf.ts16](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

**Consumer Configuration Properties**

 | Property | Value | Source | Purpose |
|--- | --- | --- | --- |
 | `clusterId` | `"aaa"` | [ |
```

```
 | src/conf.ts16](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf
 | `clientId` | `${NODE_ID}.${INSTANCE_ID}` | [ |

src/conf.ts17](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts
 | Unique consumer identifier |
|--- |

src/conf.ts18](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts
 | Kafka broker addresses |
|--- |

src/conf.ts162-165](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/index.ts8-9](https://github.com/nhsv-tradex/aaa/blob/59961d60

src/conf.ts16-18](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts162-165](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

#### Message Routing Architecture

The consumer routes incoming messages through two levels of handlers based

##### Two-Tier Routing Strategy

**Diagram: Message Routing Decision Flow**

![Chart 3](aaa-images/aaa-section19-3.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/a

#### AAA Request Handler

The `handleAAAMessage` function in[](https://github.com/nhsv-tradex/aaa/blo

[src/consumer/HandleAAARequest.ts34-126](https://github.com/nhsv-tradex/aaa

##### Direct URI Routing
```

```
The handler uses a series of conditional statements to match exact URI patt

**URI Pattern Mapping Table**

 | URI Pattern | Service Method | Purpose |
|--- | --- | --- |
 | `post:/api/v1/login` | `authenticate()` | Standard password authenticati
 | `post:/api/v1/login/sec` | `authenticate()` | Security-enhanced login |
 | `post:/api/v1/login/domain` | `authenticate()` | Domain-based login |
 | `post:/api/v1/login/social` | `authenticate()` | Social OAuth login |
 | `post:/api/v1/loginCA` | `authenticate()` | Certificate authority login
 | `post:/api/v1/login/biometric` | `authenticate()` | Biometric authentica
 | `post:/api/v1/user/SocialLogin` | `authenticate()` | Social login altern
 | `post:/api/v1/login/organization` | `authenticate()` | Organization-base
 | `post:/api/v1/login/social/organization` | `authenticate()` | Social + o
 | `post:/api/v1/login/partnerCredential` | `loginPartnerCredential()` | Pa
 | `post:/api/v1/refreshToken` | `refreshAccessToken()` | Access token refr
 | `post:/api/v1/revokeToken` | `revokeToken()` | Token revocation |
 | `post:/api/v1/login/sec/verifyOTP` | `verifyOtp()` or `loginVerifyOtp()`

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts35-58](https://github.com/nhsv-tradex/aaa/

##### Link Account URIs

The handler includes extensive support for account linking operations acros

**Link Account Operations**

 | URI Pattern | Service Method | Purpose |
|--- | --- | --- |
 | `get:/api/v1/partners` | `findAllPartners()` | List available partner sy
 | `get:/api/v1/linkAccounts` | `findAllLinkAccount()` | Get user's linked
 | `post:/api/v1/linkAccounts` | `createLinkAccount()` | Create new account
 | `delete:/api/v1/linkAccounts/{partnerId}` | `deleteLinkAccount()` | Dele
 | `post:/api/v1/linkAccounts/login` | `loginLinkAccount()` | Login via lin
 | `post:/api/v1/linkAccounts/confirm` | `confirmLinkAccount()` | Confirm l
 | `post:/api/v1/linkAccounts/unlink` | `unlink()` | Unlink accounts |
 | `post:/api/v1/linkAccounts/init` | `initLinkAccount()` | Initialize link
 | `/api/v1/linkAccounts/notifyOtpPartner` | `notifyOtpPartner()` | Send OT
 | `/api/v1/linkAccounts/notifyOtpFromPartner` | `notifyOtpFromPartner()` |
 | `/api/v1/linkAccounts/changeUsername` | `changeUsername()` | Update link
 | `post:/api/v1/linkAccounts/approve` | `createLinkAccountApprove()` | App
 | `/api/v1/linkAccounts/getUserIdByPartnerName` | `getUserIdByPartnerName(
```

```
 | `/api/v1/linkAccounts/leaderboard/settings` | `putLeaderboardSetting()`

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts59-122](https://github.com/nhsv-tradex/aaa

##### OTP and Biometric URIs

**Diagram: OTP and Biometric Message Flow**

![Chart 4](aaa-images/aaa-section19-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts90-119](https://github.com/nhsv-tradex/aaa

##### Additional Service URIs

 | URI Pattern | Service Method | Purpose |
|--- | --- | --- |
 | `get:/api/v1/scopes/search` | In-line scope lookup | Search scopes by gr
 | `/api/v1/updateProfile` | `updateProfile()` | Update user profile |
 | `/api/v1/importDbData` | `importDbJsonService.jsonDbImport()` | Import d
 | `put:/api/v1/client/{id}/changeSecret` | `changeClientSecret()` | Update
 | `/api/v1/client/updateAppVersion` | `updateAppVersion()` | Update app ve

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts52-124](https://github.com/nhsv-tradex/aaa

##### Fallback to Forward Handler

If no direct URI pattern matches, the message is passed to `handleMessage()

[src/consumer/HandleAAARequest.ts125](https://github.com/nhsv-tradex/aaa/bl

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleAAARequest.ts125](https://github.com/nhsv-tradex/aaa/bl

#### Forward Request Handler

The `handleMessage` function in[](https://github.com/nhsv-tradex/aaa/blob/5

[src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/
```

```
##### Scope-Based Forwarding Flow

**Diagram: Forward Request Processing**

![Chart 5](aaa-images/aaa-section19-5.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts10-37](https://github.com/nhsv-tradex/

##### Token and Transaction Processing

The handler extracts authentication tokens from message headers and creates

**Transaction Object Structure**

The `ITransaction` object created in[](https://github.com/nhsv-tradex/aaa/b

[src/consumer/HandleForwardRequest.ts15-25](https://github.com/nhsv-tradex/

 | Field | Source | Purpose |
|--- | --- | --- |
 | `user_id` | `token.userId` | User identifier |
 | `scope_id` | `null` (set later) | Matched scope identifier |
 | `service_user_id` | `token.serviceUserId` | Service-specific user ID |
 | `client_id` | `token.clientId` | OAuth client ID |
 | `refresh_token_id` | `token.refreshTokenId` | Associated refresh token |
 | `transaction_id` | `msg.transactionId` | Message transaction ID |
 | `data` | `JSON.stringify(msg)` | Full message payload |
 | `to_topic` | `null` (set later) | Target topic for forwarding |
 | `to_uri` | `null` (set later) | Target URI for forwarding |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts15-25](https://github.com/nhsv-tradex/

##### URI Pattern Matching with nanomatch

The handler uses the `nanomatch` library to match incoming URIs against sco

**Pattern Matching Logic**

1.  Extract scope group IDs from token (or empty array for unauthenticated
```

```
    src/consumer/HandleForwardRequest.ts26](https://github.com/nhsv-tradex/
2.  Retrieve all scopes associated with those groups[

    src/consumer/HandleForwardRequest.ts27](https://github.com/nhsv-tradex/
3.  Iterate through scopes and test URI patterns[

    src/consumer/HandleForwardRequest.ts28-34](https://github.com/nhsv-trad
4.  Forward to first matching scope[

    src/consumer/HandleForwardRequest.ts33](https://github.com/nhsv-tradex/
5.  Return `false` if no patterns match[

    src/consumer/HandleForwardRequest.ts36](https://github.com/nhsv-tradex/

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/consume

[src/consumer/HandleForwardRequest.ts26-36](https://github.com/nhsv-tradex/

#### Configuration Update Flow

The configuration update process integrates both scope and client updates t

![Chart 6](aaa-images/aaa-section19-6.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/service

[src/services/UpdateClientService.ts107-113](https://github.com/nhsv-tradex

src/conf.ts107-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Configuration Topics and URIs

The system uses standardized Kafka topics and URI patterns for configuratio

-   **Topic**: `configuration`[

    src/conf.ts99](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

    src/conf.ts109](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
-   **Client URI**: `/api/v1/system/client`[

    src/conf.ts111](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
-   **Scope URI**: `/api/v1/admin/scope`[

    src/conf.ts101](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
```

```
-   **Scope Group URI**: `/api/v1/admin/scopeGroup`[

    src/conf.ts102](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts95-115](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

##### Error Handling and Retry Logic

Configuration updates include retry mechanisms through `Utils.asyncWithRetr

[src/services/UpdateClientService.ts17](https://github.com/nhsv-tradex/aaa/

src/conf.ts23](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/service

[src/services/UpdateClientService.ts15-18](https://github.com/nhsv-tradex/a

src/conf.ts23](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

---

## External Integrations

Relevant source files

-   [

    README.md](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md)
-   [

    src/conf.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.
-   [

    src/constants/OtpTypeUriMap.ts](https://github.com/nhsv-tradex/aaa/blob

### Purpose and Scope

This document describes the external systems that the AAA service integrate

For internal service components and business logic, see [Authentication Ser

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L
```

```
[README.md30-36](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md

README.md64-76](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#

* * *

### Integration Overview

The AAA service integrates with six categories of external systems:

 | Integration Type | System | Purpose | Configuration Location |
|--- | --- | --- | --- |
 | **Database** | MySQL 8.0+ | User data, tokens, clients persistence | [ |
 | src/conf.ts47-55](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/c
 | **Cache** | Redis 3.0+ | Token cache, scope cache, session cache | [ |

src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co
| **Message Broker** | Apache Kafka | Asynchronous microservice communicati

src/conf.ts18-23](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf
| **Service Discovery** | Apache ZooKeeper | Cluster coordination, service

README.md83](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L83
| **OAuth Providers** | Google, Facebook, Apple | Social authentication | [

src/conf.ts32-35](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf
| **Partner Services** | MAS, KIS | OTP verification routing | [

src/constants/OtpTypeUriMap.ts2-22](https://github.com/nhsv-tradex/aaa/blob

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md30-36](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md

src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

* * *

### MySQL Database Integration

#### Connection Configuration

The AAA service connects to MySQL 8.0+ for persistent data storage. Databas
```

```
![Chart 1](aaa-images/aaa-section20-1.svg)

**Database Configuration Parameters**

 | Parameter | Environment Variable | Default | Description |
|--- | --- | --- | --- |
 | `host` | `TRADEX_ENV_MYSQL_HOST` | Required | MySQL server hostname |
 | `port` | `TRADEX_ENV_MYSQL_PORT` | `3306` | MySQL server port |
 | `user` | `TRADEX_ENV_MYSQL_USER` | Required | Database username |
 | `password` | `TRADEX_ENV_MYSQL_PASSWORD` | Required | Database password
 | `database` | \- | `tradex-aaa` | Database name (hardcoded) |
 | `connectionLimit` | \- | `10` | Max connections in pool |
 | `timezone` | \- | `UTC` | Database timezone |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts47-55](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

README.md213-225](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

#### Database Schema

The MySQL database contains the following key tables:

 | Table | Purpose | Related DAO |
|--- | --- | --- |
 | `Users` | User accounts and profiles | UserDao |
 | `Clients` | OAuth2 client applications | ClientDao |
 | `LoginMethods` | Available authentication methods | LoginMethodDao |
 | `Scopes` | Permission definitions | ScopeDao |
 | `ScopeGroups` | Permission groupings | ScopeGroupDao |
 | `Tokens` | Access and refresh tokens | RefreshTokenDao |
 | `OTPs` | One-time password records | OtpDao |
 | `Biometrics` | Biometric authentication data | BiometricDao |
 | `LinkAccounts` | Multi-provider account links | LinkAccountDao |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md213-225](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.

* * *

### Redis Cache Integration

#### Cache Configuration
```

```
Redis provides high-performance caching for tokens, scopes, and session dat

![Chart 2](aaa-images/aaa-section20-2.svg)

**Redis Configuration Parameters**

 | Parameter | Default | Environment Override | Description |
|--- | --- | --- | --- |
 | `host` | `127.0.0.1` | `TRADEX_ENV_REDIS_HOST` | Redis server hostname |
 | `port` | `6379` | `TRADEX_ENV_REDIS_PORT` | Redis server port |
 | `options` | `{}` | \- | Additional connection options |

**Cache Duration Settings**

 | Setting | Value | Purpose |
|--- | --- | --- |
 | `defaultLongRedisDuration` | `604800` seconds (7 days) | Default TTL for
 | `cacheTimeout` | `180000` ms (3 minutes) | TTL for scope and configurati

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts130-134](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

#### Cache Usage Patterns

The Redis cache is used throughout the AAA service for:

1.  **Token Storage**: Refresh tokens are cached with 7-day TTL
2.  **Scope Caching**: Permission scopes are cached with 3-minute TTL
3.  **Session Data**: User session information cached with 7-day TTL
4.  **Rate Limiting**: OTP request counting uses Redis

For detailed cache implementation, see [Redis Caching Service](#redis-cachi

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md33](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L3

src/conf.ts24](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts
```

```
src/conf.ts127](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

### Apache Kafka Integration

#### Kafka Configuration

Kafka serves as the primary message broker for asynchronous microservice co

![Chart 3](aaa-images/aaa-section20-3.svg)

**Kafka Configuration Parameters**

 | Parameter | Environment Variable | Default | Description |
|--- | --- | --- | --- |
 | `kafkaUrls` | `TRADEX_ENV_KAFKA_URLS` | Required | Array of Kafka broker
 | `clusterId` | \- | `aaa` | Kafka cluster identifier |
 | `clientId` | `TRADEX_ENV_NODE_ID` |
 | `TRADEX_ENV_INSTANCE_ID` | Dynamic | Unique client ID: `{nodeId}.{instan
 | `retryTimes` | \- | `3` | Number of retry attempts for failed operations
 | `defaultKafkaTimeout` | \- | `10000` ms | Default timeout for Kafka oper

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts16-26](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts162-169](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

#### Topic Configuration

The AAA service subscribes to and publishes on the following Kafka topics:

 | Topic | Direction | Purpose | Message Handler |
|--- | --- | --- | --- |
 | `user` | Consumer | Authentication requests, token operations | HandleAA
 | `configuration` | Consumer | Scope and client configuration loading | Ha
 | `mas-rest-bridge` | Producer | Matrix card OTP verification (MAS) | Hand
 | `mas-ws-bridge` | Producer | Smart/Hardware/SMS OTP verification (MAS) |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts56-58](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts99-114](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con
```

```
src/constants/OtpTypeUriMap.ts8-11](https://github.com/nhsv-tradex/aaa/blob

#### Timeout Configuration

Service-specific Kafka timeouts are configured for various operations:

 | Operation | Timeout Key | Default | Description |
|--- | --- | --- | --- |
 | Facebook Login | `loginFacebook` | `15000` ms | Timeout for Facebook OAu
 | Google Login | `loginGoogle` | `15000` ms | Timeout for Google OAuth |
 | Apple Login | `loginApple` | `15000` ms | Timeout for Apple Sign-In |
 | TechX Login | `loginTechx` | `15000` ms | Timeout for TechX OAuth |
 | Password + OTP | `loginPasswordOtp` | `45000` ms | Extended timeout for
 | OTP Service | `otpService` | `10000` ms | Timeout for OTP operations (de
 | Direct Service Login | `loginDirectToService` | `10000` ms | Timeout for
 | Domain Login | `loginDomain` | `10000` ms | Timeout for domain-based log
 | Scope Loading | `loadScope` | `30000` ms | Timeout for loading scopes fr

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts31-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

src/conf.ts178-182](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/co

* * *

### Apache ZooKeeper Integration

#### Service Discovery

Apache ZooKeeper provides service discovery and cluster coordination for th

![Chart 4](aaa-images/aaa-section20-4.svg)

**Service Discovery Features**

 | Feature | Description |
|--- | --- |
 | **Broker Discovery** | ZooKeeper maintains the registry of available Kaf
 | **Cluster Coordination** | Coordinates distributed Kafka consumer groups
 | **Leader Election** | Manages partition leader election for Kafka topics
 | **Configuration Management** | Stores Kafka cluster configuration |

Each AAA service instance identifies itself with a unique `clientId` compos
```

```
**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md34](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L3

README.md83](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L83

src/conf.ts17](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

* * *

### External OAuth Provider Integration

#### OAuth Provider Configuration

The AAA service integrates with three major OAuth providers for social auth

![Chart 5](aaa-images/aaa-section20-5.svg)

**OAuth Provider Timeout Configuration**

 | Provider | Grant Type Constant | Timeout Configuration | Default Timeout
|--- | --- | --- | --- |
 | **Google** | `GrantType.ACCESS_GOOGLE` | `conf.timeouts.loginGoogle` | `
 | **Facebook** | `GrantType.ACCESS_FACEBOOK` | `conf.timeouts.loginFaceboo
 | **Apple** | `GrantType.ACCESS_APPLE` | `conf.timeouts.loginApple` | `150

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts32-35](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

README.md13](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L13

README.md206-208](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

#### Social Authentication Flow

The social authentication flow allows users to authenticate using their exi

1.  Client requests authentication with grant type `access_google`, `access
2.  AAA service forwards the OAuth token to the respective provider for val
3.  Provider validates the token and returns user information
4.  AAA service creates or links the user account
5.  AAA service generates JWT access and refresh tokens
6.  Tokens are returned to the client
```

```
For detailed authentication flows, see [Social and Organization Login](#soc

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md206-208](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.

src/conf.ts32-35](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

* * *

### Partner Service Integration

#### OTP Verification Routing

The AAA service integrates with partner services (MAS, KIS) for OTP verific

![Chart 6](aaa-images/aaa-section20-6.svg)

**MAS Partner OTP Configuration**

 | OTP Type | API Endpoint | Kafka Topic | Purpose |
|--- | --- | --- | --- |
 | `MATRIX_CARD` | `/api/v1/services/eqt/authCardMatrix` | `mas-rest-bridge
 | `SMART_OTP` | `/api/v2/bridge/verifyOtp` | `mas-ws-bridge` | Smart OTP v
 | `HARDWARE_OTP` | `/api/v2/bridge/verifyOtp` | `mas-ws-bridge` | Hardware
 | `SMS_OTP` | `/api/v2/bridge/verifySMSOTP` | `mas-ws-bridge` | SMS OTP ve

**KIS Partner OTP Configuration**

 | OTP Type | API Endpoint | Kafka Topic | Purpose |
|--- | --- | --- | --- |
 | `MATRIX_CARD` | `/api/v1/auth/matrix/verifyKisCard` | `mas-rest-bridge`
 | `SMART_OTP` | `/api/v1/otp/verify` | `mas-ws-bridge` | KIS smart OTP ver

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

src/conf.ts128](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

#### Partner Service Communication

The AAA service uses the `OTP_TYPE_URI_MAP` constant to route OTP verificat
```

```
1.  **Configuration Lookup**: The `conf.otp_type_uri_map` references the `O
2.  **OTP Type Mapping**: When an OTP verification is required, the service
3.  **Topic Selection**: The appropriate Kafka topic is selected (either `m
4.  **Message Publishing**: The verification request is published to the pa
5.  **Response Handling**: The partner service responds with verification r

This architecture allows the AAA service to support multiple partners with

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

src/conf.ts128](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

* * *

### Integration Architecture Summary

![Chart 7](aaa-images/aaa-section20-7.svg)

**Integration Summary Table**

 | Integration | Type | Protocol | Configuration | Timeout | Retry |
|--- | --- | --- | --- | --- | --- |
 | MySQL | Database | TCP | `conf.db.*` | Default connection timeout | Conn
 | Redis | Cache | TCP | `conf.redis.*` | Default connection timeout | Conn
 | Kafka | Message Broker | TCP | `conf.kafkaUrls` | `10000` ms default | `
 | ZooKeeper | Service Discovery | TCP | Managed by Kafka | N/A | N/A |
 | Google OAuth | Authentication | HTTPS | `conf.timeouts.loginGoogle` | `1
 | Facebook OAuth | Authentication | HTTPS | `conf.timeouts.loginFacebook`
 | Apple OAuth | Authentication | HTTPS | `conf.timeouts.loginApple` | `150
 | MAS/KIS Partners | OTP Verification | Kafka | `conf.otp_type_uri_map` |

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts1-184](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/con

README.md30-36](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#

README.md64-76](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#

* * *

### Environment Variable Reference
```

```
All external integrations are configured through environment variables. The

 | Environment Variable | Purpose | Required | Example |
|--- | --- | --- | --- |
 | `TRADEX_ENV_MYSQL_HOST` | MySQL server hostname | Yes | `mysql.example.c
 | `TRADEX_ENV_MYSQL_PORT` | MySQL server port | No (default: 3306) | `3306
 | `TRADEX_ENV_MYSQL_USER` | MySQL username | Yes | `aaa_user` |
 | `TRADEX_ENV_MYSQL_PASSWORD` | MySQL password | Yes | `secure_password` |
 | `TRADEX_ENV_REDIS_HOST` | Redis server hostname | No (default: 127.0.0.1
 | `TRADEX_ENV_REDIS_PORT` | Redis server port | No (default: 6379) | `6379
 | `TRADEX_ENV_KAFKA_URLS` | Kafka broker URLs (comma-separated) | Yes | `b
 | `TRADEX_ENV_NODE_ID` | Node identifier for distributed deployment | Yes
 | `TRADEX_ENV_INSTANCE_ID` | Instance identifier for horizontal scaling |
 | `TRADEX_ENV_DOMAIN` | Service domain for multi-tenant support | Yes | `t
 | `TRADEX_ENV_DOMAINS` | Additional domains (comma-separated) | No | `vcsc

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.ts

[src/conf.ts13](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf.t

src/conf.ts17-18](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

src/conf.ts48-51](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/conf

README.md162-172](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.m

* * *

### Integration Health Monitoring

The AAA service provides a health check endpoint to verify integration stat

 | Endpoint | Method | Purpose | Success Response |
|--- | --- | --- | --- |
 | `/currentTime` | GET | Health check and timestamp | Current server time

This endpoint can be used by orchestration systems (Kubernetes, Docker Swar

**Health Check Interval**: 30 seconds (configurable in deployment orchestra

**Sources**:[](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.md#L

[README.md228-234](https://github.com/nhsv-tradex/aaa/blob/59961d60/README.

---
```

```
## Error Handling

Relevant source files

-   [

    src/constants/errors.ts](https://github.com/nhsv-tradex/aaa/blob/59961d
-   [

    src/consumer/HandleAAARequest.ts](https://github.com/nhsv-tradex/aaa/bl
-   [

    src/errors/MultipleLoginMethodError.ts](https://github.com/nhsv-tradex/
-   [

    src/errors/NoGrantTypeError.ts](https://github.com/nhsv-tradex/aaa/blob

This document covers the error handling mechanisms, request validation patt

For information about authentication flows that may generate errors, see [A

### Error Hierarchy and Base Classes

The AAA service uses a standardized error handling approach built on the `t

#### Error Class Structure

![Chart 1](aaa-images/aaa-section21-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/UnAuthenticationError.ts1-10](https://github.com/nhsv-tradex/aa

src/errors/ServiceDownError.ts1-10](https://github.com/nhsv-tradex/aaa/blob

src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blob

src/errors/NoLoginMethodError.ts1-10](https://github.com/nhsv-tradex/aaa/bl

src/errors/TerminateError.ts1-6](https://github.com/nhsv-tradex/aaa/blob/59

### Specific Error Types

The service defines several specialized error classes for different failure
```

```
| Error Class | Error Code | Purpose | Constructor Parameters |
|--- | --- | --- | --- |
| `UnAuthenticationError` | `UN_AUTHENTICATION` | Authentication failures
| `ServiceDownError` | `SERVICE_DOWN` | Service unavailability | None |
| `NoGrantTypeError` | `NO_GRANT_TYPE` | Missing grant type in requests |
| `NoLoginMethodError` | `NO_LOGIN_METHOD` | Missing or invalid login meth
| `TerminateError` | N/A | Process termination with error context | `messa

#### UnAuthenticationError

Used when authentication attempts fail due to invalid credentials, expired
```

// Usage pattern
throw new UnAuthenticationError();

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/UnAuthenticationError.ts3-7](https://github.com/nhsv-tradex/aaa

#### ServiceDownError

Indicates that external services or dependencies are unavailable, used for

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/ServiceDownError.ts3-7](https://github.com/nhsv-tradex/aaa/blob

#### NoGrantTypeError

Thrown when OAuth2 grant type is missing or invalid in authentication reque

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoGrantTypeError.ts3-7](https://github.com/nhsv-tradex/aaa/blob

#### NoLoginMethodError

Indicates that a login method is missing or invalid in authentication reque

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoLoginMethodError.ts3-7](https://github.com/nhsv-tradex/aaa/bl

#### TerminateError

A specialized error class that wraps another error with additional context

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/TerminateError.ts1-6](https://github.com/nhsv-tradex/aaa/blob/5

### Request Validation Patterns

The AAA service implements comprehensive request validation using the `trad

#### Validation Flow Architecture

![Chart 2](aaa-images/aaa-section21-2.svg)
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blo

src/errors/NoLoginMethodError.ts1-10](https://github.com/nhsv-tradex/aaa/bl

#### Validation Layers

 | Validation Layer | Purpose | Error Types |
|--- | --- | --- |
 | Request Structure | Validates JSON structure and required fields | Stand
 | Grant Type Validation | Ensures valid OAuth2 grant type | `NoGrantTypeEr
 | Login Method Validation | Validates authentication method configuration
 | Credential Format | Validates credential format and requirements | Stand
 | Business Logic | Domain-specific validation rules | `UnAuthenticationErr

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoGrantTypeError.ts3-7](https://github.com/nhsv-tradex/aaa/blob

src/errors/NoLoginMethodError.ts3-7](https://github.com/nhsv-tradex/aaa/blo

src/errors/UnAuthenticationError.ts3-7](https://github.com/nhsv-tradex/aaa/

#### Validation Error Propagation

![Chart 3](aaa-images/aaa-section21-3.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blo

src/errors/NoLoginMethodError.ts1-10](https://github.com/nhsv-tradex/aaa/bl

### Error Handling in Authentication Flows

#### Authentication Error Flow

![Chart 4](aaa-images/aaa-section21-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/UnAuthenticationError.ts1-10](https://github.com/nhsv-tradex/aa

src/errors/ServiceDownError.ts1-10](https://github.com/nhsv-tradex/aaa/blob
```

```
src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blob

src/errors/NoLoginMethodError.ts1-10](https://github.com/nhsv-tradex/aaa/bl

### Error Propagation Through Message Systems

#### Kafka Message Error Handling

The service processes authentication requests through Kafka messaging, requ

![Chart 5](aaa-images/aaa-section21-5.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/ServiceDownError.ts1-10](https://github.com/nhsv-tradex/aaa/blo

src/errors/UnAuthenticationError.ts1-10](https://github.com/nhsv-tradex/aaa

### Error Handling Patterns by Component

#### Grant Type and Login Method Validation

Grant type and login method validation occur early in the authentication fl

![Chart 6](aaa-images/aaa-section21-6.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blo

src/errors/NoLoginMethodError.ts1-10](https://github.com/nhsv-tradex/aaa/bl

#### External Service Integration Errors

When integrating with external services for user data, notifications, or pa

 | Integration Point | Error Scenario | Error Type |
|--- | --- | --- |
 | Database connections | Connection timeout | `ServiceDownError` |
 | Redis cache | Cache unavailable | `ServiceDownError` |
 | Partner APIs | API endpoint down | `ServiceDownError` |
 | Notification services | Push service unavailable | `ServiceDownError` |
 | OTP delivery | SMS/Email service down | `ServiceDownError` |
```

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/ServiceDownError.ts1-10](https://github.com/nhsv-tradex/aaa/blo

### Error Response Structure

All errors follow a consistent response structure when returned through the
```

interface ErrorResponse {
  code: string;   // Error code (e.g., "UN_AUTHENTICATION")
  data: any;   // Additional error context
  message: string;   // Human-readable message
  stack: string;   // Stack trace for debugging
}

```
#### Error Code Standards

 | Error Code | HTTP Equivalent | Usage Context |
|--- | --- | --- |
 | `UN_AUTHENTICATION` | 401 Unauthorized | Invalid credentials, expired to
 | `SERVICE_DOWN` | 503 Service Unavailable | External service failures |
 | `NO_GRANT_TYPE` | 400 Bad Request | Missing OAuth2 grant type |
 | `NO_LOGIN_METHOD` | 400 Bad Request | Missing or invalid login method |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/UnAuthenticationError.ts5](https://github.com/nhsv-tradex/aaa/b

src/errors/ServiceDownError.ts5](https://github.com/nhsv-tradex/aaa/blob/59

src/errors/NoGrantTypeError.ts5](https://github.com/nhsv-tradex/aaa/blob/59

src/errors/NoLoginMethodError.ts5](https://github.com/nhsv-tradex/aaa/blob/

### Debugging and Troubleshooting

#### Error Identification

Each error class provides a unique error code that can be used for:

-   Client-side error handling logic
-   Monitoring and alerting systems
-   Log aggregation and analysis
-   Performance metrics and SLA tracking

#### Common Error Scenarios

 | Scenario | Root Cause | Error Type | Resolution |
|--- | --- | --- | --- |
 | Invalid client credentials | Misconfigured OAuth2 client | `UnAuthentica
 | Database connection issues | Network or DB server problems | `ServiceDow
 | Missing grant type in request | Malformed authentication request | `NoGr
 | External API timeouts | Third-party service degradation | `ServiceDownEr

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/errors/

[src/errors/UnAuthenticationError.ts1-10](https://github.com/nhsv-tradex/aa

src/errors/ServiceDownError.ts1-10](https://github.com/nhsv-tradex/aaa/blob
```

```
src/errors/NoGrantTypeError.ts1-10](https://github.com/nhsv-tradex/aaa/blob

---

## Development Setup

Relevant source files

-   [

    .babelrc](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc)
-   [

    .eslintrc.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/.eslin
-   [

    .gitignore](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore
-   [

    .nvmrc](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc)
-   [

    package-lock.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/pac
-   [

    package.json](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.

This page documents the development environment requirements, setup procedu

* * *

### Prerequisites

#### Node.js and npm

The AAA service requires **Node.js 16.13.0** as specified in[](https://gith

[.nvmrc1](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc#L1-L1)

 | Tool | Version | Purpose |
|--- | --- | --- |
 | Node.js | 16.13.0 | JavaScript runtime environment |
 | npm | 7.x+ | Package manager (bundled with Node.js 16) |
 | nvm | Latest | Node version manager (recommended) |
```

```
To install the correct Node.js version using nvm:
```

nvm install
nvm use

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc#L1-L

[.nvmrc1](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc#L1-L1)[

package.json1-66](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.

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#### External Dependencies

The AAA service requires access to an external key repository for RSA and J

 | Repository | Purpose | Default Path |
|--- | --- | --- |
 | `tradex-key` | RSA password encryption keys and JWT signing keys | `../t

The key repository must be cloned as a sibling directory to the `aaa` proje
```

parent-directory/ ├── aaa/ # This repository └── tradex-key/ # External key repository └── dev/ └── aaa/ ├── rsa-private.key ├── rsa-public.key └── jwt-*.key

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json10](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.j

.gitignore4](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore#L4

* * *

### Initial Setup

#### Clone Repository
```

git clone https://github.com/nhsv-tradex/aaa (https://github.com/nhsv-tradex/aaa)
cd aaa

```
#### Install Dependencies

Install production and development dependencies:
```

npm install

```
This installs all packages from[](https://github.com/nhsv-tradex/aaa/blob/5

[package.json18-41](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag

**Production Dependencies:**

-   `mysql@^2.16.0` - MySQL database driver
-   `redis@^3.0.2` - Redis cache client
-   `jsonwebtoken@^8.3.0` - JWT token generation/validation
-   `tradex-common` - Shared utilities from private GitHub repository
-   Additional dependencies for Kafka, ZooKeeper, OTP generation

**Development Dependencies:**

-   `typescript@^5.2.2` - TypeScript compiler
-   `@typescript-eslint/eslint-plugin@^6.7.2` - ESLint TypeScript support
-   `@typescript-eslint/parser@^6.7.2` - TypeScript parser for ESLint
-   `eslint@^8.50.0` - Code linting
-   `prettier@^3.0.3` - Code formatting
-   `@types/*` - TypeScript type definitions

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json18-53](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag

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### Configuration Files

#### TypeScript Configuration

TypeScript compilation is controlled by `tsconfig.json` (implied from build

-   Transpiles TypeScript to JavaScript
-   Outputs to `build/src/` directory
-   Preserves source directory structure

#### Babel Configuration

[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc#L1-L8)

[.babelrc1-8](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc#L1-
```

{
  "presets": ["es2015"],
  "plugins": ["transform-class-properties"]
}

```
 | Configuration | Purpose |
|--- | --- |
 | `es2015` preset | Transpile ES2015/ES6 syntax to ES5 |
 | `transform-class-properties` | Support for class property syntax |

#### ESLint Configuration

[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.eslintrc.json#L1-L20)

[.eslintrc.json1-20](https://github.com/nhsv-tradex/aaa/blob/59961d60/.esli
```

{
  "env": {
    "browser": true,
    "es2021": true
  },
  "extends": [
    "eslint:recommended",
    "plugin:@typescript-eslint/recommended"
  ],
  "parser": "@typescript-eslint/parser",
  "parserOptions": {
    "ecmaVersion": "latest",
    "sourceType": "module"
  }
}

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc#L1

[.babelrc1-8](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc#L1-

.eslintrc.json1-20](https://github.com/nhsv-tradex/aaa/blob/59961d60/.eslin

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### Build Process

The build process involves multiple stages to produce production-ready Java

![Chart 1](aaa-images/aaa-section22-1.svg)

#### Build Scripts

[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L7-L14)

[package.json7-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/package

 | Script | Command | Purpose |
|--- | --- | --- |
 | `clean` | `rm -rf build && mkdir build` | Remove existing build artifact
 | `build` | `npm run clean && tsc -p tsconfig.json` | Full production buil
 | `start` | `tsc -w -p tsconfig.json` | Watch mode compilation |
 | `copyKeys` | `mkdir -p build/src/keys && cp -R ../tradex-key/dev/aaa bui
 | `run-local` | `cp src/env.js build/src/env.js && npm run copyKeys && cd
 | `build-local` | `npm run build && npm run run-local` | Full local build

#### Build Workflow

![Chart 2](aaa-images/aaa-section22-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json7-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/package

* * *

### Development Workflow

#### Watch Mode Development

For active development with automatic recompilation on file changes:
```

npm start

```
This runs `tsc -w -p tsconfig.json`[](https://github.com/nhsv-tradex/aaa/bl

[package.json9](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.js

1.  Compiles TypeScript to JavaScript
2.  Watches for file changes
3.  Incrementally recompiles modified files
4.  Outputs to `build/src/`

In a separate terminal, manually copy keys and run:
```

npm run copyKeys
cp src/env.js build/src/env.js
cd build/src && node index.js

```
#### Full Local Build and Run

For a complete build and execution cycle:
```

npm run build-local

```
This script[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json

[package.json13](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.j

1.  Cleans build directory
2.  Compiles all TypeScript files
3.  Copies environment configuration
4.  Copies external RSA/JWT keys
5.  Launches the application

#### Directory Structure After Build

![Chart 3](aaa-images/aaa-section22-3.svg)

#### Ignored Files

[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore#L1-L10)

[.gitignore1-10](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignor

 | Pattern | Reason |
|--- | --- |
 | `aaa` | Compiled binary output |
 | `env.js` | Local environment configuration |
 | `node_modules/**` | npm dependencies |
 | `keys/**` | RSA/JWT keys (security) |
 | `.idea/**`, `.vscode/**` | IDE configuration |
 | `build/**` | Compiled build artifacts |
 | `*.iml` | IntelliJ module files |
 | `src/env.js` | Source environment config |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json7-14](https://github.com/nhsv-tradex/aaa/blob/59961d60/package

.gitignore1-10](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore

* * *

### Testing

#### Unit Tests

Run unit tests with:
```

```
npm run utest
```

```
This script[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json

[package.json14](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.j

1.  Verifies Node version with `node --version | grep v1[0-9]*`
2.  Compiles TypeScript: `tsc -p .`
3.  Executes tests: `mocha build/src/tests/*.test.js`

Test files are located in `src/tests/` and compiled to `build/src/tests/`.

#### Testing Workflow

![Chart 4](aaa-images/aaa-section22-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json14](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.j

* * *

### Environment Configuration

#### Local Environment File

During local development, create `src/env.js` to override default configura

-   Is excluded from version control[

    .gitignore9](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignor
-   Gets copied to `build/src/env.js` during local runs[

    package.json11](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag
-   Executes in a VM sandbox to set environment variables (see [Configurati

#### Environment Variables

The AAA service reads configuration from environment variables with the `TR

 | Variable | Purpose | Default |
|--- | --- | --- |
 | `TRADEX_DOMAIN` | Primary domain identifier | Set in code |
 | `TRADEX_ENV_DOMAINS` | Multi-domain configuration | Optional |
 | Database connection variables | MySQL host, port, user, password | From
 | Redis connection variables | Cache host, port | From conf.ts |
```

```
 | Kafka connection variables | Broker URLs, topics | From conf.ts |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore#

[.gitignore2-9](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore

package.json11](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.js

* * *

### Package Binary Generation

For creating standalone executables using `pkg`:

The[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.json#L56-L65

[package.json56-65](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag
```

{
  "pkg": {
    "assets": [],
    "options": ["experimental-modules"],
    "targets": ["node16-alpine-x64"],
    "output": "dist/entrypoint"
  }
}

```
This generates a single executable file for Alpine Linux (x64 architecture)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json56-65](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag

* * *

### Common Development Tasks

#### Adding New Dependencies
```

npm install --save

npm install --save-dev

```
#### Code Formatting
```

npx prettier --write "src/**/*.ts"

```
#### Linting
```

npx eslint "src/**/*.ts"

```
#### Clean Build

To remove all build artifacts and start fresh:
```

npm run clean
npm run build

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json42-53](https://github.com/nhsv-tradex/aaa/blob/59961d60/packag

* * *

### Troubleshooting

#### Missing Keys Error

**Problem:** Application fails to start with key loading errors.

**Solution:** Ensure the `tradex-key` repository is cloned and the `copyKey
```

cd .. && git clone tradex-key
cd aaa && npm run copyKeys

```
#### TypeScript Compilation Errors

**Problem:** TypeScript compilation fails with type errors.

**Solution:** Ensure all type definition packages are installed:
```

npm install

```
Check that `tsconfig.json` is properly configured for the project structure

#### Node Version Mismatch

**Problem:** Application behaves unexpectedly or dependencies fail to insta

**Solution:** Use the exact Node.js version specified:
```

nvm use 16.13.0

```
**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json10](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.j

.nvmrc1](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc#L1-L1)

* * *

### Development Environment Summary

![Chart 5](aaa-images/aaa-section22-5.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/package.jso

[package.json1-66](https://github.com/nhsv-tradex/aaa/blob/59961d60/package

.nvmrc1](https://github.com/nhsv-tradex/aaa/blob/59961d60/.nvmrc#L1-L1)[

.babelrc1-8](https://github.com/nhsv-tradex/aaa/blob/59961d60/.babelrc#L1-L

.eslintrc.json1-20](https://github.com/nhsv-tradex/aaa/blob/59961d60/.eslin

.gitignore1-10](https://github.com/nhsv-tradex/aaa/blob/59961d60/.gitignore

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## Constants and Type Definitions

Relevant source files

-   [

    global.d.ts](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.
-   [

    src/constants/Constants.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/constants/GrantType.ts](https://github.com/nhsv-tradex/aaa/blob/599
-   [

    src/constants/OtpTypeUriMap.ts](https://github.com/nhsv-tradex/aaa/blob
-   [
```

```
    src/constants/UpdateObjectKeys.ts](https://github.com/nhsv-tradex/aaa/b
-   [

    src/constants/biometricKeys.ts](https://github.com/nhsv-tradex/aaa/blob
-   [

    src/constants/errors.ts](https://github.com/nhsv-tradex/aaa/blob/59961d
-   [

    src/constants/index.ts](https://github.com/nhsv-tradex/aaa/blob/59961d6
-   [

    src/constants/messages.ts](https://github.com/nhsv-tradex/aaa/blob/5996

### Purpose and Scope

This page provides comprehensive reference documentation for all constants,

For information about how grant types are implemented and processed, see [G

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### Grant Type Enumeration

The `GrantType` enum defines all supported OAuth2 grant types and authentic

#### Grant Type Definitions

 | Grant Type | Value | Purpose |
|--- | --- | --- |
 | `PASSWORD` | `"password"` | Standard password-based authentication |
 | `DEMO` | `"demo"` | Demo/testing authentication |
 | `PASSWORD_TRADEX` | `"password_tradex"` | Tradex-specific password authe
 | `PASSWORD_OTP` | `"password_otp"` | Password + OTP two-factor authentica
 | `PASSWORD_FACEID` | `"password_faceid"` | Password + Face ID authenticat
 | `PASSWORD_CA` | `"password_ca"` | Password with certificate authority |
 | `ACCESS_GOOGLE` | `"access_google"` | Google OAuth social login |
 | `ACCESS_FACEBOOK` | `"access_facebook"` | Facebook OAuth social login |
 | `CLIENT_CREDENTIALS` | `"client_credentials"` | Service-to-service authe
 | `ACCESS_DOMAIN` | `"access_domain"` | Domain-based authentication |
 | `LOGIN_BIOMETRIC` | `"biometric"` | Biometric-only authentication |
 | `ACCESS_APPLE` | `"access_apple"` | Apple Sign-In social login |
 | `SOCIAL_LOGIN` | `"social_login"` | Generic social login |
 | `LINK_ACCOUNT` | `"link_account"` | Account linking between providers |
 | `KB_FINA` | `"kbfina"` | KB Fina partner-specific authentication |
```

```
 | `ORGANIZATION_LOGIN` | `"organization_login"` | Organization-based authe
 | `ORGANIZATION_SOCIAL_LOGIN` | `"organization_social_login"` | Organizati
 | `BIOMETRIC_OTP` | `"biometric_otp"` | Biometric + OTP combination authen

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/GrantType.ts1-20](https://github.com/nhsv-tradex/aaa/blob/59

#### Grant Type Flow Mapping

![Chart 1](aaa-images/aaa-section23-1.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/GrantType.ts1-20](https://github.com/nhsv-tradex/aaa/blob/59

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### OTP Configuration Constants

The `OTP_TYPE_URI_MAP` constant defines partner-specific URI patterns and K

#### OTP Type URI Mapping Structure

![Chart 2](aaa-images/aaa-section23-2.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blo

#### MAS Partner OTP Types

 | OTP Type | URI | Kafka Topic |
|--- | --- | --- |
 | `MATRIX_CARD` | `/api/v1/services/eqt/authCardMatrix` | `mas-rest-bridge
 | `SMART_OTP` | `/api/v2/bridge/verifyOtp` | `mas-ws-bridge` |
 | `HARDWARE_OTP` | `/api/v2/bridge/verifyOtp` | `mas-ws-bridge` |
 | `SMS_OTP` | `/api/v2/bridge/verifySMSOTP` | `mas-ws-bridge` |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts2-12](https://github.com/nhsv-tradex/aaa/blo

#### KIS Partner OTP Types
```

```
 | OTP Type | URI | Kafka Topic |
|--- | --- | --- |
 | `MATRIX_CARD` | `/api/v1/auth/matrix/verifyKisCard` | `mas-rest-bridge`
 | `SMART_OTP` | `/api/v1/otp/verify` | `mas-ws-bridge` |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/OtpTypeUriMap.ts14-21](https://github.com/nhsv-tradex/aaa/bl

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### Biometric Authentication Constants

Biometric authentication uses several constant groups to manage device regi

#### Biometric Error Keys

The `BIOMETRIC_KEYS` object defines error codes and status messages for bio

 | Constant | Value | Purpose |
|--- | --- | --- |
 | `NOT_FOUND` | `"LOGIN_BIOMETRIC_NOT_FOUND"` | Biometric registration not
 | `VERIFY_FAILED` | `"LOGIN_BIOMETRIC_SIGNATURE_VERIFICATION_FAILED"` | Si
 | `PASSWORD_NOT_MATCH` | `"LOGIN_BIOMETRIC_PASSWORD_NOT_MATCH"` | Password
 | `PUBLIC_KEY_EXISTED` | `"LOGIN_BIOMETRIC_PUBLIC_KEY_EXISTED"` | Public k
 | `OTP_NOT_VERIFIED` | `"LOGIN_BIOMETRIC_OTP_DOES_NOT_VERIFIED"` | OTP ver
 | `OTP_VERIFY_FAILED` | `"LOGIN_BIOMETRIC_OTP_VERIFIED_FAILED"` | OTP veri
 | `VERIFY_BIOMETRIC_FAILED` | `"VERIFY_BIOMETRIC_FAILED"` | General biomet
 | `BIOMETRIC_UNREGISTER_SUCCESS` | `"BIOMETRIC_UNREGISTER_SUCCESS"` | Biom
 | `BIOMETRIC_ACTIVE_ON_ANOTHER_DEVICE` | `"BIOMETRIC_ACTIVE_ON_ANOTHER_DEV

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/biometricKeys.ts1-11](https://github.com/nhsv-tradex/aaa/blo

#### Biometric Status Values

The `BIOMETRIC_STATUS` object defines the activation state of biometric reg

 | Status | Value | Description |
|--- | --- | --- |
 | `INACTIVE` | `"INACTIVE"` | Biometric authentication disabled or cancell
 | `ACTIVE` | `"ACTIVE"` | Biometric authentication enabled and operational

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan
```

```
[src/constants/biometricKeys.ts13-16](https://github.com/nhsv-tradex/aaa/bl

#### Biometric Cancellation Reasons

The `BIOMETRIC_CANCEL_REASON` object defines Vietnamese-language reasons fo

 | Reason | Value | English Translation |
|--- | --- | --- |
 | `CHANGE_DEVICE` | `"hủy do user đổi device"` | Cancelled due to device c
 | `CHANGE_PASS` | `"hủy do user đổi password"` | Cancelled due to password
 | `CANCEL` | `"hủy do user"` | Cancelled by user |
 | `OTP_VERIFY` | `"huỷ do user không xác thực OTP"` | Cancelled due to OTP

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/biometricKeys.ts18-23](https://github.com/nhsv-tradex/aaa/bl

#### Partner-Specific Biometric Login Errors

The `BIOMETRIC_LOGIN_ERROR` object maps partners to their specific error co

 | Partner | Error Code | Description |
|--- | --- | --- |
 | `kbsv` | `"INVALID_CLIENT_CREDENTIAL"` | KBSV invalid credential error |
 | `vcsc` | `"LOGIN_WRONG_PASSWORD"` | VCSC wrong password error |
 | `kbfinance` | `"INVALID_CLIENT_CREDENTIAL"` | KB Finance invalid credent
 | `kis` | `"LOGIN_WRONG_PASSWORD"` | KIS wrong password error |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/biometricKeys.ts25-30](https://github.com/nhsv-tradex/aaa/bl

#### Biometric Constant Relationships

![Chart 3](aaa-images/aaa-section23-3.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/biometricKeys.ts1-30](https://github.com/nhsv-tradex/aaa/blo

* * *

### General Service Constants
```

```
The `Constants` class defines miscellaneous service-wide constant strings u

 | Constant | Value | Purpose |
|--- | --- | --- |
 | `ACCOUNT_NUMBER_IS_EMPTY` | `"ACCOUNT_NUMBER_IS_EMPTY"` | Validation err
 | `TOKEN_REQUEST_MFA_DATA_FIELD_IS_EMPTY` | `"TOKEN_REQUEST_MFA_DATA_FIELD
 | `NHSV_OTP` | `"NHSV_OTP"` | NHSV OTP provider identifier |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/Constants.ts1-5](https://github.com/nhsv-tradex/aaa/blob/599

* * *

### Error Code Constants

The error constants module defines standard error codes used throughout the

 | Constant | Value | Context |
|--- | --- | --- |
 | `WRONG_OTP` | `"WRONG_OTP"` | OTP verification failed |
 | `WRONG_MOBILE_OTP` | `"WRONG_MOBILE_OTP"` | Mobile OTP verification fail
 | `VERIFY_TYPE_NOT_FOUND` | `"VERIFY_TYPE_NOT_FOUND"` | Unknown verificati
 | `INTERNAL_SERVER_ERROR` | `"INTERNAL_SERVER_ERROR"` | Generic server err

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/errors.ts1-4](https://github.com/nhsv-tradex/aaa/blob/59961d

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### Message Constants

The `Messages` class defines user-facing and system messages, particularly

 | Constant | Value | Purpose |
|--- | --- | --- |
 | `OTP_GENERATE_TO_FAST` | `"OTP_GENERATE_TO_FAST"` | OTP requested too qu
 | `OTP_LIMIT_GENERATE` | `"OTP_LIMIT_GENERATE"` | Daily OTP generation lim
 | `SUCCESS` | `"SUCCESS"` | Generic success message |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/messages.ts1-5](https://github.com/nhsv-tradex/aaa/blob/5996
```

```
* * *

### User Level Constants

The `USER_LEVEL` object defines user role classifications in the system.

 | Level | Value | Description |
|--- | --- | --- |
 | `CUSTOMER` | `"CUSTOMER"` | End-user customer account |
 | `USER` | `"USER"` | Standard user account |

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/index.ts1-4](https://github.com/nhsv-tradex/aaa/blob/59961d6

* * *

### Update Object Keys

The `UPDATE_OBJECT_KEYS` object defines entity type identifiers used in upd

 | Key | Value | Purpose |
|--- | --- | --- |
 | `CLIENT` | `"CLIENT"` | OAuth2 client entity identifier for update opera

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/UpdateObjectKeys.ts1-3](https://github.com/nhsv-tradex/aaa/b

* * *

### Global Type Definitions

The `global.d.ts` file extends the Node.js global namespace to allow arbitr

#### Global Namespace Extension
```

```
declare module global {
  namespace NodeJS {
    interface Global {
      [key: string]: any,
    }
  }
}
```

```
This type definition allows the application to set and access global config

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.ts

[global.d.ts1-7](https://github.com/nhsv-tradex/aaa/blob/59961d60/global.d.

* * *

### Constant Usage Architecture

![Chart 4](aaa-images/aaa-section23-4.svg)

**Sources:**[](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/constan

[src/constants/GrantType.ts1-20](https://github.com/nhsv-tradex/aaa/blob/59

src/constants/OtpTypeUriMap.ts1-22](https://github.com/nhsv-tradex/aaa/blob

src/constants/biometricKeys.ts1-30](https://github.com/nhsv-tradex/aaa/blob

src/constants/Constants.ts1-5](https://github.com/nhsv-tradex/aaa/blob/5996

src/constants/errors.ts1-4](https://github.com/nhsv-tradex/aaa/blob/59961d6

src/constants/messages.ts1-5](https://github.com/nhsv-tradex/aaa/blob/59961

src/constants/index.ts1-4](https://github.com/nhsv-tradex/aaa/blob/59961d60

src/constants/UpdateObjectKeys.ts1-3](https://github.com/nhsv-tradex/aaa/bl

* * *

### Summary Table: All Constants by Category

 | Category | File | Key Constants | Primary Usage |
|--- | --- | --- | --- |
 | Grant Types | `GrantType.ts` | 18 grant type enumerations | Authenticati
 | OTP Configuration | `OtpTypeUriMap.ts` | Partner-specific URI/topic maps
 | Biometric Keys | `biometricKeys.ts` | Error codes, status values | Biome
 | Biometric Status | `biometricKeys.ts` | `ACTIVE`, `INACTIVE` | Device re
 | Biometric Cancellation | `biometricKeys.ts` | 4 cancellation reasons | A
 | Biometric Errors | `biometricKeys.ts` | 4 partner-specific errors | Erro
 | General Constants | `Constants.ts` | Validation messages | Request valid
 | Error Codes | `errors.ts` | 4 error code strings | Error handling |
```

```
 | Messages | `messages.ts` | OTP rate limiting messages | User feedback |
 | User Levels | `index.ts` | `CUSTOMER`, `USER` | Authorization |
 | Update Keys | `UpdateObjectKeys.ts` | Entity identifiers | Kafka update
 | Global Types | `global.d.ts` | Global namespace extension | TypeScript c

**Sources:** All constant files in[](https://github.com/nhsv-tradex/aaa/blo

[src/constants/](https://github.com/nhsv-tradex/aaa/blob/59961d60/src/const

---
```